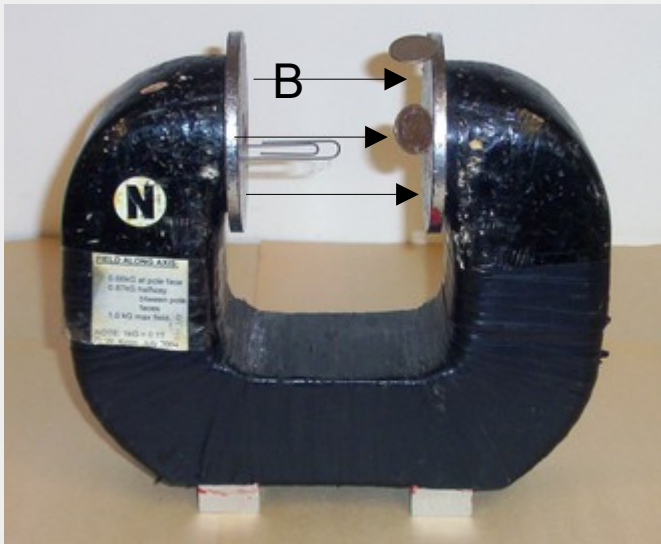
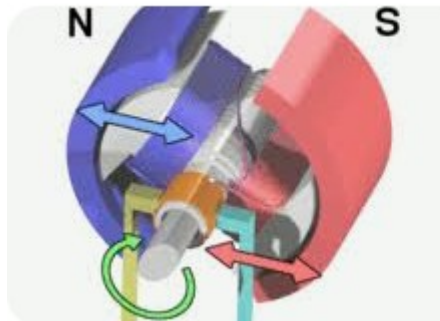


@ veronique lankar 2023

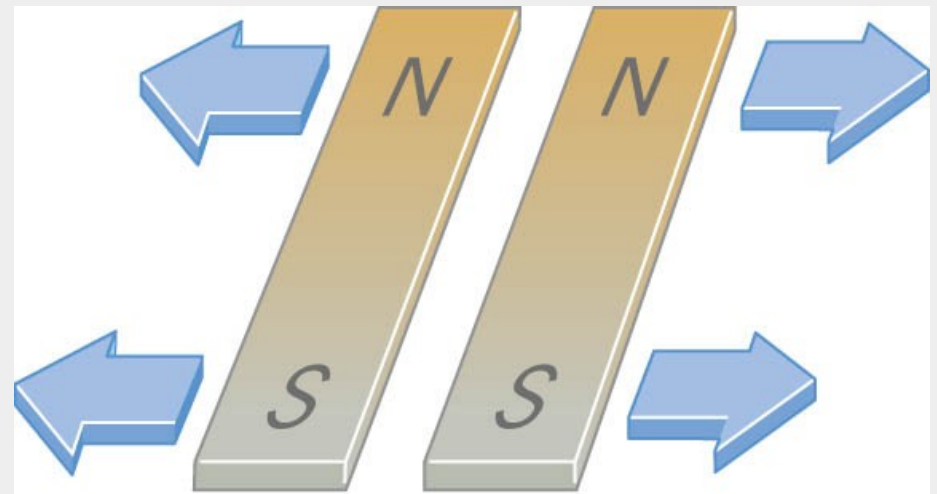
Magnetic Forces and Magnetic Fields



- historical notes
- magnetic field of magnets, Earth
- magnetic force on a wire carrying current
- magnetic force on a moving charge =
Lorentz force
- applications: cyclotron, motor,
mass spectrometers
- torque on a current loop

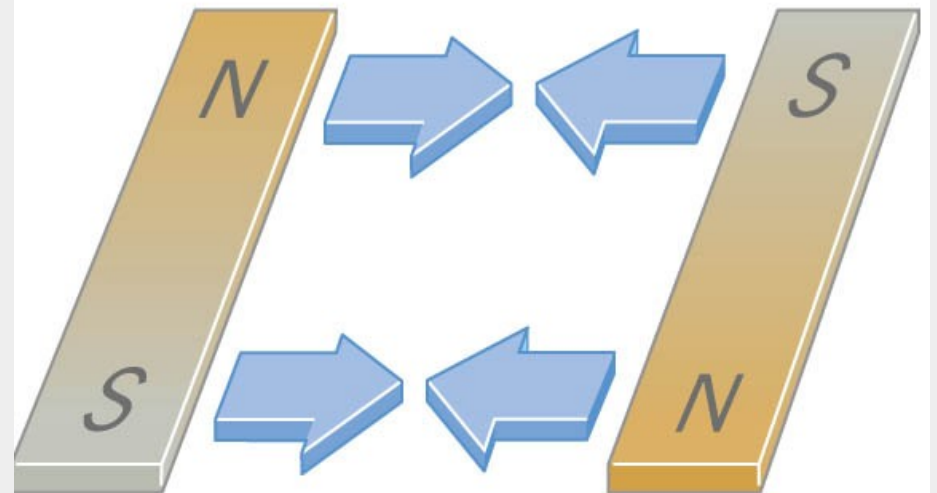


The behavior of magnetic poles is similar to that of like and unlike electric charges.



Like poles repel

(a)



Unlike poles attract

(b)

The word magnet was adopted in **Middle English** from **Latin** **magnetum** "**lodestone**", **ultimately from Greek** **μαγνήτις [λίθος]** (**magnētis [lithos]**) meaning "**[stone] from Magnesia**", a part of ancient Greece where lodestones were found. Lodestones, suspended so they could turn, were the first magnetic compasses.



Lodestone Photograph by Phil DEGGING...
fineartamerica.com · [In stock](#)

https://www.researchgate.net/figure/True-navigation-by-magnetic-parameters-indicated-in-spiny-lobsters-The-lobsters-were_fig6_246743980

See website. They stimulated a “fake” magnetic field. Got the animals lost and confused



Orientation, Homing, and Navigation in Caribbean Spiny Lobsters. ... This remarkable ability, known as a “magnetic map sense”, endows the lobsters with a sensitive navigational system rivaling that of sea turtles and salmon. Precisely how lobsters and other animals detect magnetic fields is not known

<https://pages.vassar.edu/sensoryecology/lost-and-confused-the-effect-of-magnetic-pulses-on-orientation-of-the-caribbean-spiny-lobster/>

William Gilbert



William Gilbert

Born	24 May 1544 Colchester, England
Died	30 November 1603 (aged 59) London, England
Nationality	English
Education	St John's College, Cambridge (M 1569)
Known for	Studies of magnetism, De Magnetete
	Scientific career
Fields	Physician

- contemporary of Kepler, Galileo,
- physician of Queen Elizabeth 1st (died same year)
She gave him money for his experiments.
- Found that magnetism can be destroyed
By heating and then restored.
- a magnet can get stronger if it is stoke by
Another magnet
- you can magnetize an iron bar
- Earth is a giant magnet

Before:

1269

Petrus Peregrinus

Wrote about magnets

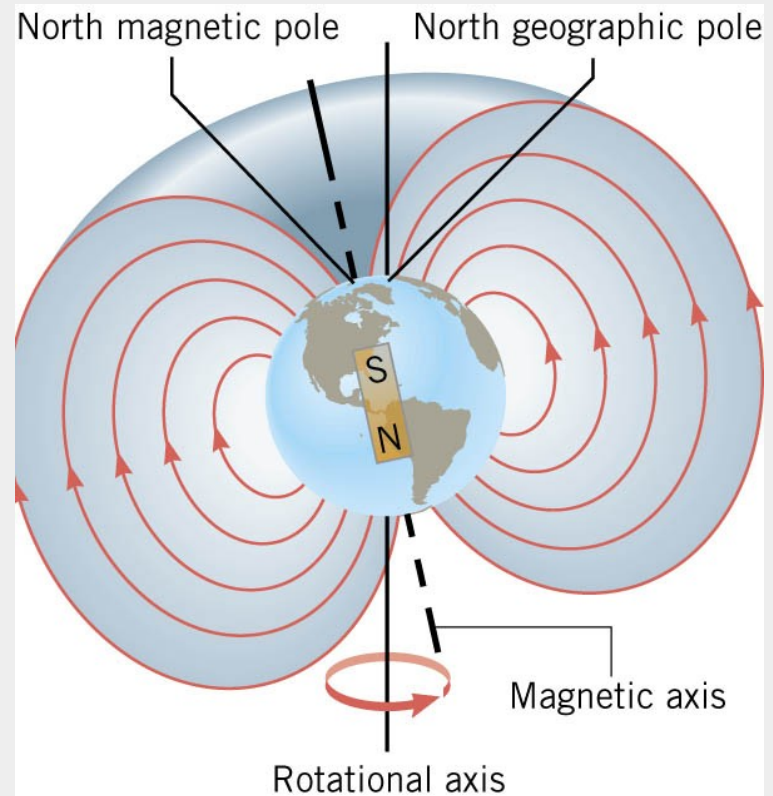
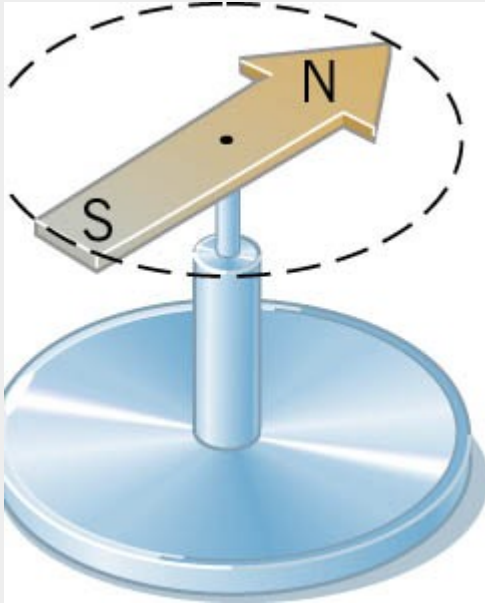
In a letter. Little is known
About him.

Video:

Destroying magnetism



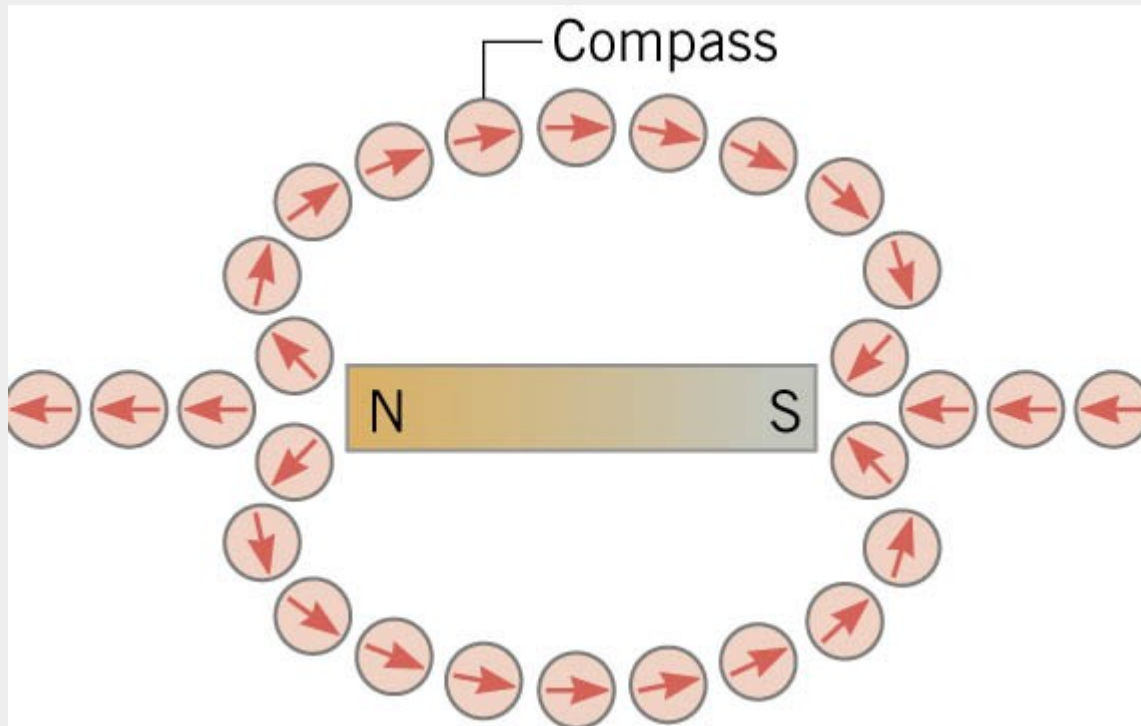
21.1 Magnetic Fields



The needle of a compass is permanent magnet that has a north magnetic pole (N) at one end and a south magnetic pole (S) at the other.

Movie magnetic field gauss law and simulation
Some calculus. Flux of magnetic field.

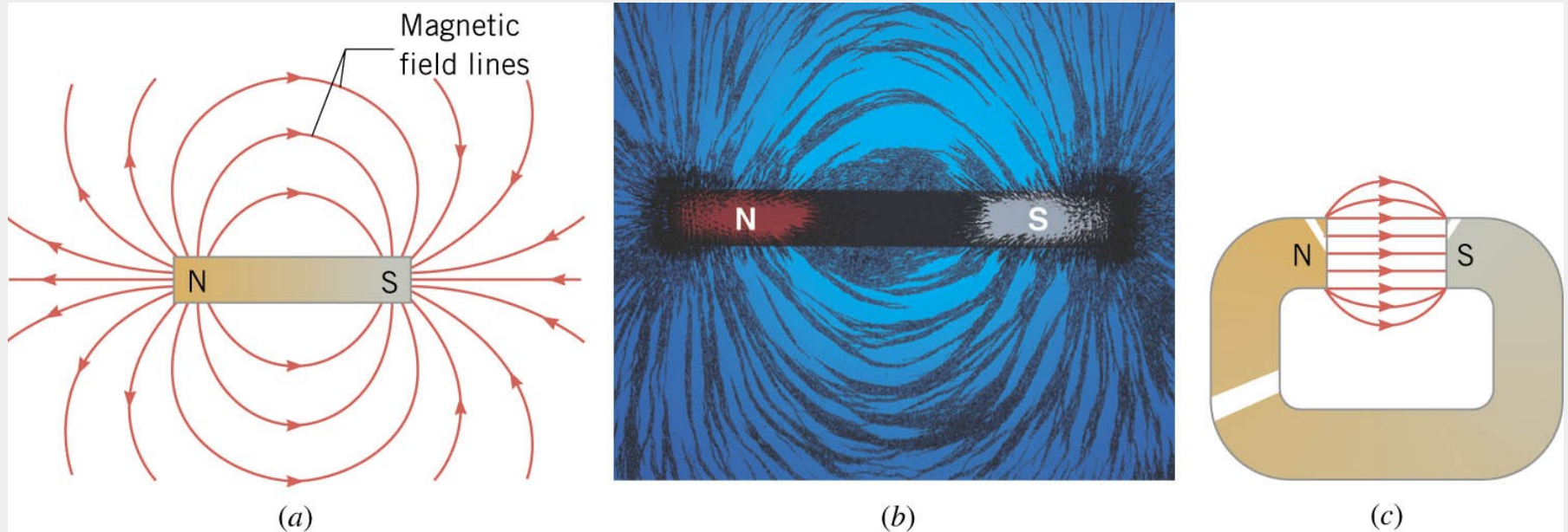
Surrounding a magnet there is a ***magnetic field***. The direction of the magnetic field at any point in space is the direction indicated by the north pole of a small compass needle placed at that point.



See exploration of physical sciences magnetic field

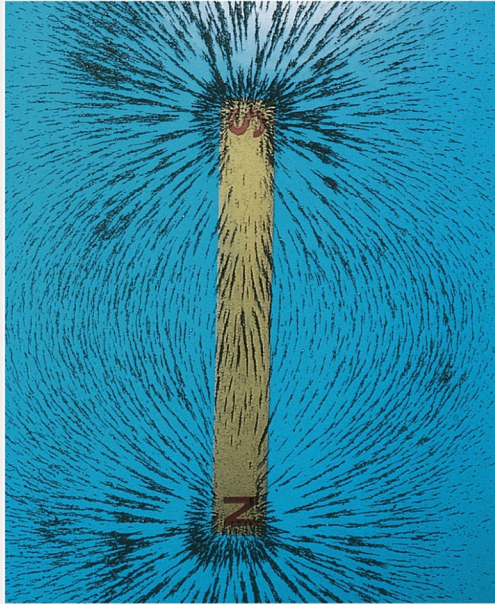
<http://phet.colorado.edu/en/simulation/magnets-and-electromagnets>

The magnetic field lines and pattern of iron filings in the vicinity of a bar magnet and the magnetic field lines in the gap of a horseshoe magnet. Note that the pattern is like the electric dipole



Notice how the horseshoe magnet produces an homogeneous Magnetic field. Same direction and same magnitude if we neglect The fringes

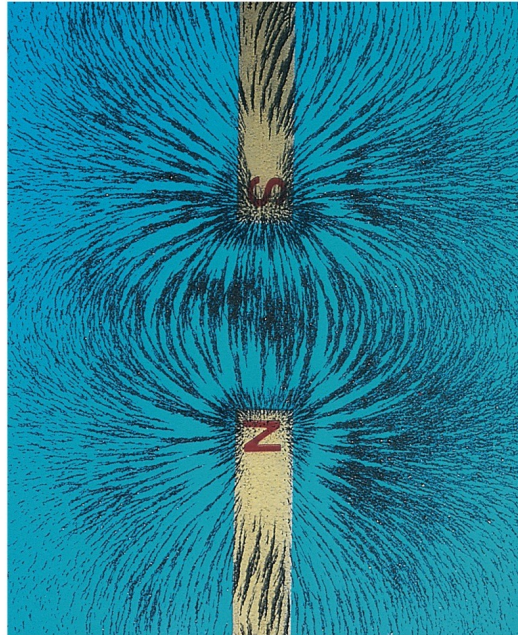
Magnetic field pattern surrounding a bar magnet



a

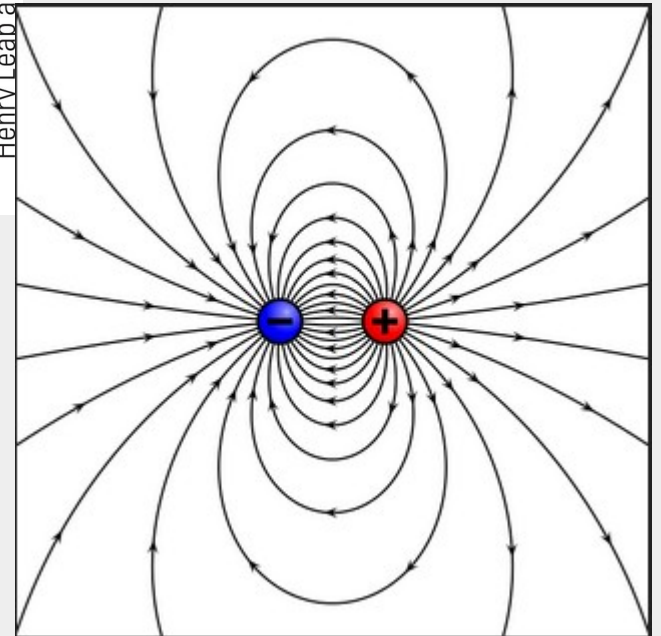
Henry Leap and Jim Lehman

Magnetic field pattern between *opposite* poles (N-S) of two bar magnets

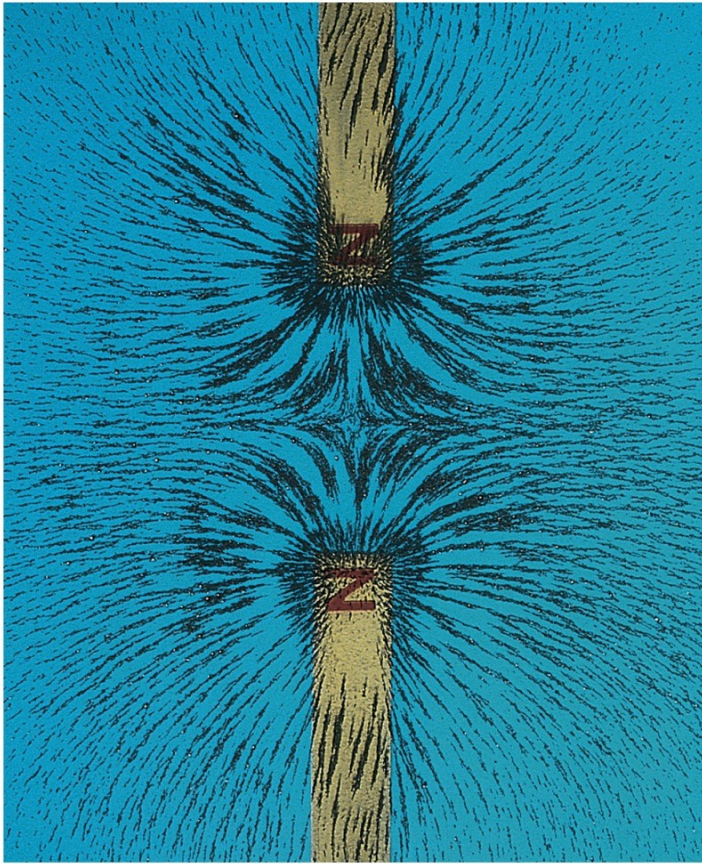


b

Henry Leap and Jim Lehman



Magnetic field pattern
between *like* poles (N-N)
of two bar magnets



c

Henry Leap and Jim Lehman

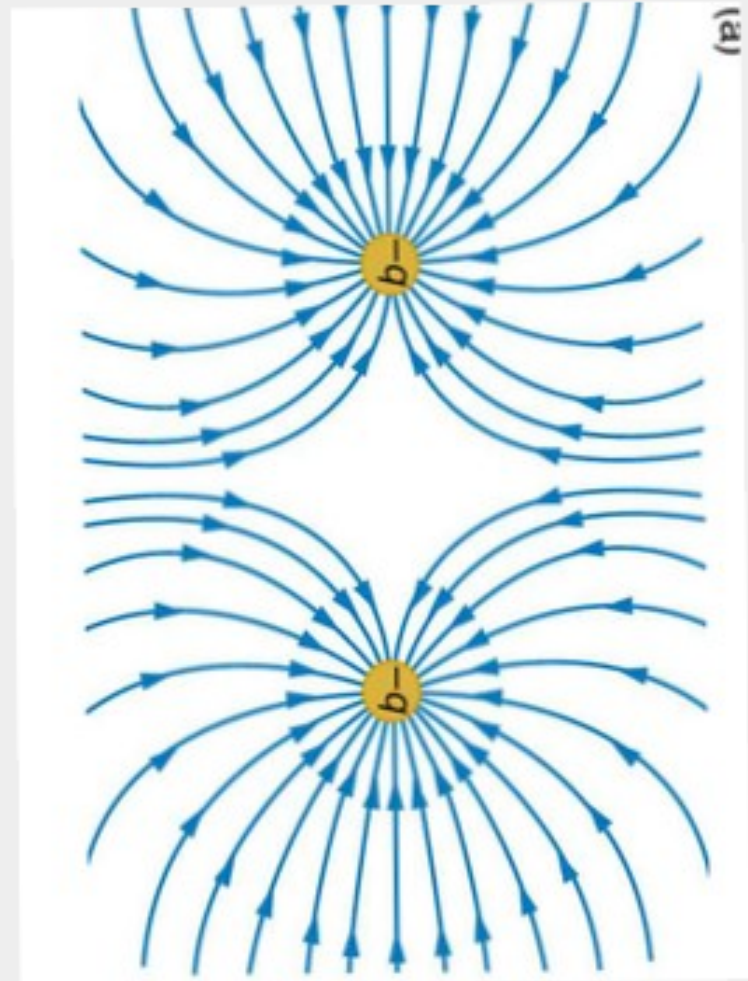


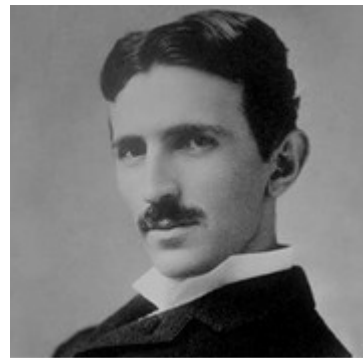
Table 29.1**Some Approximate Magnetic Field Magnitudes**

Source of Field	Field Magnitude (T)
Strong superconducting laboratory magnet	30
Strong conventional laboratory magnet	2
Medical MRI unit	1.5
Bar magnet	10^{-2}
Surface of the Sun	10^{-2}
Surface of the Earth	0.5×10^{-4}
Inside human brain (due to nerve impulses)	10^{-13}

© Cengage Learning. All Rights Reserved.

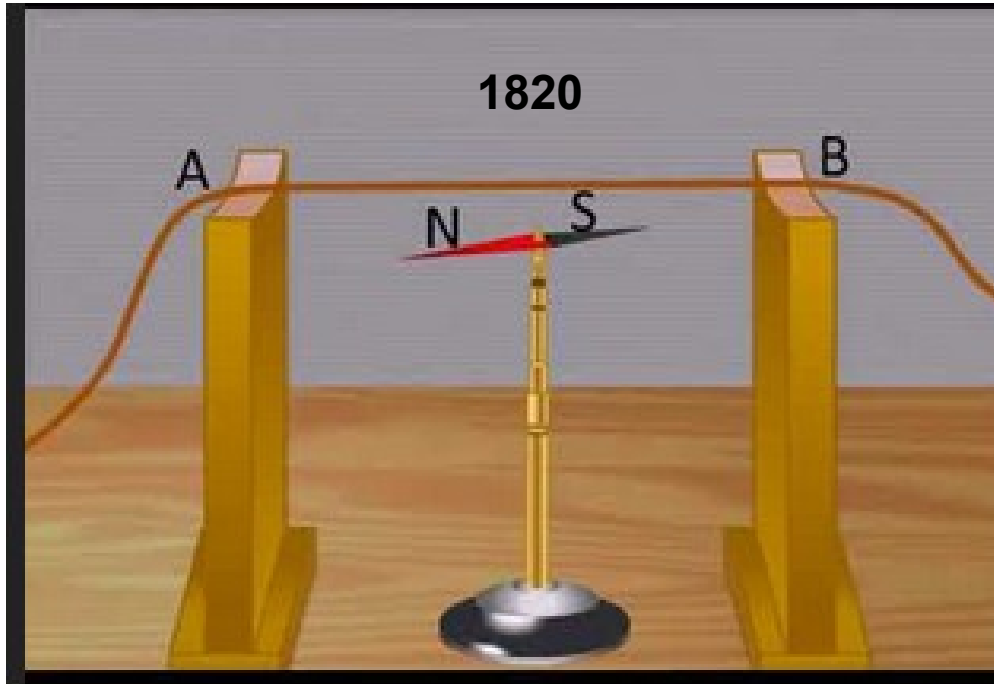
1 Telsa = 10,000 Gauss

Nikola Tesla (1856-1943)



Skeletal muscles $\rightarrow 10^{-10}$ T.

Oersted experiment



Hans Christian Ørsted was a Danish physicist and chemist who discovered that electric currents create magnetic fields, which was the first connection found between electricity and magnetism. Oersted's law



Simulation exploration

Electric Currents Produce Magnetic Fields

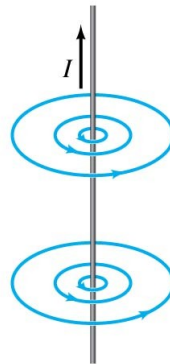
Experiment shows that an electric current produces a magnetic field.



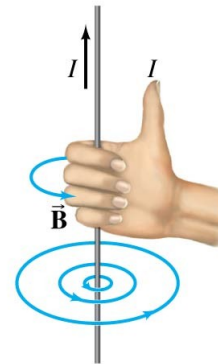
(a)



(b)



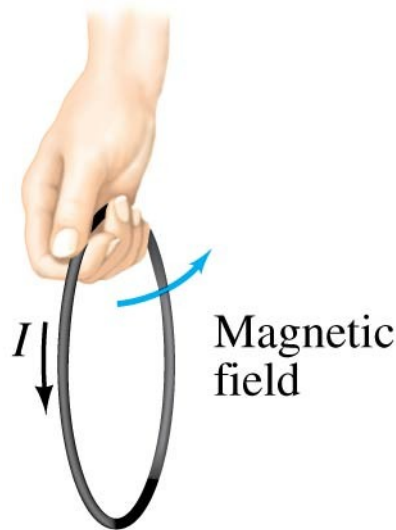
(c)

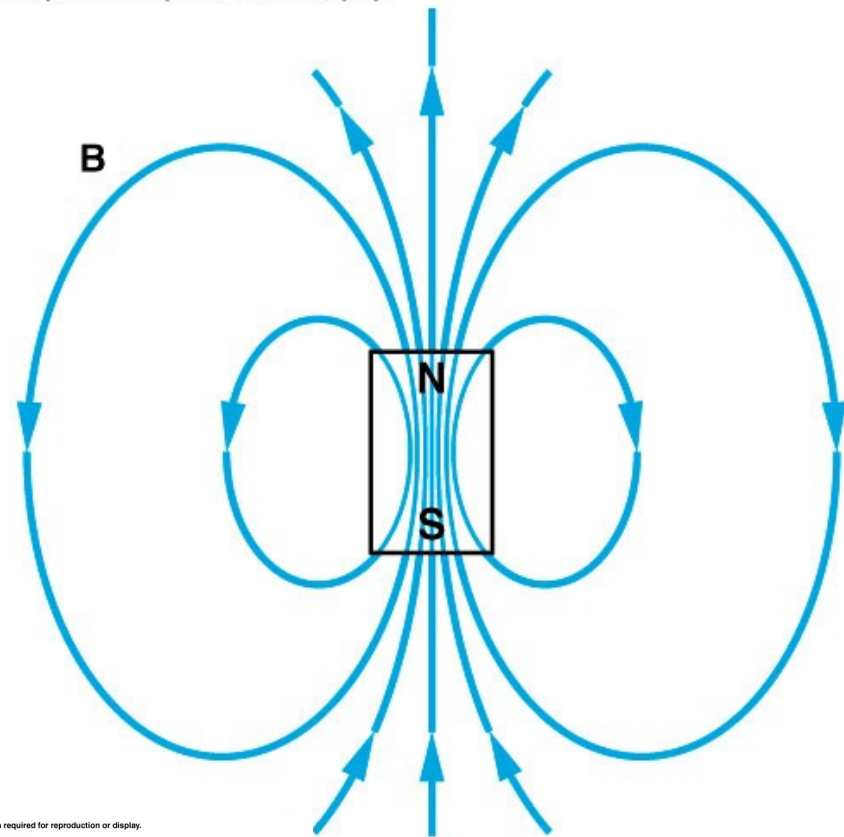
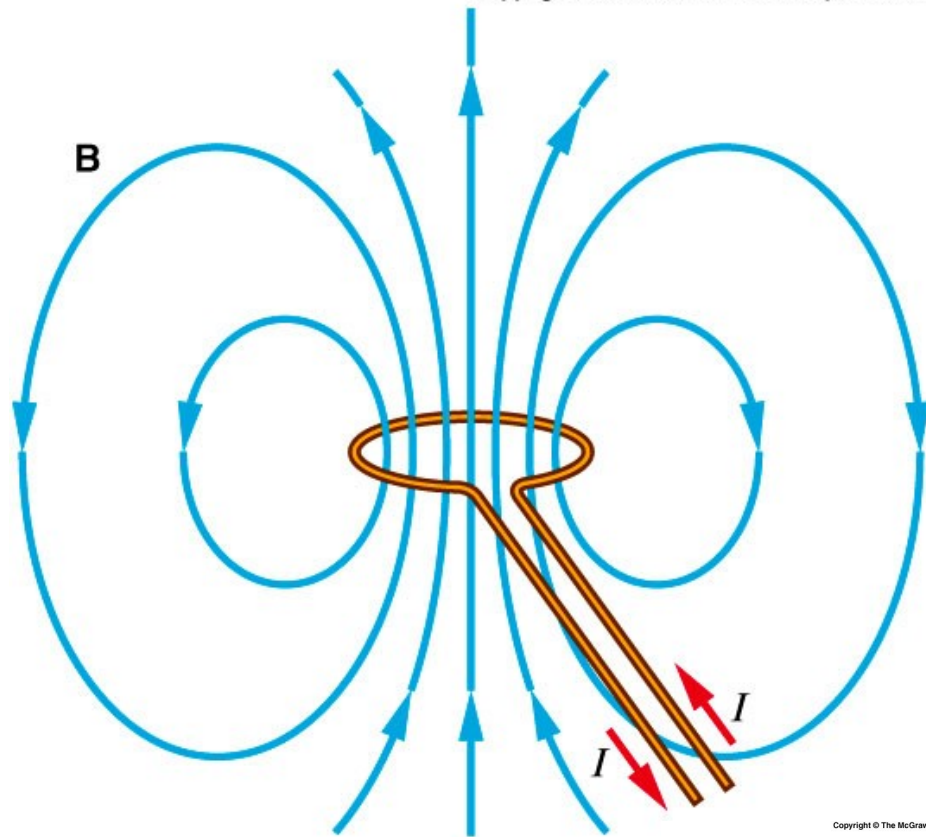


(d)

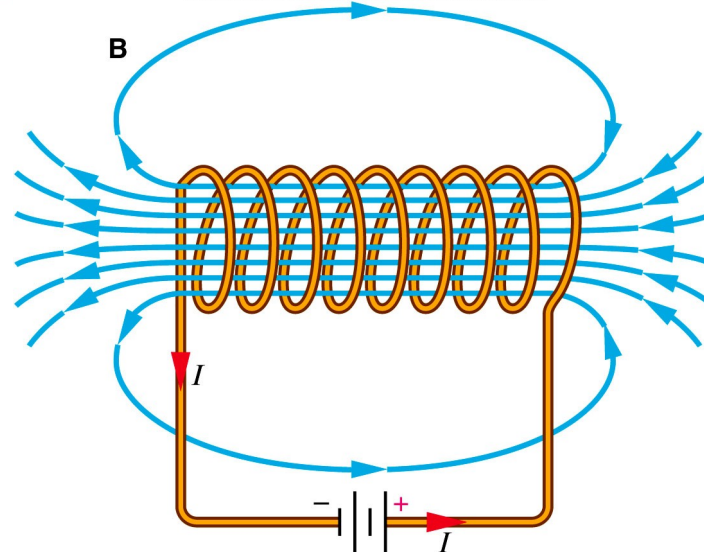
Electric Currents Produce Magnetic Fields

The direction of the field is given by a right-hand rule.

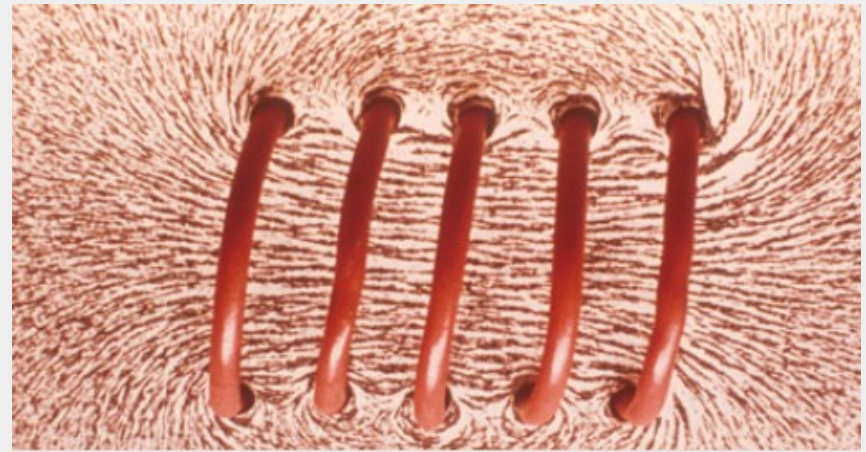
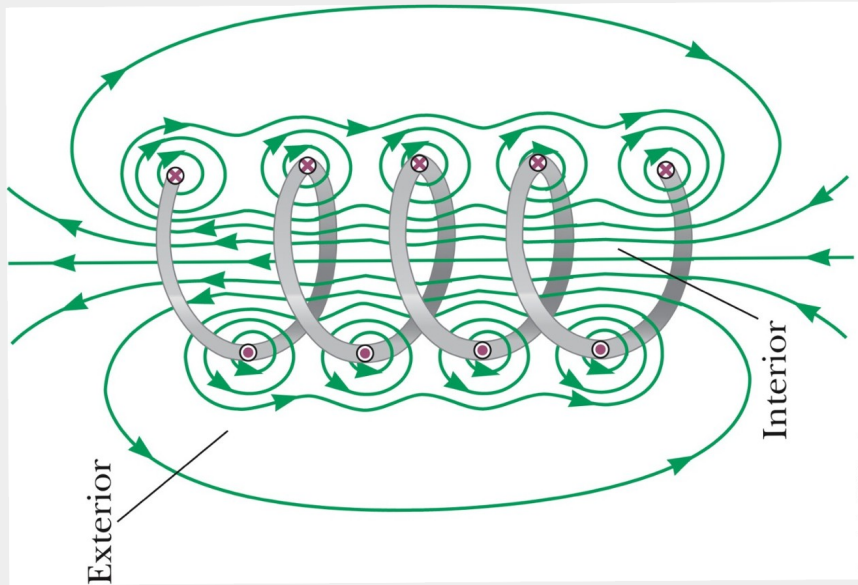




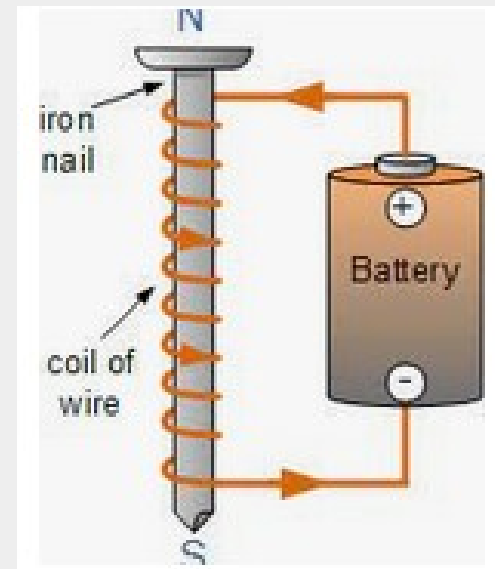
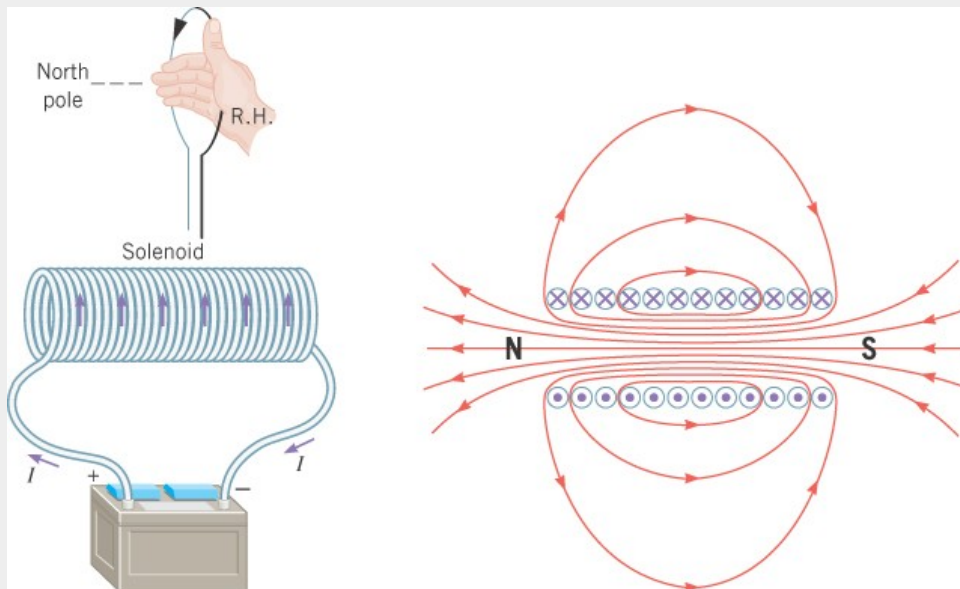
Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

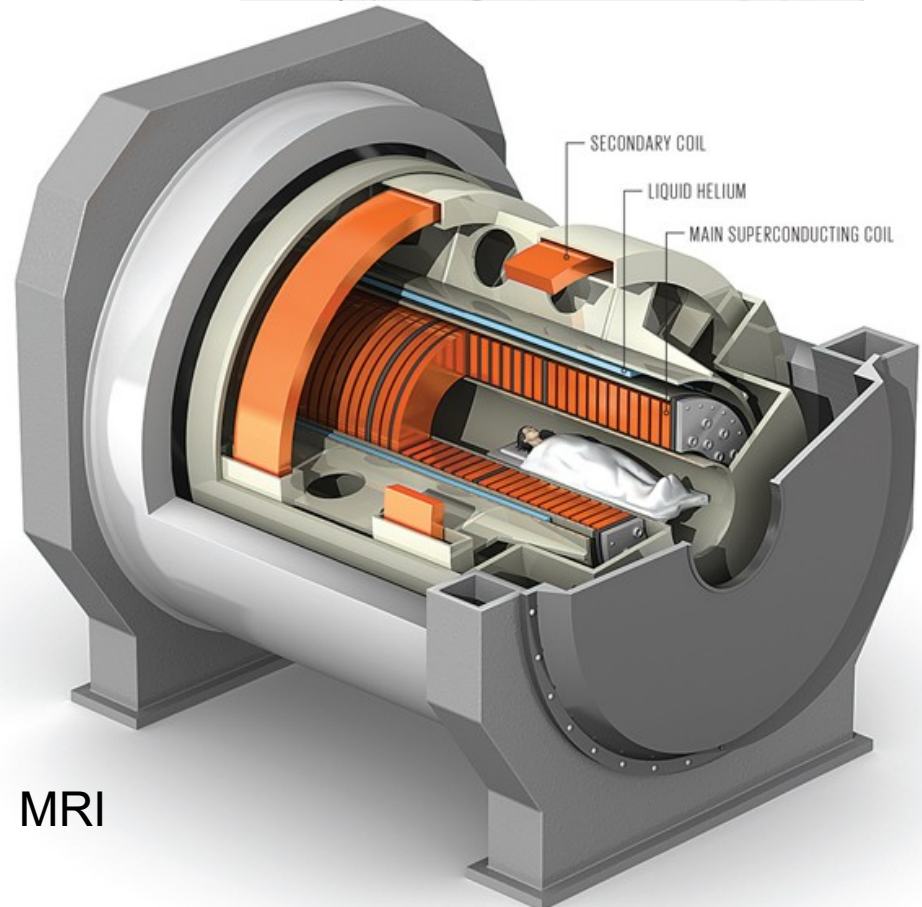
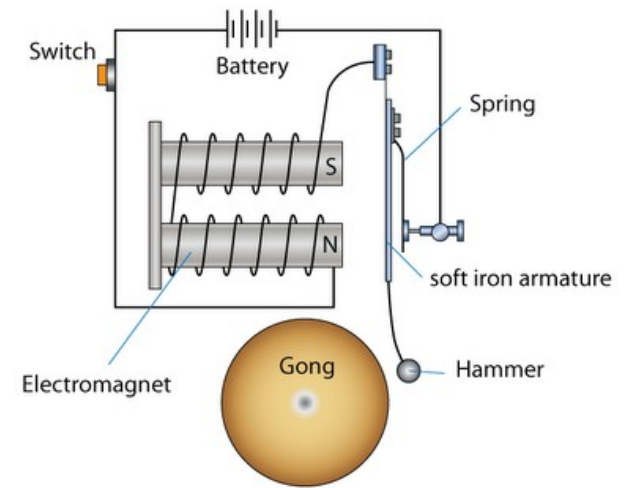


Application → electromagnet

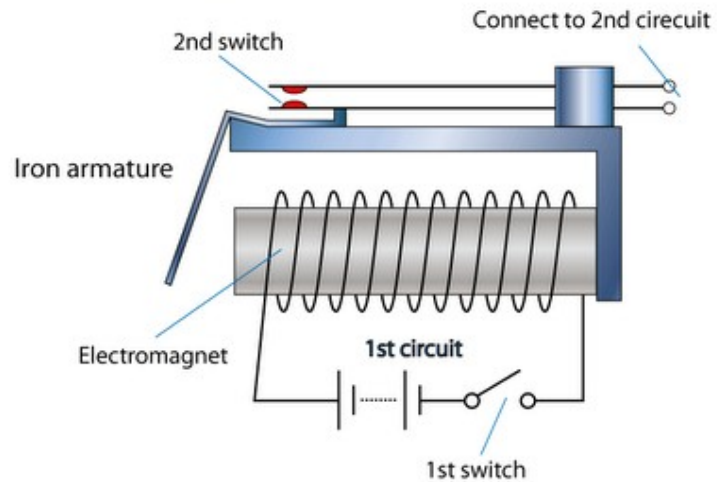


Application: electromagnet

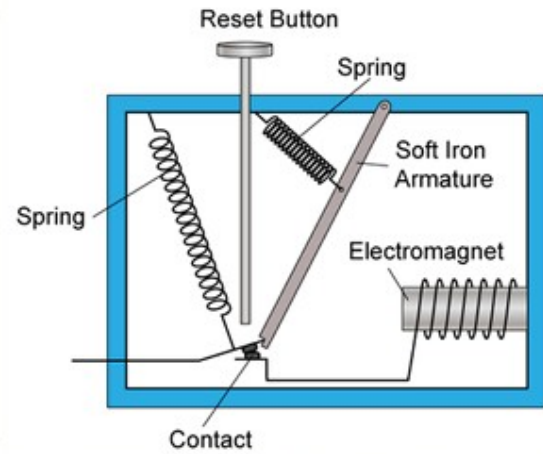




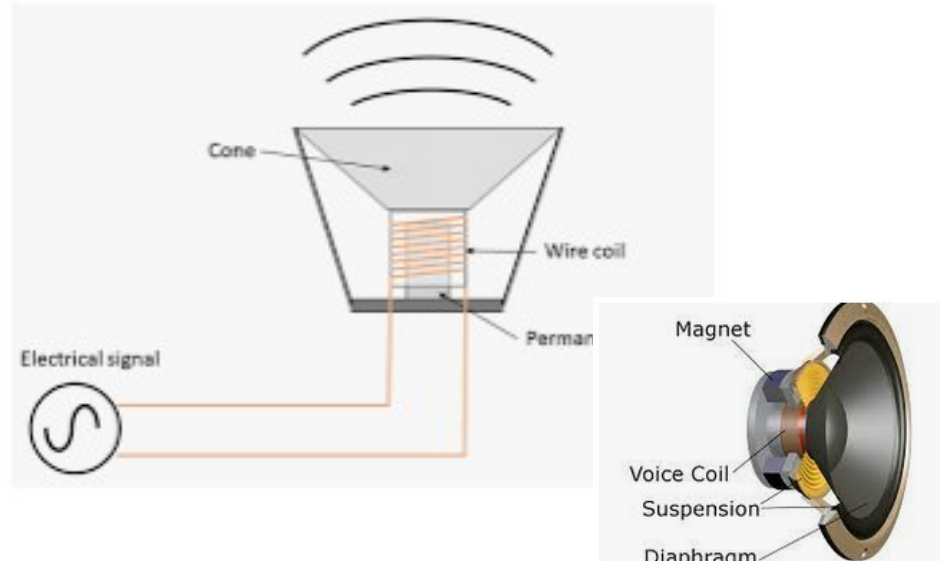
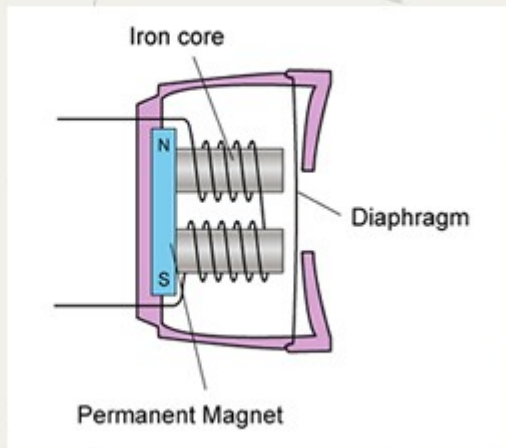
Electromagnetic Relay



Circuit Breaker



Telephone Earpiece



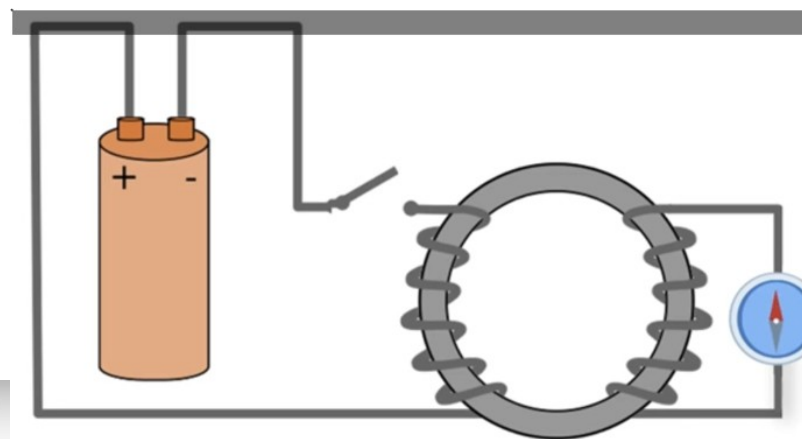
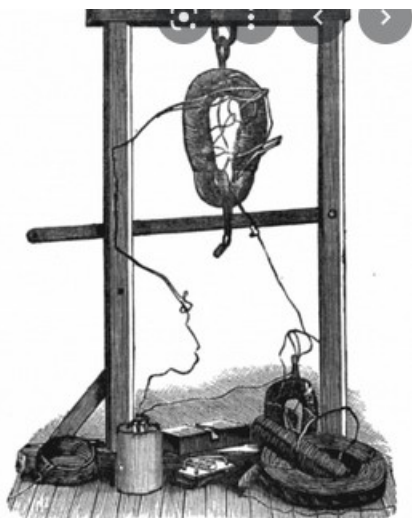
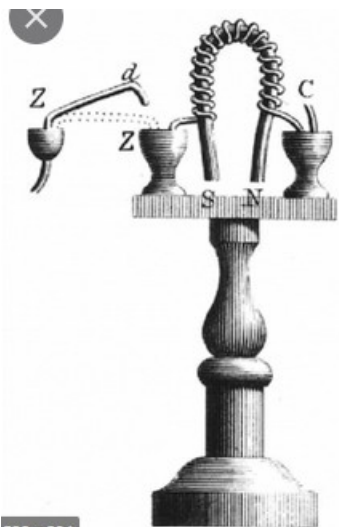


William
Sturgeon

He made the first electromagnet

Faraday's used electromagnets to
Show how to produce electricity
With electromagnets.

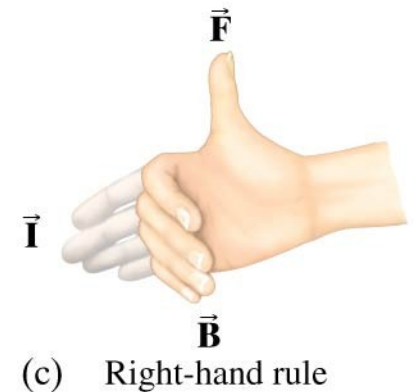
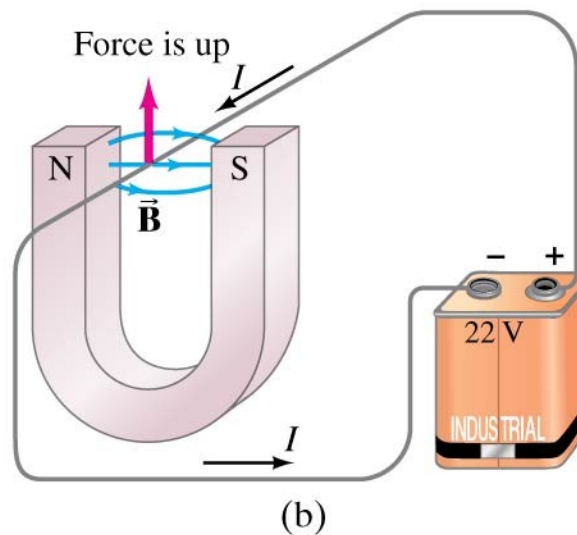
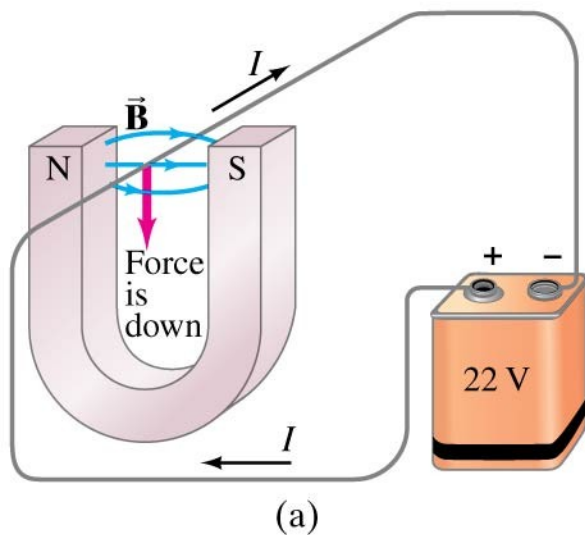
Michael Callan improved the
Electromagnets



Watch lat...

Force on an Electric Current in a Magnetic Field; Definition of $\vec{B}$

A magnet exerts a force on a current-carrying wire. The direction of the force is given by a right-hand rule.



Force on an Electric Current in a Magnetic Field; Definition of B

The force on the wire depends on the current, the length of the wire, the magnetic field, and its orientation.

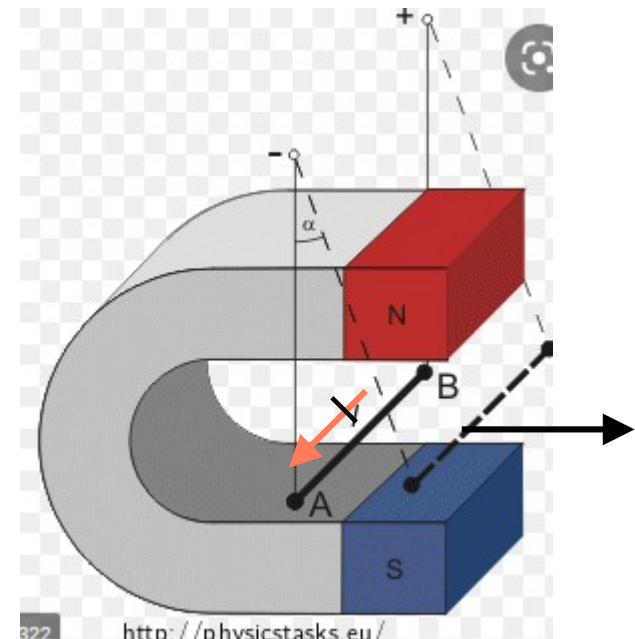
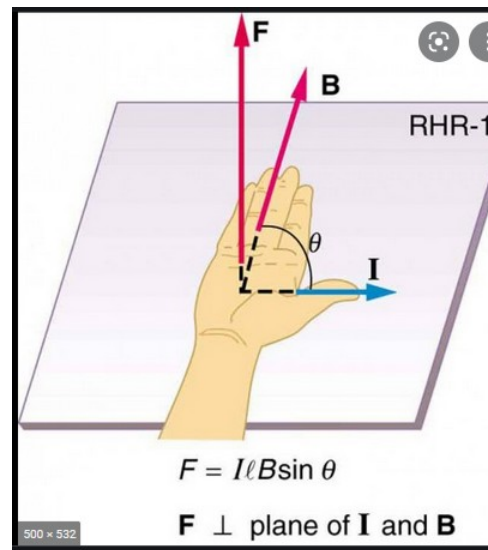
$$F = I\ell B \sin \theta.$$

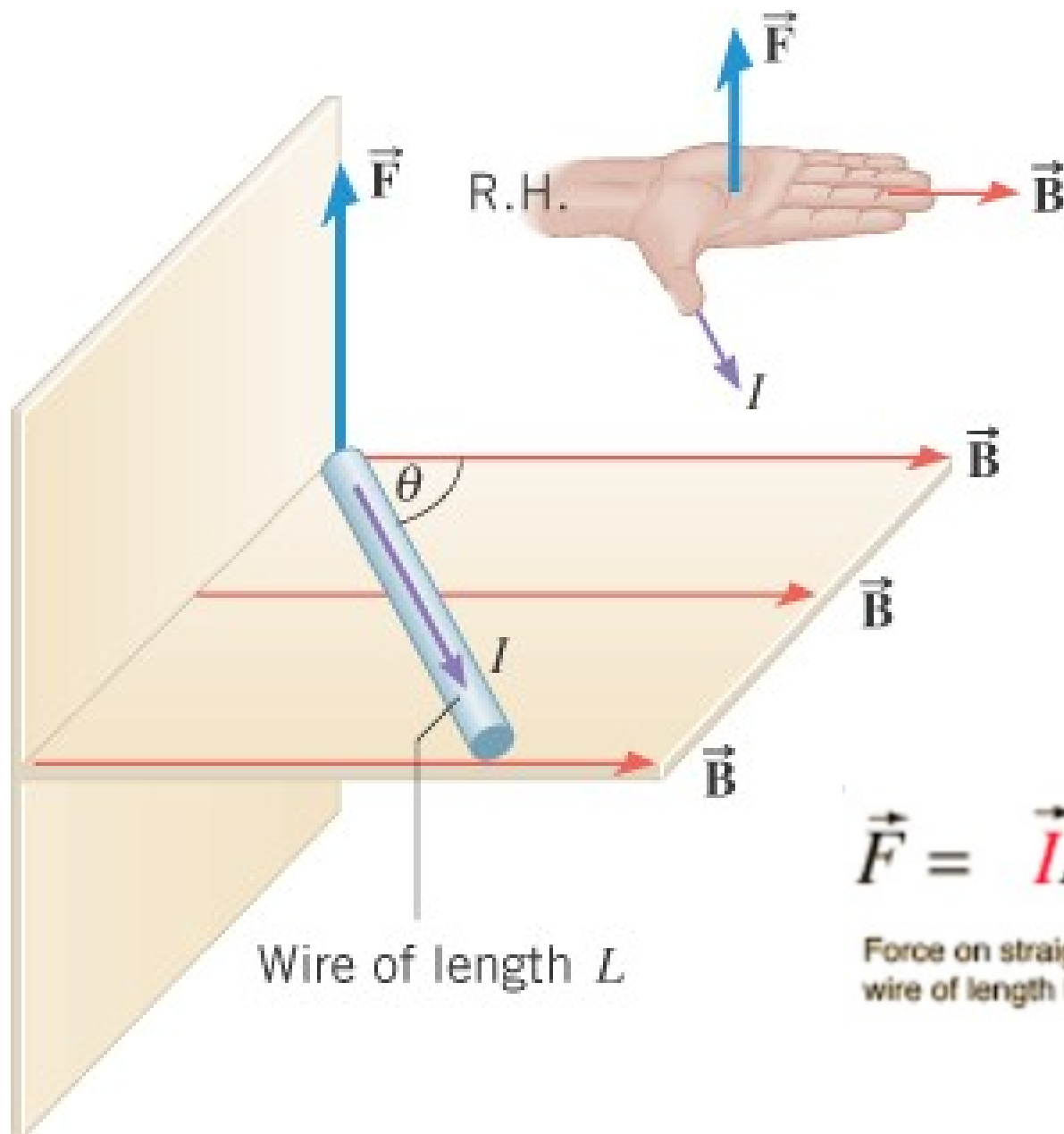
This equation defines the magnetic field B.₍₂₀₋₁₎

θ is the angle between B and I.

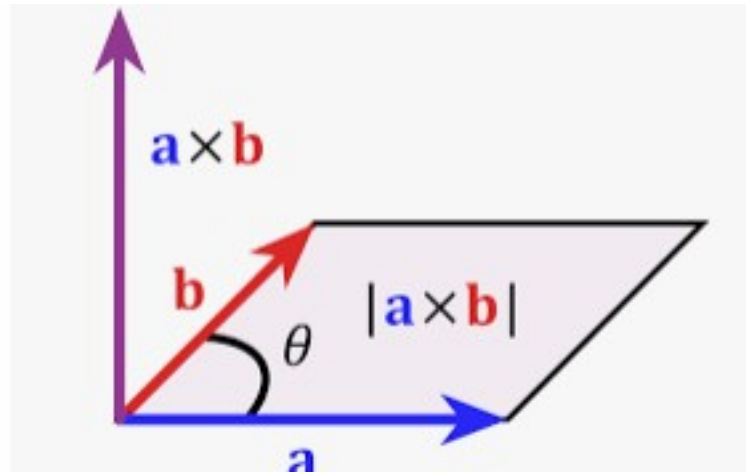
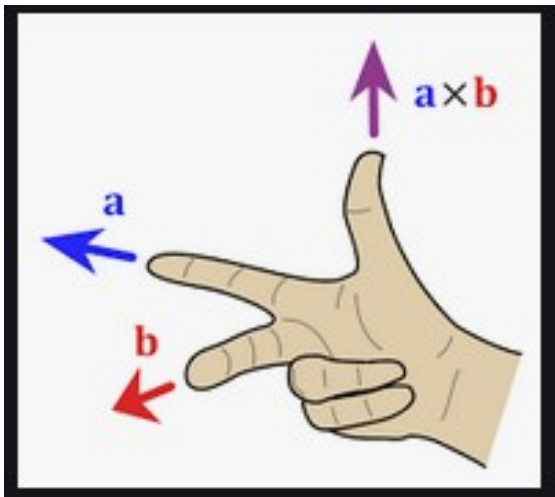
If they are parallel

→ No force.





$$\vec{F} = \vec{I}L \times \vec{B}$$



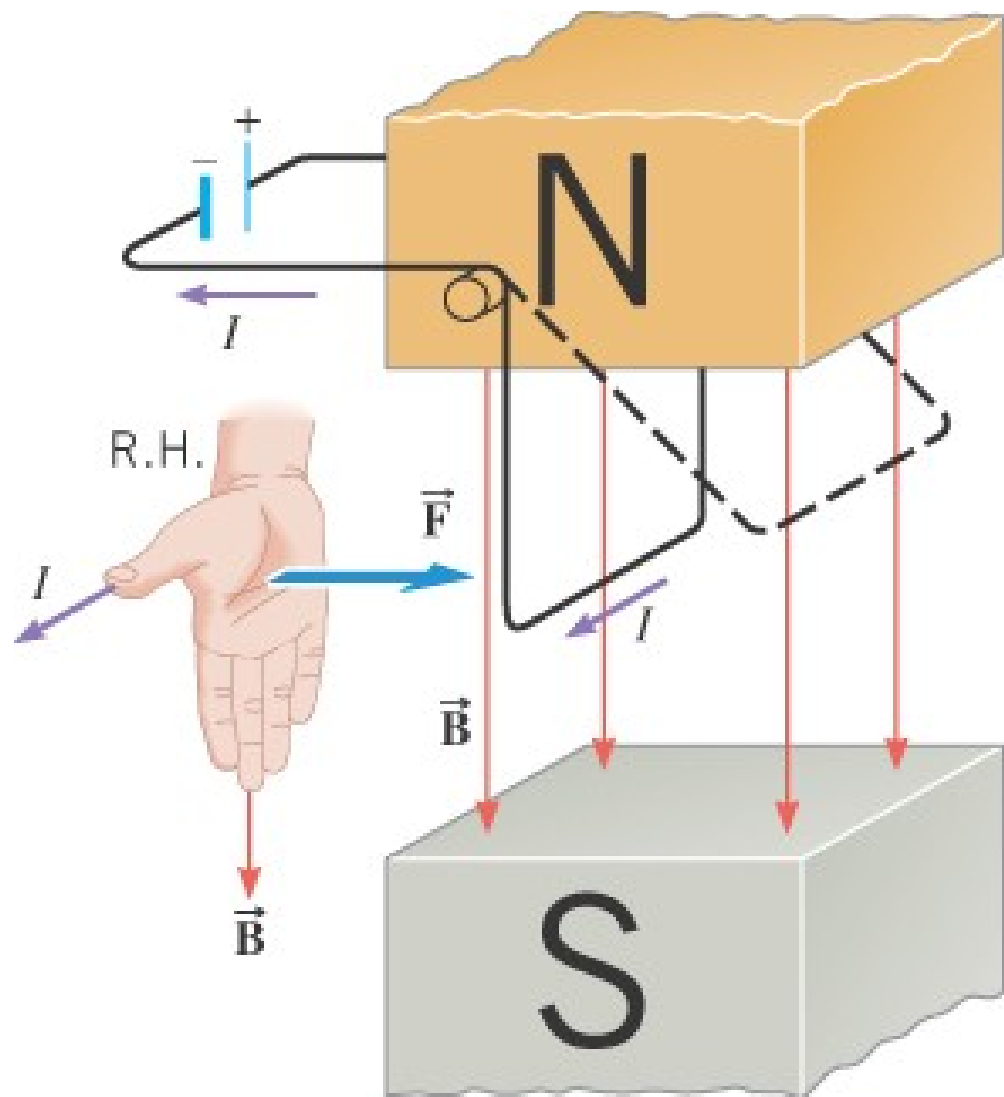
Vector Cross Product Formula

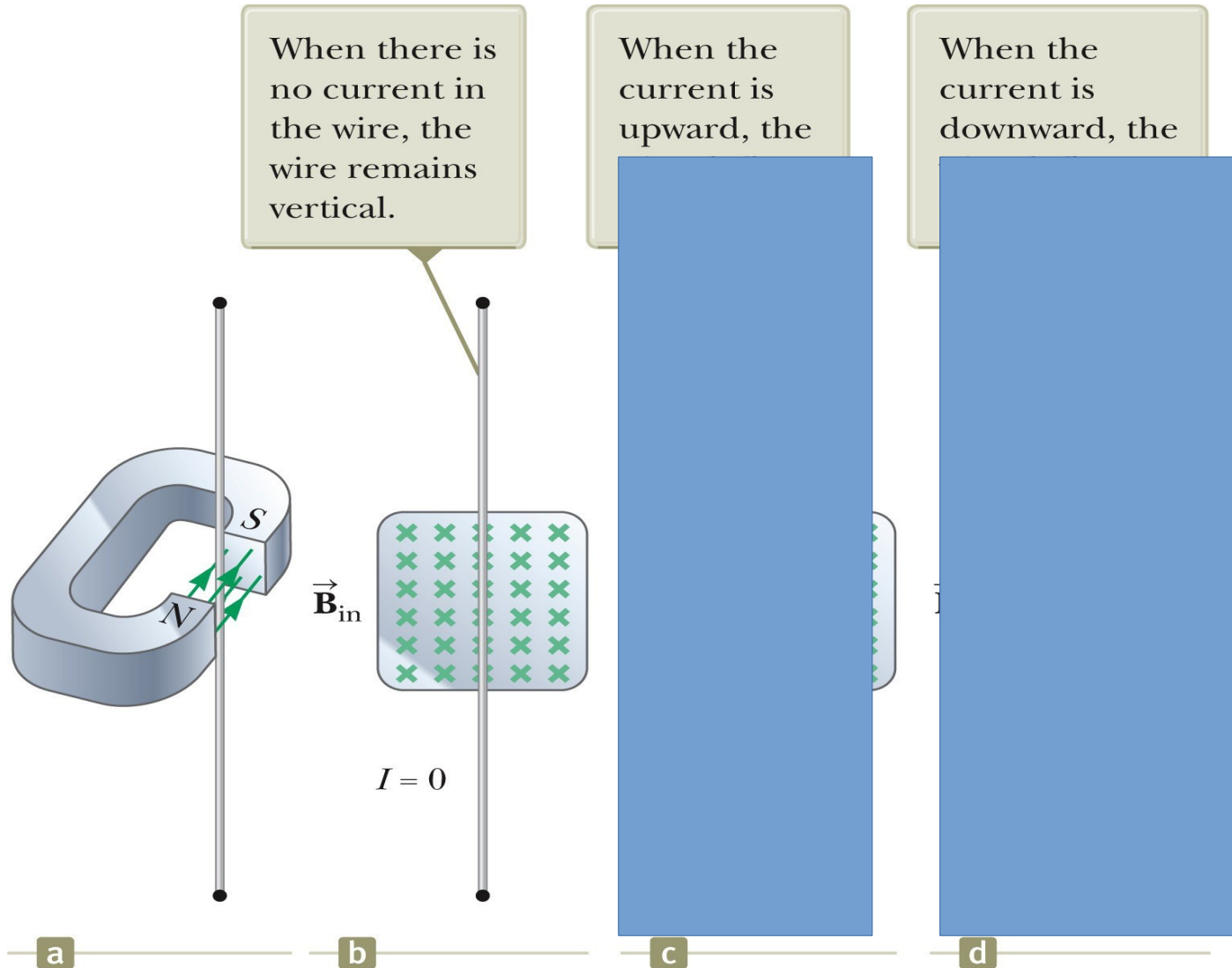


$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

$$\vec{a} \times \vec{b} = i (a_2 b_3 - a_3 b_2) + j (a_3 b_1 - a_1 b_3) + k (a_1 b_2 - a_2 b_1)$$

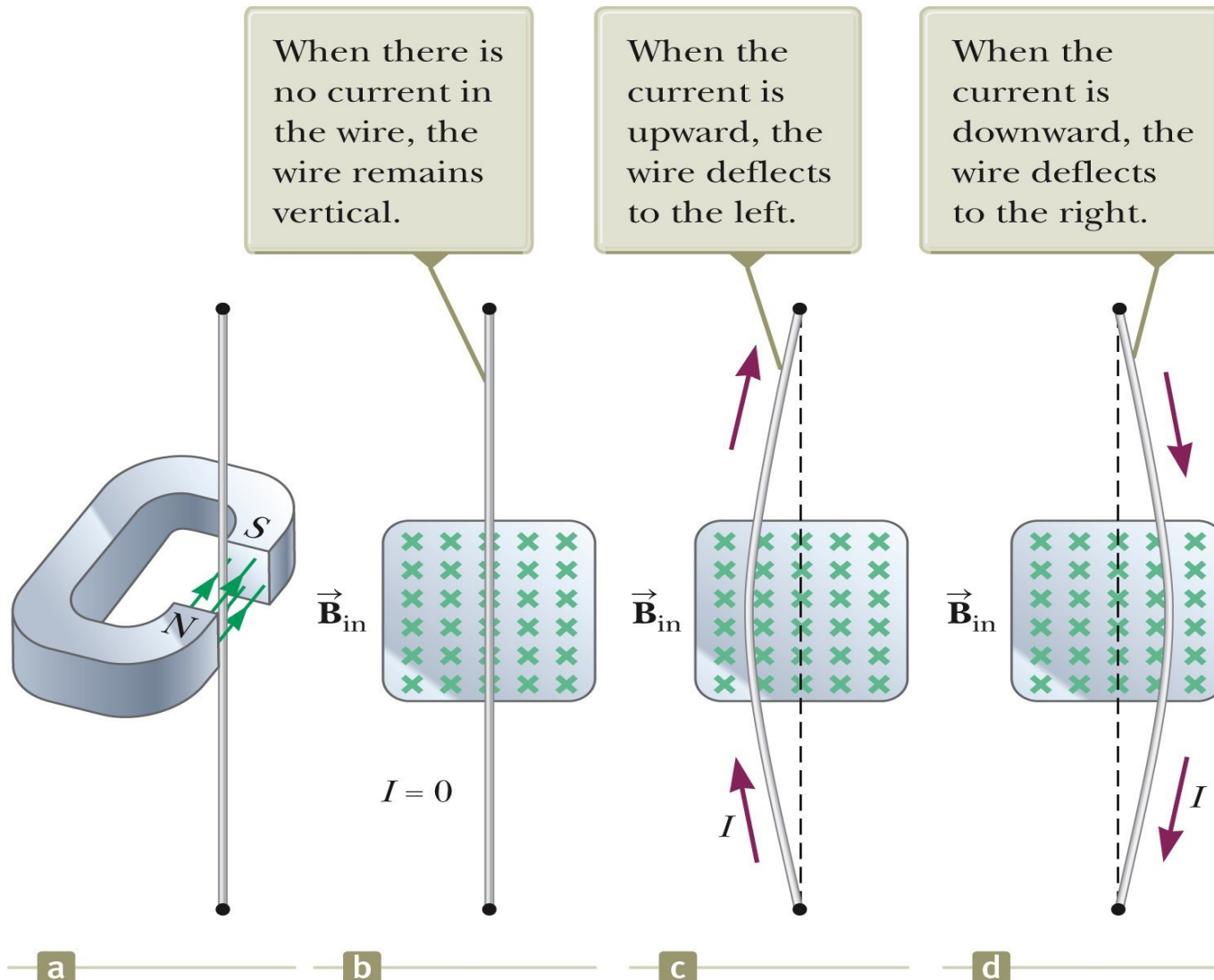
The magnetic force on the moving charges pushes the wire to the right.





© Cengage Learning. All Rights Reserved.

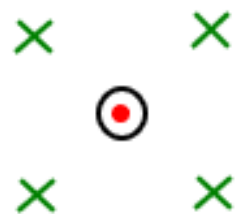
The force is called the Lorentz force (demo)
See next page for demo



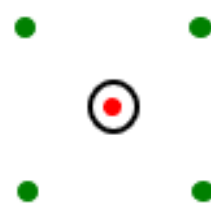
© Cengage Learning. All Rights Reserved.

The force is called the Lorentz force (demo)
See next page for demo

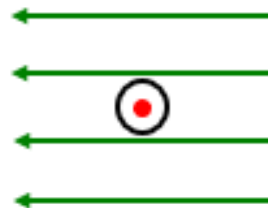
6. A straight, horizontal length of copper wire is immersed in a uniform **magnetic field**. The current through the wire is out of page. Which magnetic field can possibly suspend this wire to balance the gravity?



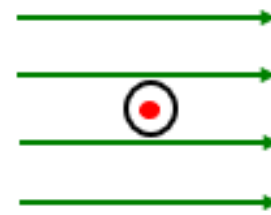
A



B



C



D

Force on an Electric Current in a Magnetic Field; Definition of B

Unit of B: the tesla, T.

$$1 \text{ T} = 1 \text{ N/A} \cdot \text{m}$$

Another unit sometimes used: the gauss (G).

$$1 \text{ G} = 10^{-4} \text{ T}$$

What is the magnetic field that exerts a force of 2.4×10^{-4} newtons on a current of 12 amperes in a wire 30 centimeters long set perpendicular to the field?

A wire carrying 1.5 amperes has a length of 20 centimeters in a magnetic field of 40 milliteslas. If the wire is perpendicular to the field, how much force does the field exert on the wire?

— 2 —

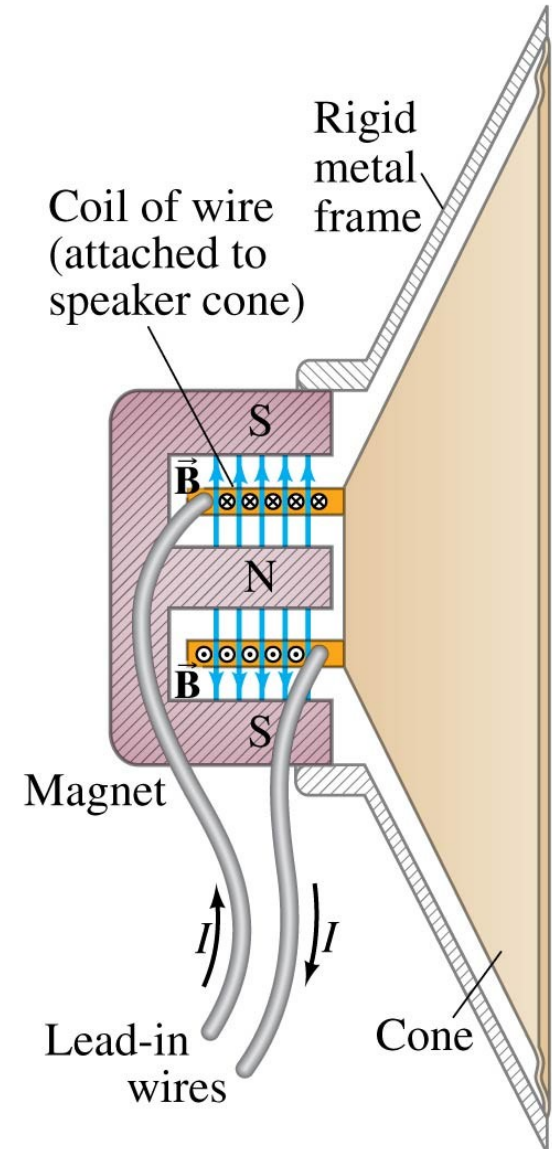
What are the direction and magnitude of the force that an upward-directed magnetic field of 2.0 milliteslas exerts on a north-flowing current of 10 amperes in a wire 15 centimeters long?

The force on a 0.80m wire that is perpendicular to Earth's magnetic field is 0.12 N . What is the current in the wire? (Bearth= 5×10^{-5} T)

A copper wire 40cm long carries a current of 6.0A and weighs 0.35N. A certain magnetic field is strong enough to balance the force of gravity on the wire. What is the strength of the magnetic field?

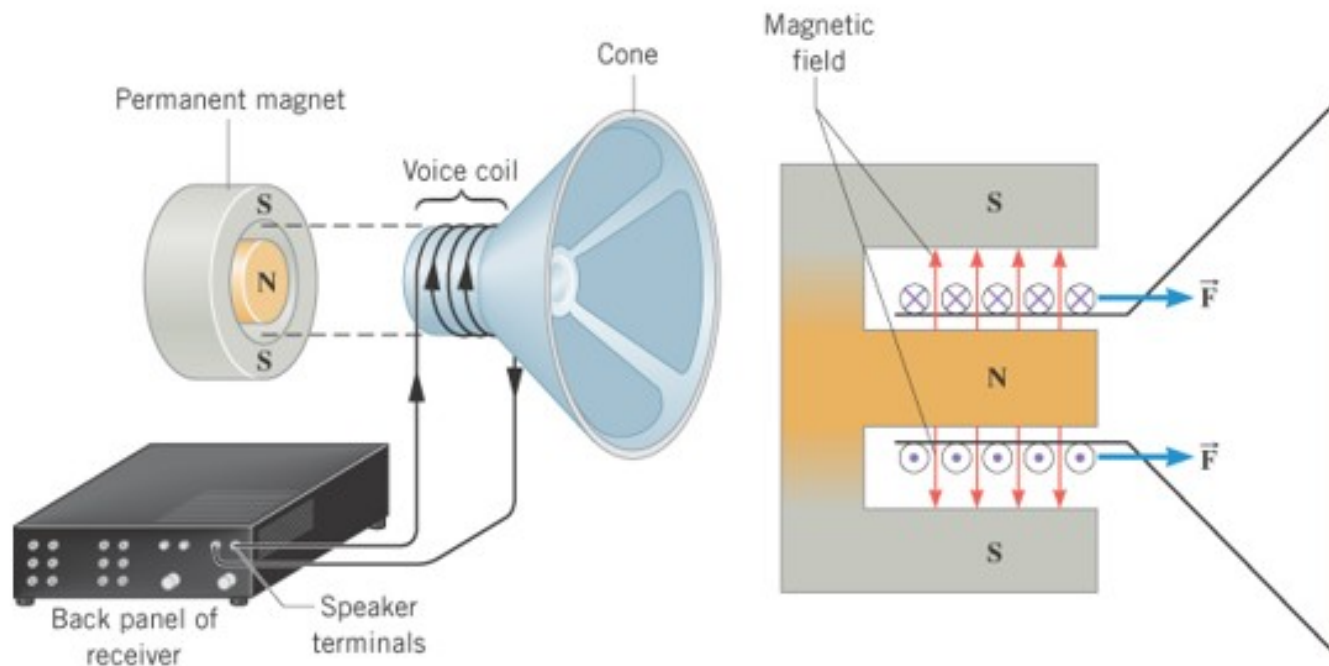
20-10 Applications: Galvanometers, Motors, Loudspeakers

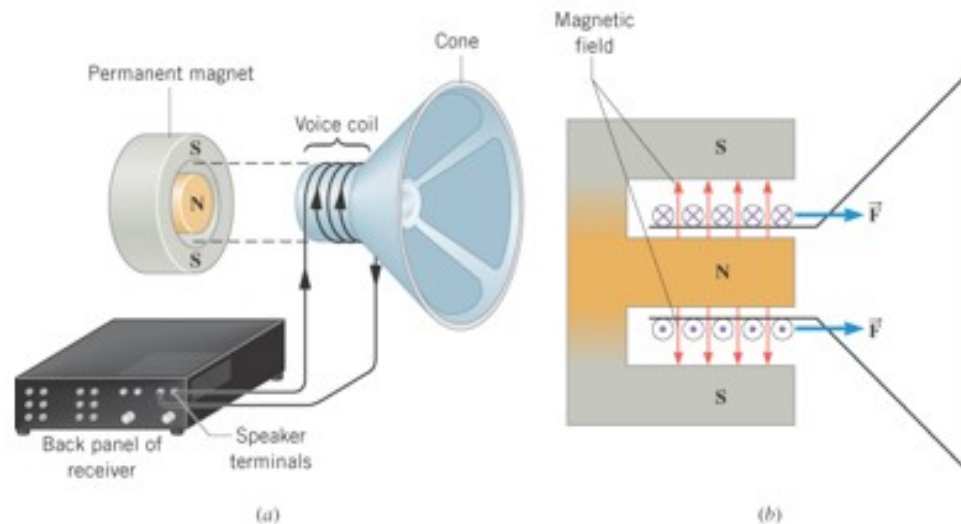
Loudspeakers use the principle that a magnet exerts a force on a current-carrying wire to convert electrical signals into mechanical vibrations, producing sound.



Example 5 The Force and Acceleration in a Loudspeaker

The voice coil of a speaker has a diameter of 0.0025 m, contains 55 turns of wire, and is placed in a 0.10-T magnetic field. The current in the voice coil is 2.0 A. (a) Determine the magnetic force that acts on the coil and the cone. (b) The voice coil and cone have a combined mass of 0.0200 kg. Find their acceleration.





(a)
$$F = ILB \sin \theta$$

$$= (2.0 \text{ A}) [55\pi(0.0025 \text{ m})] (0.10 \text{ T}) \sin 90^\circ$$

$$= 0.86 \text{ N}$$

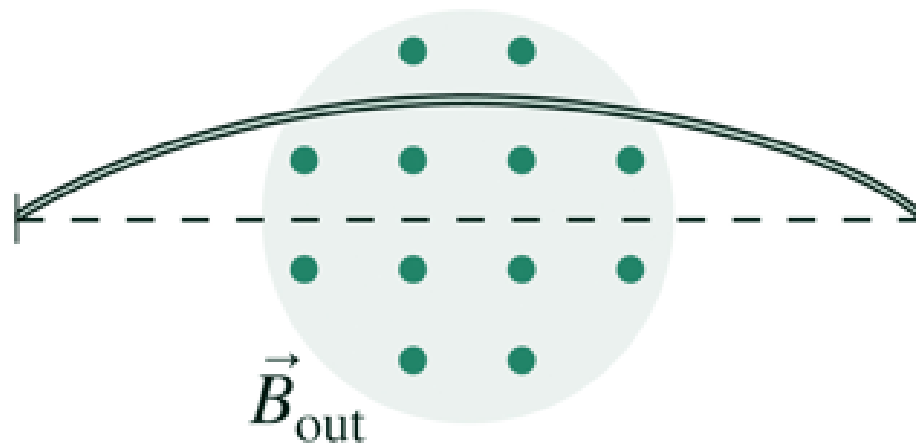
(b)
$$a = \frac{F}{m} = \frac{0.86 \text{ N}}{0.020 \text{ kg}} = 43 \text{ m/s}^2$$

1) The magnetic field in a loudspeaker is 0.15T. The wire consists of 250 turns wound on a 2.5 cm diameter cylindrical form. The resistance of the wire is 8 ohms. Find the force exerted on the wire when 15V is placed across the wire.

2) A wire carrying 15A of current has a length of 25cm in a magnetic field of 0.85T. The force on a current-carrying wire in a uniform magnetic field can be found using the equation $F = BIL\sin(\text{angle})$, angle between wire and B. Find the force on the wire if it makes an angle with the magnetic field lines of :
90 , 45 , 0 degrees

- The figure shows a flexible conducting wire passing through a magnetic field that points out of the page. The wire is deflected upward, as shown. In which direction is current flowing in the wire?

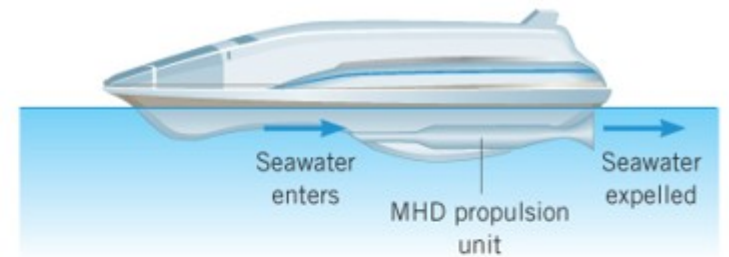
- A. To the left
- B. To the right



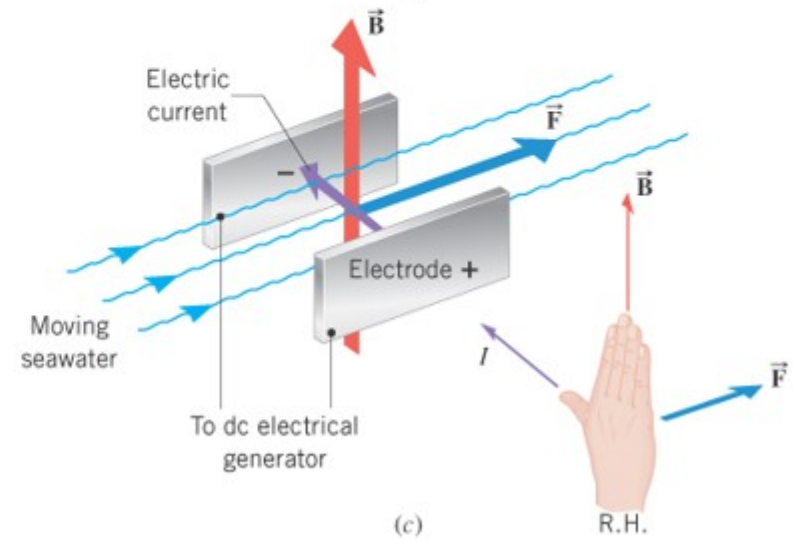
Remember the current is moving positive charges



(a)



(b)



(c)

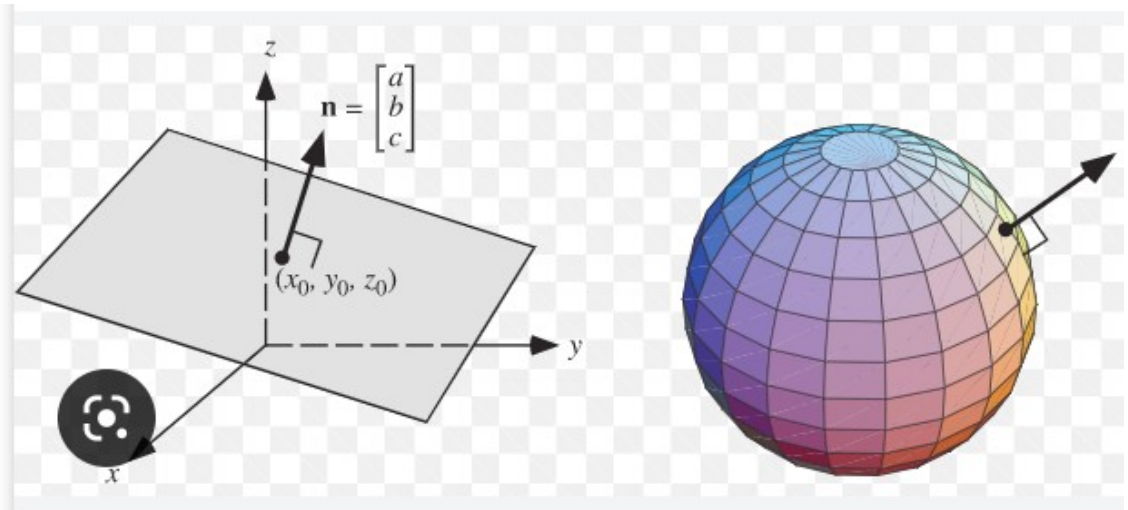
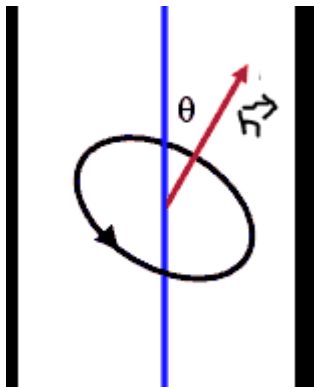
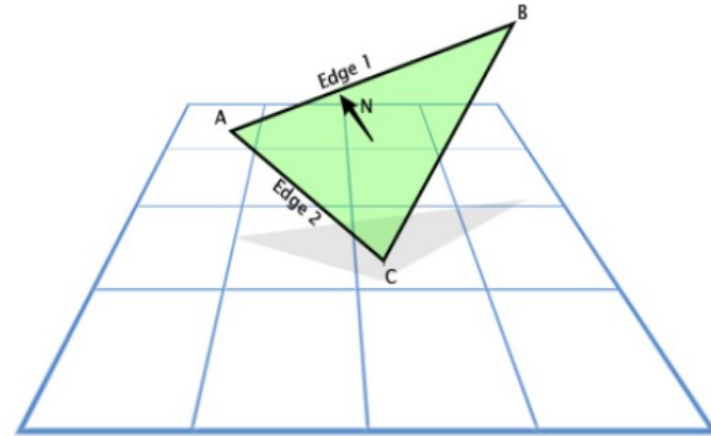
Calculus part

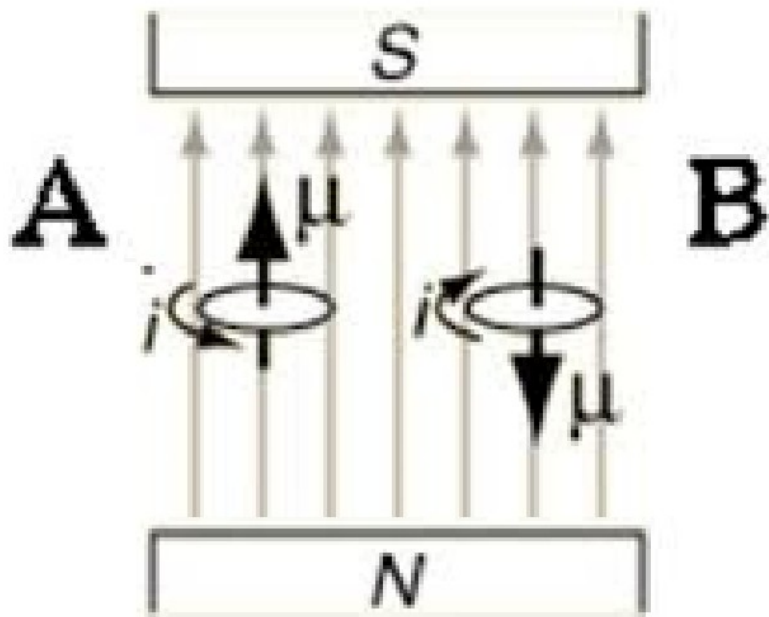
Remember the normal vector ?

Gives the orientation of a element of area

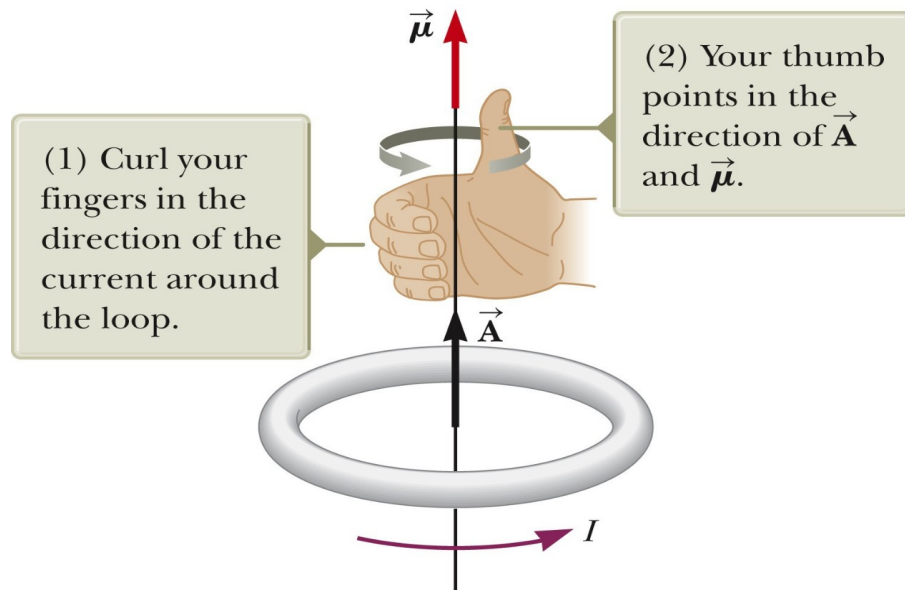
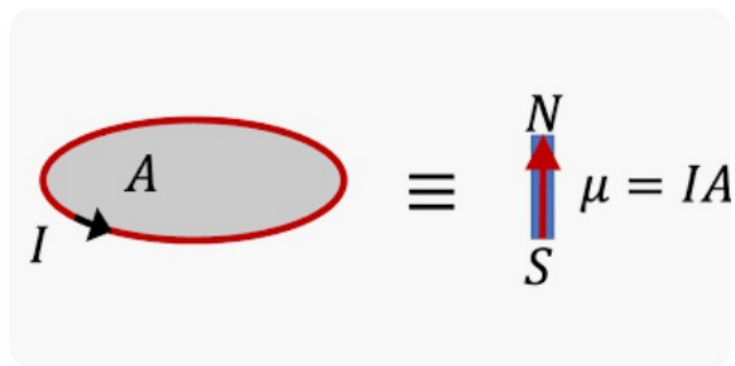
With a loop of current,
The direction of $\mathbf{n}$ (unit vector)
Is given by the right hand rule.

$\mathbf{A}$ is a vector $\mathbf{A} = A \mathbf{n}$





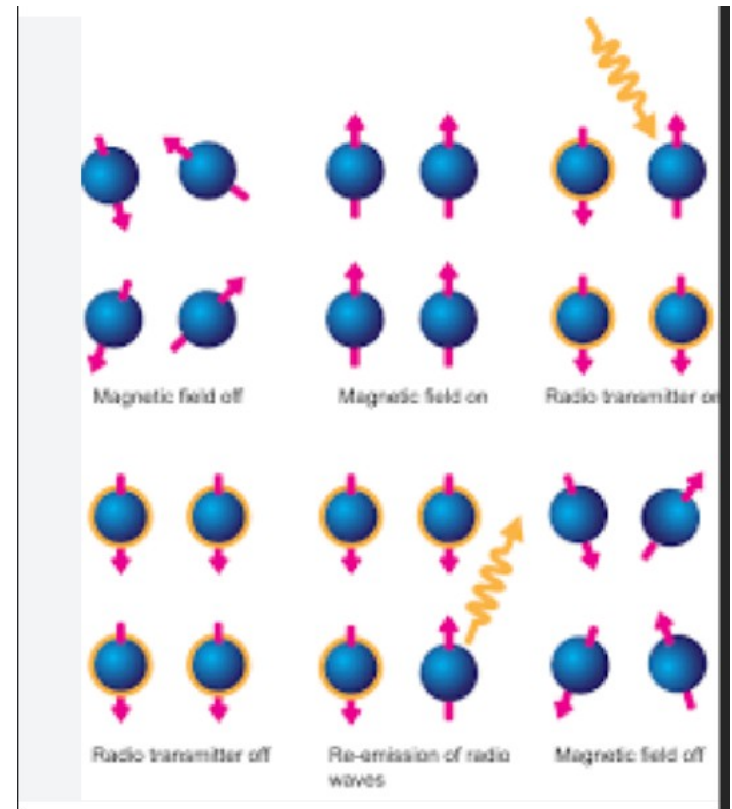
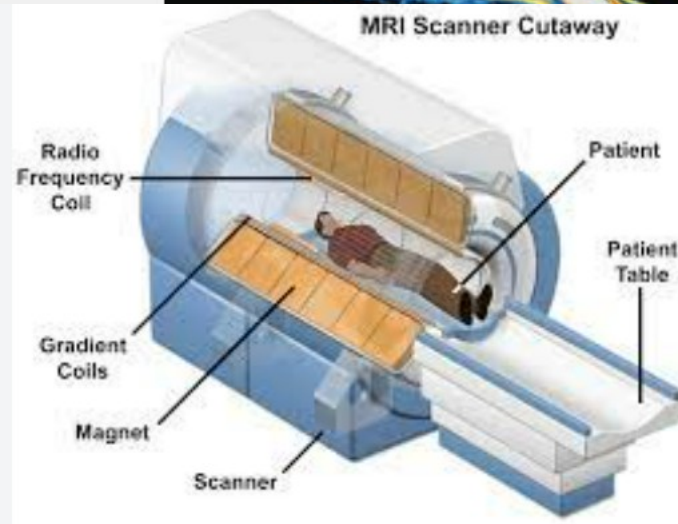
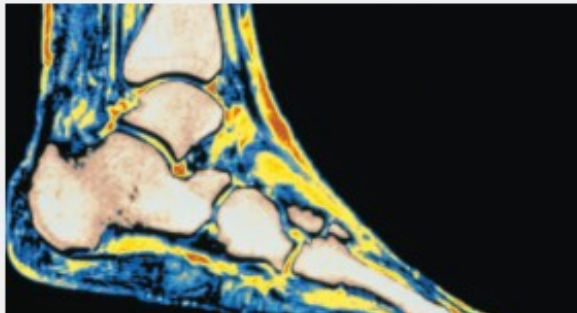
Magnetic dipole for one loop = $I A \hat{n}$
 It is a vector. Unit is Amps m²



BIO Application

Magnetic Resonance Imaging

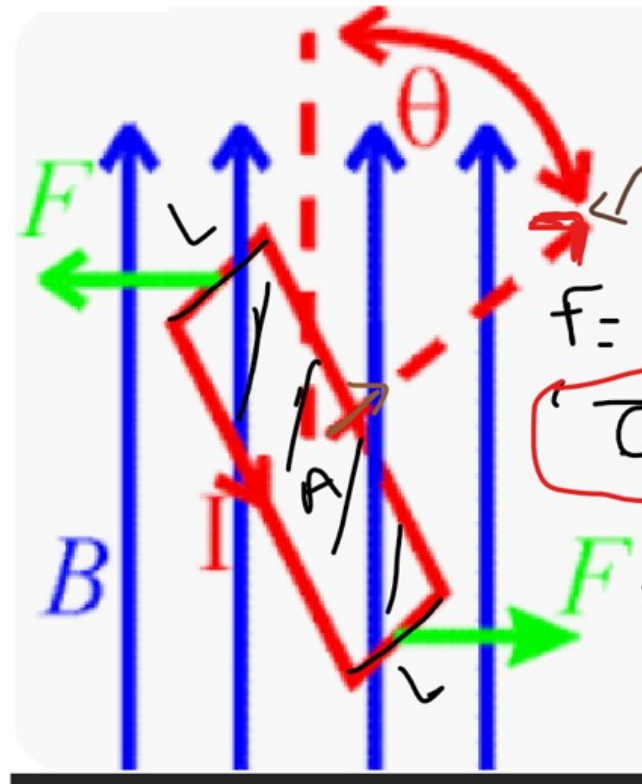
In magnetic resonance imaging (MRI), a patient is placed in a strong magnetic field. Each hydrogen nucleus in the patient acts like a miniature current loop with a magnetic dipole moment that tends to align with the applied field. Radio waves of just the right frequency then flip these magnetic moments out of alignment. The extent to which the radio waves are absorbed is proportional to the amount of hydrogen present. This makes it possible to image details in hydrogen-rich soft tissue that cannot be seen in x-ray images. (X rays are superior to MRI for imaging bone, which is hydrogen deficient.)



From Khan Academy.
The MCAT series

$$\vec{\tau}_{mag. dipole} = \vec{\mu} \times \vec{B}$$

$$\vec{\mu} = I\vec{A}$$



Vector μ
Wants to align with B

$$F = N I L B \sin(\theta)$$

$$\tau = N I B A \sin(\theta)$$

f. (force) in N

τ (torque) in N.m

A (area loop) in m^2

N (number loops)

Torque on a loop carrying current in a magnetic field.

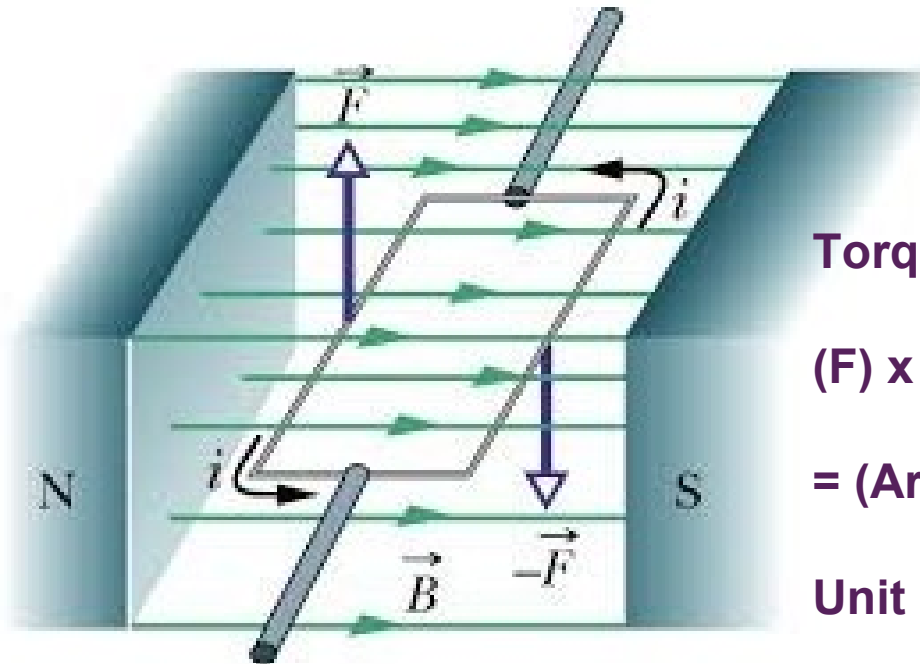
See equation sheet. A torque is a vector. Its direction is

Given by the direction of rotation. Use your right hand.

Is the loop rotation CW or CCW? Is the vector torque toward you or away?

(in screen or out?)

Force on loop currents → electric motor



Torque max=

$(F) \times \text{width} = (I \times \text{length} \times B) \times \text{width}$

$= (\text{Area loop}) \times I \times B$

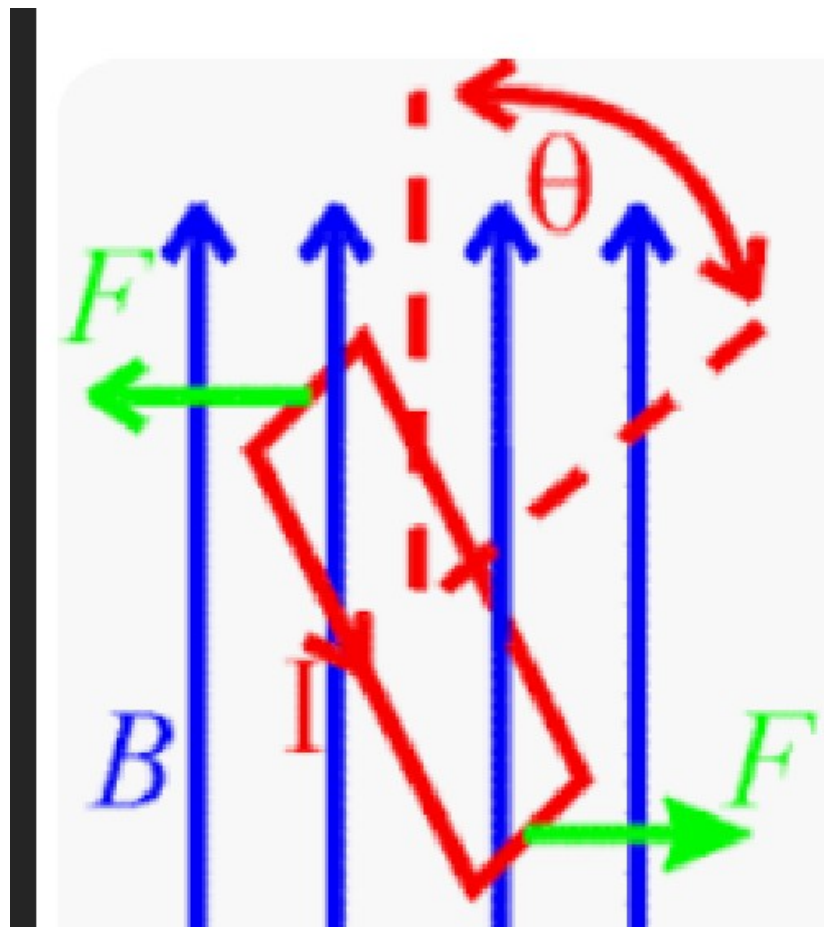
Unit is N.m

A rectangular loop 6cm high and 8cm wide is suspended as shown
In a horizontal magnetic field of 0.1 T and it carries a current of 0.5A.

- A) How much is the force on the horizontal sides? (as shown)
- B).What is the torque exerted on the loop
- C)What happens when the plane of the loop has turned 90 degrees

SHOW the normal vector. The loop rotates unit $\vec{n} \parallel \vec{B}$

$$\tau = \overbrace{NIA}^{\text{magnetic moment}} B \sin \phi$$



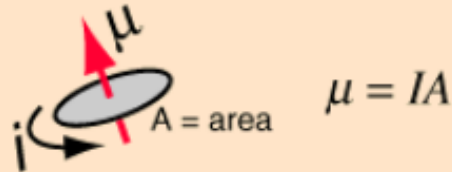
—3—

The loop of Figure 11-7 is 10 centimeters high and 20 centimeters wide, and it carries a current of 5 amperes in a field of 30 milliteslas. Find the torque on the loop.

Angle is 30 degrees

Magnetic Dipole Moment

From the expression for the [torque on a current loop](#), the characteristics of the current loop are summarized in its magnetic moment



The magnetic moment can be considered to be a vector quantity with direction perpendicular to the current loop in the right-hand-rule direction. The torque is given by

$$\tau = \mu \times B$$

As seen in the geometry of a current loop, this torque tends to line up the magnetic moment with the [magnetic field](#) B , so this represents its lowest energy configuration. The [potential energy](#) associated with the magnetic moment is

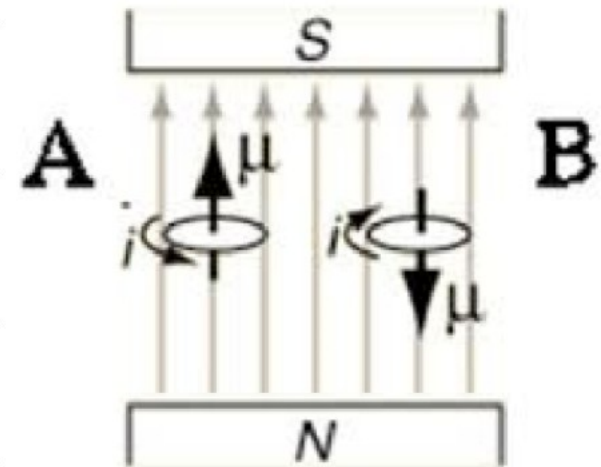
$$U(\theta) = -\mu \cdot B$$

so that the difference in energy between aligned and anti-aligned is

$$\Delta U = 2\mu B$$

If N loops,
Multiply by N .

$$\text{Torque} = N I A B \sin(\theta)$$



A: minimum $U = -NIAB$

B: maximum $U = NIAB$

In between $U = 0$

(μ perpendicular to B)

Example 6 The Torque Exerted on a Current-Carrying Coil

A coil of wire has an area of $2.0 \times 10^{-4} \text{ m}^2$, consists of 100 loops or turns, and contains a current of 0.045 A. The coil is placed in a uniform magnetic field of magnitude 0.15 T. (a) Determine the magnetic moment of the coil. (a) Find the maximum torque that the magnetic field can exert on the coil.

Example 6 The Torque Exerted on a Current-Carrying Coil

A coil of wire has an area of $2.0 \times 10^{-4} \text{ m}^2$, consists of 100 loops or turns, and contains a current of 0.045 A. The coil is placed in a uniform magnetic field of magnitude 0.15 T. (a) Determine the magnetic moment of the coil.

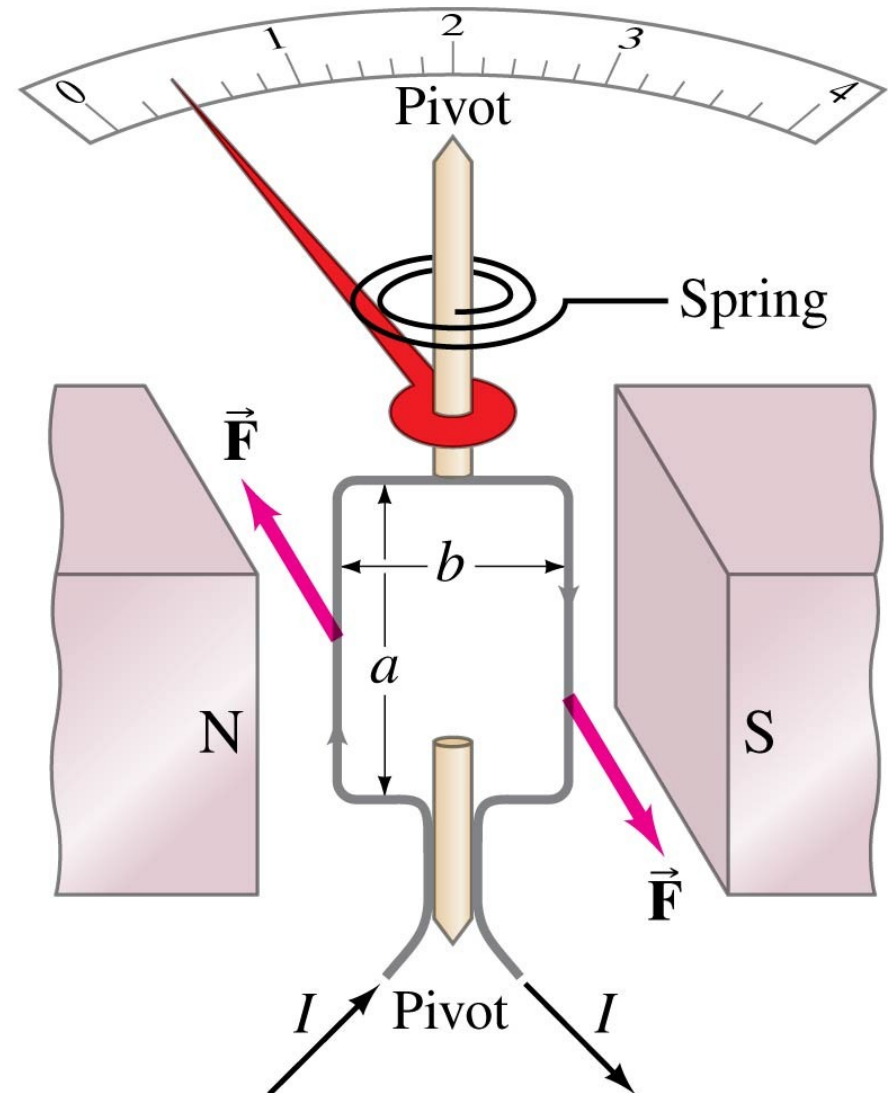
(a) Find the maximum torque that the magnetic field can exert on the coil.

$$(a) \quad \overbrace{NIA}^{\text{magnetic moment}} = (100)(0.045 \text{ A})(2.0 \times 10^{-4} \text{ m}^2) = 9.0 \times 10^{-4} \text{ A} \cdot \text{m}^2$$

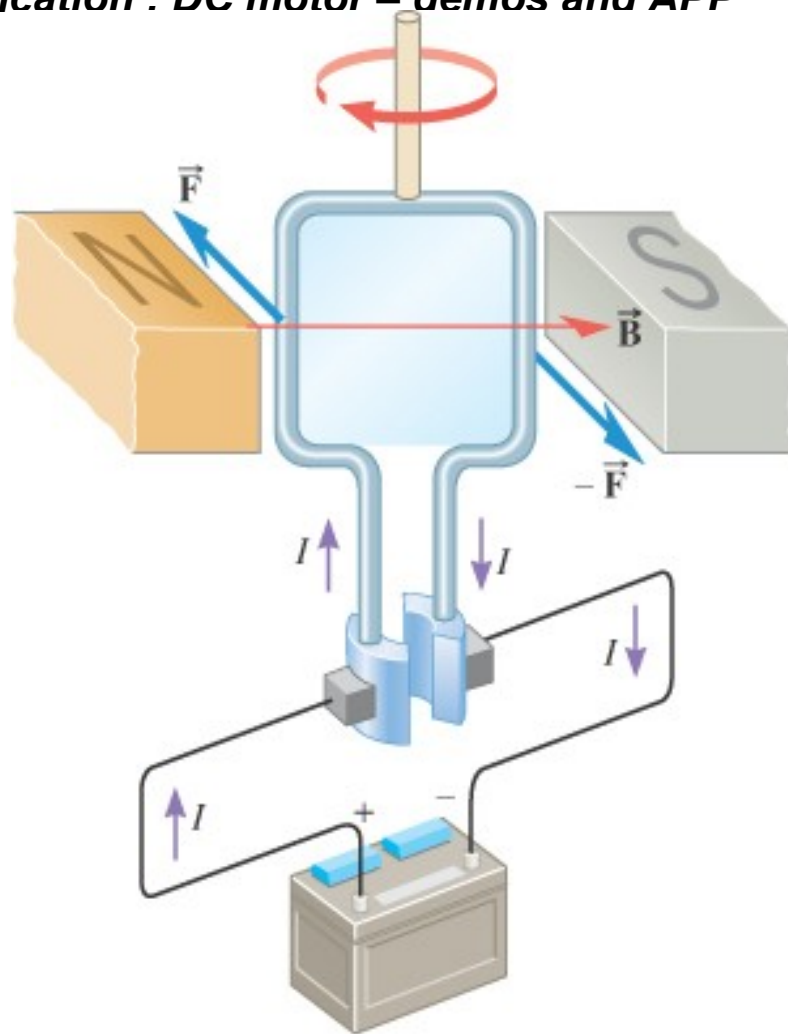
$$(b) \quad \tau = \overbrace{NIA}^{\text{magnetic moment}} B \sin \phi = (9.0 \times 10^{-4} \text{ A} \cdot \text{m}^2)(0.15 \text{ T}) \sin 90^\circ = 1.4 \times 10^{-4} \text{ N} \cdot \text{m}$$

Applications: Galvanometers,

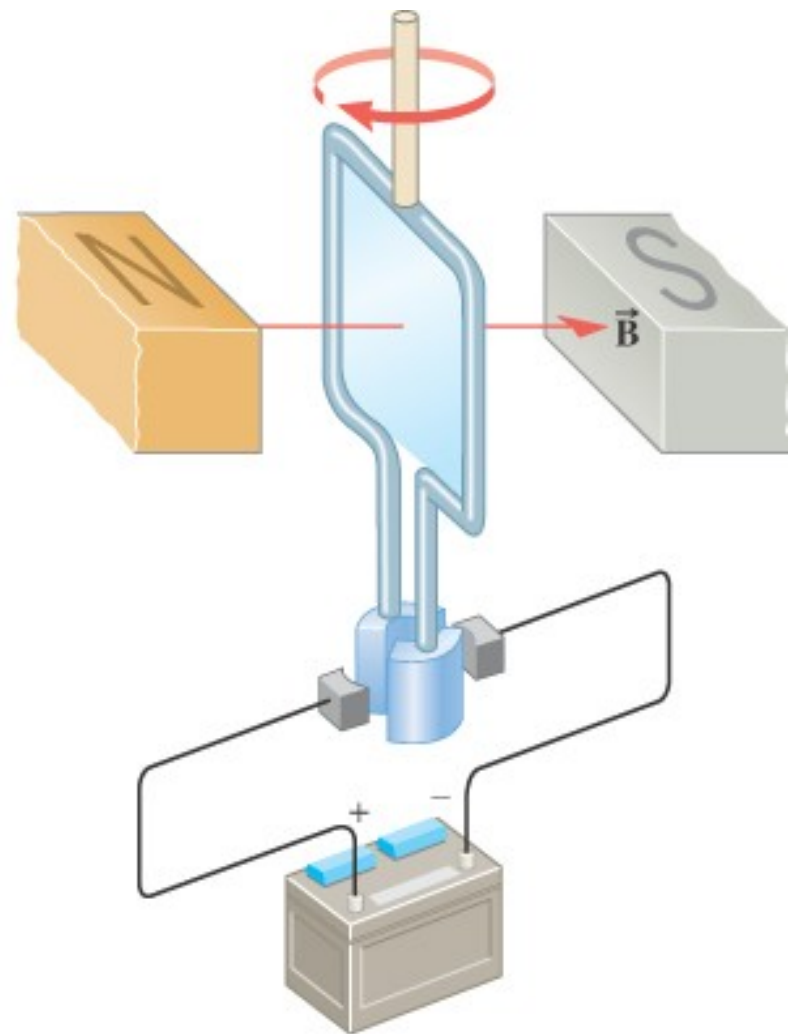
A galvanometer takes advantage of the torque on a current loop to measure current.



Application : DC motor – demos and APP



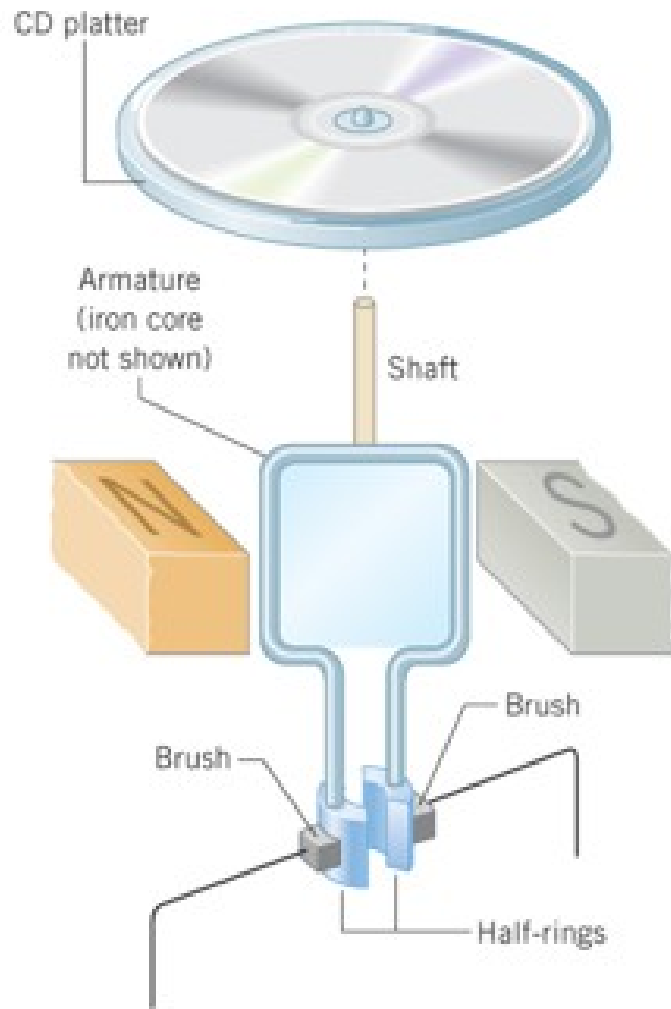
(a)



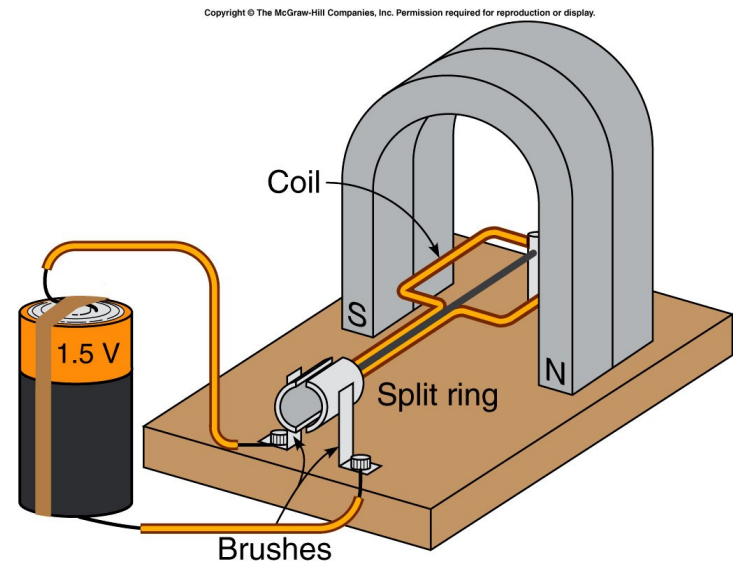
(b)

A loop in a magnetic field will experience a torque that wants to align $\vec{n}$

(normal vector $\vec{A} = A \vec{n}$) and $\vec{B}$



The basic components of a dc motor.



The Magnetic Dipole Moment

- Magnetic dipole moment

$$\vec{\mu} = i \vec{A}$$

- SI unit: Am^2 , $\text{Nm/T} = \text{J/T}$

- A coil of wire has N loops of the same area:

$$\vec{\mu}_{\text{coil}} = Ni \vec{A}$$

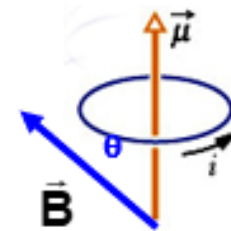
- Torque

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

- Magnetic potential

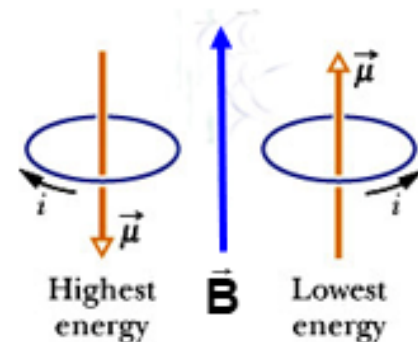
$$U = -\vec{\mu} \cdot \vec{B}$$

- Electric dipole and magnetic dipole



Small bar magnet	5 J/T
Earth	8.0×10^{22} J/T
Proton	1.4×10^{-26} J/T
Electron	9.3×10^{-24} J/T

	Electric Dipole	Magnetic Dipole
Moment	$p = qd$	$\mu = NiA$
Torque	$\vec{\tau} = \vec{p} \times \vec{E}$	$\vec{\tau} = \vec{\mu} \times \vec{B}$
Potential Energy	$U = -\vec{p} \cdot \vec{E}$	$U = -\vec{\mu} \cdot \vec{B}$



Example 27.9 Magnetic torque on a circular coil

A circular coil 0.0500 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a counterclockwise (as viewed from above) current of 5.00 A. The coil is in a uniform 1.20 T magnetic field directed toward the right. Find the magnitudes of the magnetic moment and the torque on the coil.

Draw the situation. Draw the magnetic dipole $\vec{\mu}$
Which way it is pointing. Draw the forces that will
make the loop spin.
Torque is a vector. What is its direction.

Example 27.10 Potential energy for a coil in a magnetic field

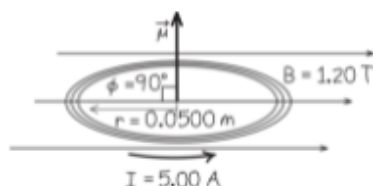
If the coil in [Example 27.9](#) rotates from its initial orientation to one in which its magnetic moment $\vec{\mu}$ is parallel to $\vec{B}$, what is the change in potential energy?

Example 27.9 Magnetic torque on a circular coil

A circular coil 0.0500 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a counterclockwise (as viewed from above) current of 5.00 A. The coil is in a uniform 1.20 T magnetic field directed toward the right. Find the magnitudes of the magnetic moment and the torque on the coil.

IDENTIFY and SET UP This problem uses the definition of magnetic moment and the expression for the torque on a magnetic dipole in a magnetic field. Figure 27.35 shows the situation. Equation (27.24) gives the magnitude μ of the magnetic moment of a single turn of wire; for N turns, the magnetic moment is N times greater. Equation (27.25) gives the magnitude τ of the torque.

Figure 27.35



Our sketch for this problem.

Do textbook 27.9 and 27.10

Example 27.10 Potential energy for a coil in a magnetic field

If the coil in Example 27.9 rotates from its initial orientation to one in which its magnetic moment $\vec{\mu}$ is parallel to $\vec{B}$, what is the change in potential energy?



Miniature



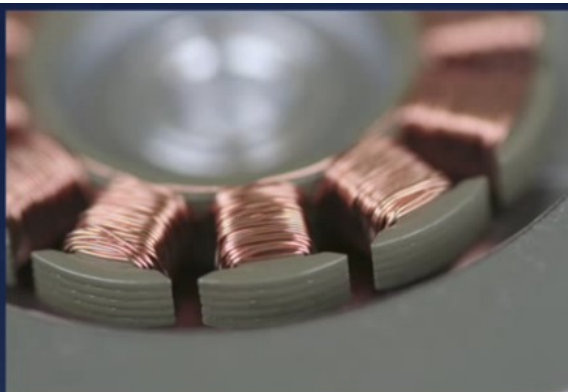
Bicycle assist



Washing machine



Industrial



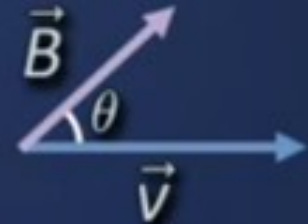
Disk drive



Toyota Prius
hybrid

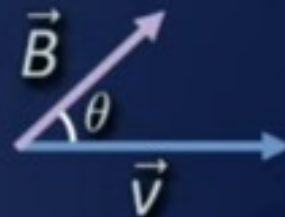
The Magnetic Force

- Acts on **moving** charged particles
- Force is proportional to charge q and magnetic field strength B
- Also depends on charge velocity $\vec{v}$
 - proportional to $v \sin \theta$
- Magnitude: $F = qvB \sin \theta$
 - Unit of magnetic field: $\text{N} \cdot \text{s} / \text{C} \cdot \text{m}$
 - $1 \text{ N} \cdot \text{s} / \text{C} \cdot \text{m} = 1 \text{ tesla (T)}$



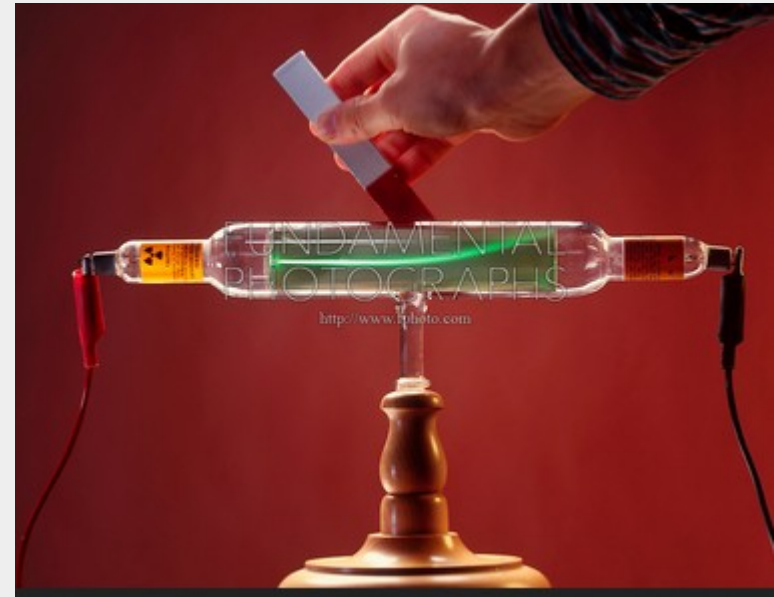
The Magnetic Force

- Acts on **moving** charged particles
- Direction: perpendicular to both $\vec{v}$ and $\vec{B}$
 - Right-hand rule: curl your right fingers in the direction that would rotate $\vec{v}$ onto $\vec{B}$
 - Your thumb then points in the direction of the magnetic force on a positive charge
 - Opposite direction for negative charge
- Mathematical shorthand: $\vec{F}_B = q\vec{v} \times \vec{B}$



Magnetism is not about magnets, Its about moving charges

→ A magnetic field (magnet) acts on a moving charge



Movie cathode ray
And light bulb



Magnetic Fields

- Magnetic force

$$\vec{F}_B = q \vec{v} \times \vec{B}$$

- Right-hand rule determine the direction of magnetic force. So the magnetic force is always perpendicular to $\vec{v}$ and $\vec{B}$.
- The magnitude of the magnetic force is

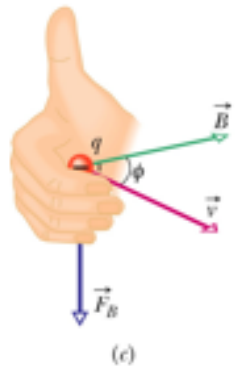
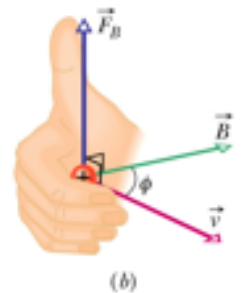
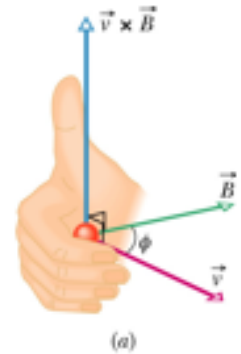
$$F_B = |q|vB \sin \theta$$

$$\vec{F}_E = q \vec{E}$$



$$\vec{F}_B = q \vec{v} \times \vec{B}$$

- The electric force is along the direction of the electric field, the magnetic force is perpendicular to the magnetic field.
- The electric force acts on a charged particle regardless of whether the particle is moving, the magnetic force acts on a charged particle only when the particle is in motion.
- The electric force does work in displacing a charged particle, the magnetic force does no work when a particle is displaced.



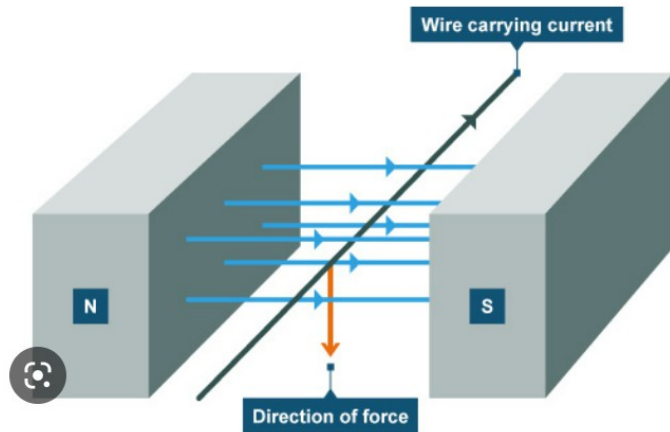
2 equations for the magnetic force

→ on a current carrying wire

$$F = I L B \sin(\theta)$$

L is the length of the wire inside the field(m)

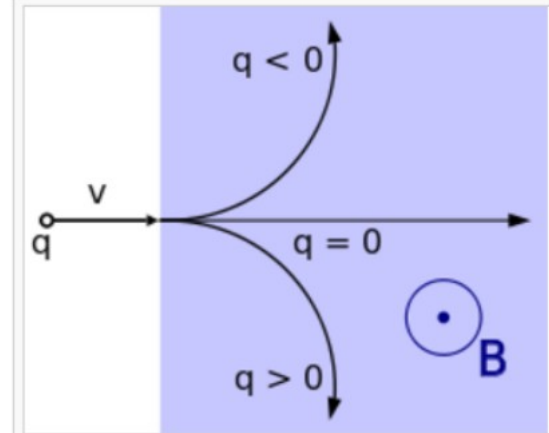
B is the magnetic field (T)



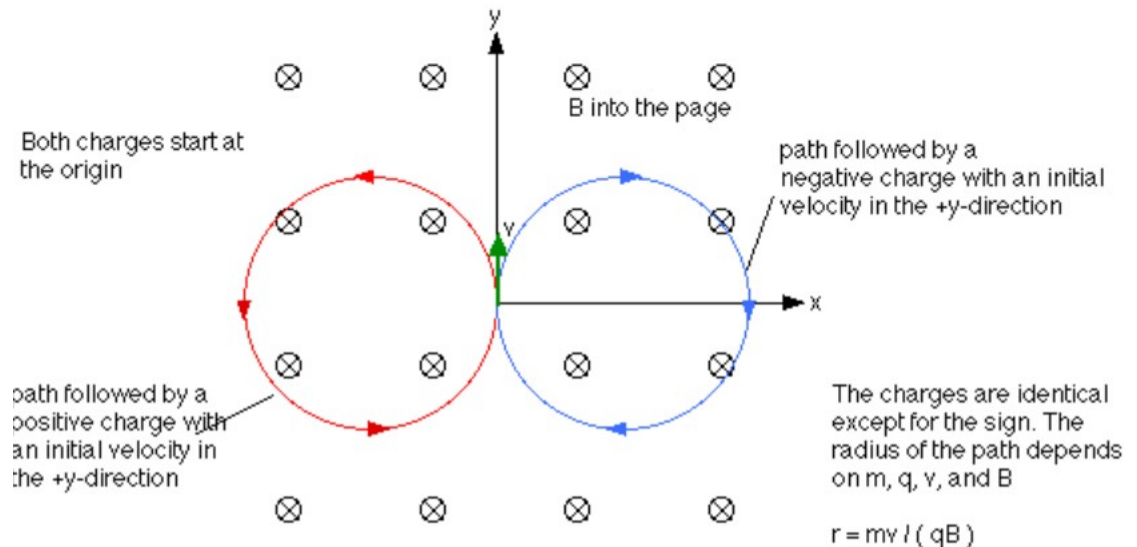
→ on a moving charge

$$F = qVB \sin(\theta)$$

V is the velocity of the charge (m/s)



Trajectory of a particle with a positive or negative charge q under the influence of a magnetic field B , which is directed perpendicularly out of the screen.



1. A particle in a magnetic field is found to have zero magnetic force on it. Which situation is impossible to happen?

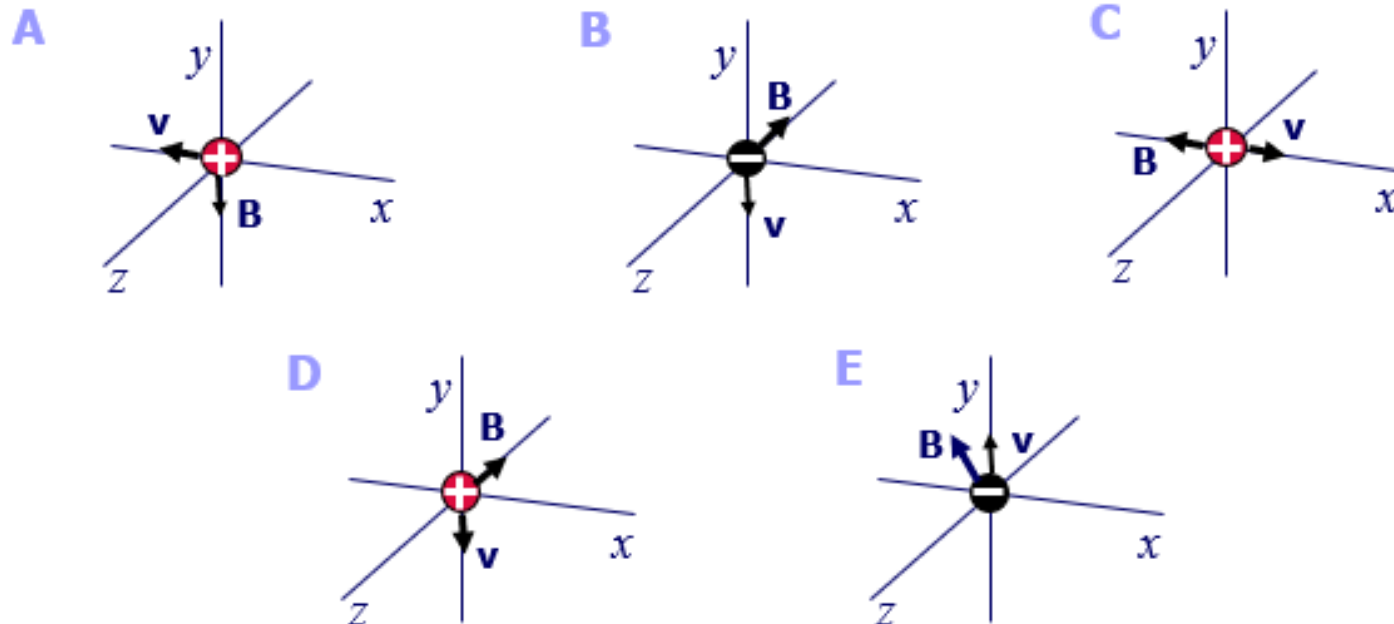
$$F_B = |q|vB \sin \theta$$

- A. The particle is neutral.
- B. The particle is stationary.
- C. The motion of the particle is along magnetic field.
- D. The motion of the particle is opposite to magnetic field.
- E. All of them are possible.

Magnetic force and Earth magnetic field.

Direction of Magnetic Force

2. The figures shows five situations in which a charged particle with velocity $\mathbf{v}$ travels through a uniform magnetic field $\mathbf{B}$. In which situation, is the direction of the magnetic force along $+x$ axis ?

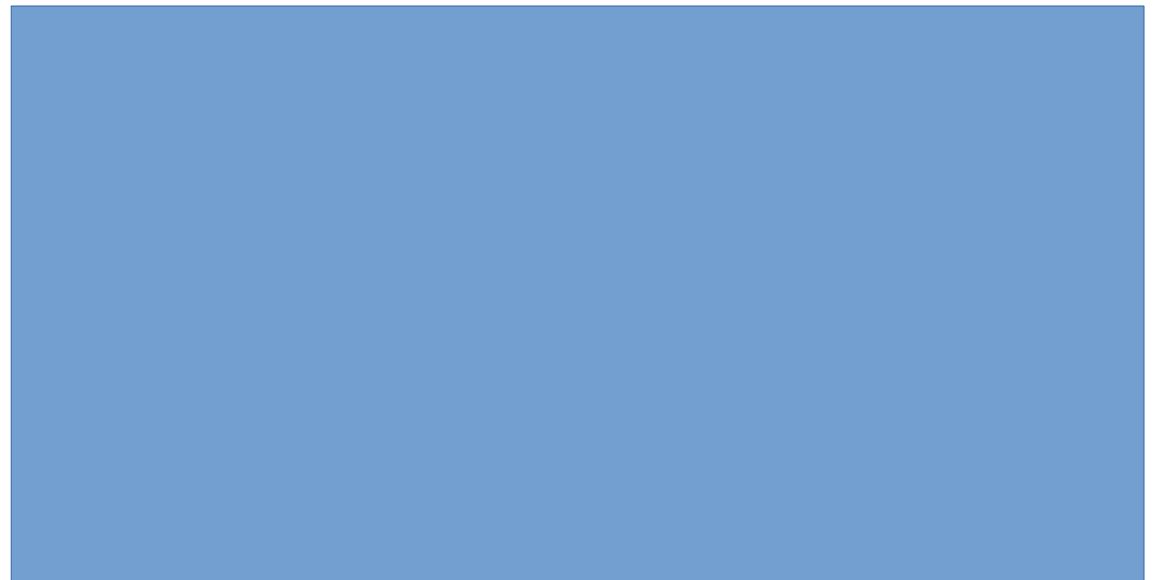
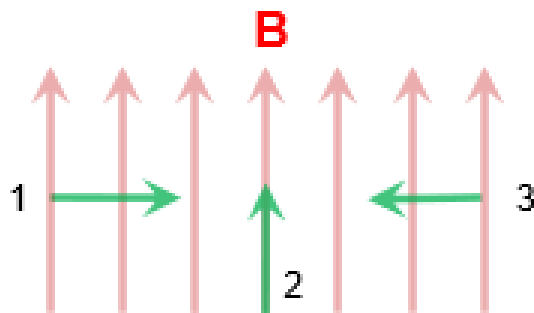


Which what is the force along X-axis for these cases?

!! negative charge moves other way.

Example 26.1. Steering Protons

Figure shows 3 protons entering a 0.10-T magnetic field.
All three are moving at 2.0 Mm/s.
Find the magnetic force on each.

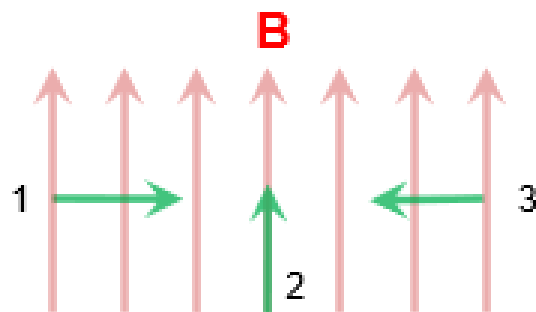


Example 26.1. Steering Protons

Figure shows 3 protons entering a 0.10-T magnetic field.

All three are moving at 2.0 Mm/s.

Find the magnetic force on each.



$$\mathbf{F} = q \mathbf{v} \times \mathbf{B}$$

$$\begin{aligned}\mathbf{F}_1 &= q v B \hat{\mathbf{i}} = (1.6 \times 10^{-19} \text{ C})(2.0 \times 10^6 \text{ m/s})(0.10 \text{ T}) \hat{\mathbf{i}} \\ &= 3.2 \times 10^{-14} \text{ N} \hat{\mathbf{i}} = 32 \text{ fN} \hat{\mathbf{i}}\end{aligned}$$

$$\mathbf{F}_2 = 0$$

$$\mathbf{F}_3 = -q v B \hat{\mathbf{i}} = -32 \text{ fN} \hat{\mathbf{i}}$$

1) A pith ball with a charge of 30nC is traveling at 12m/s through a magnetic field of 50mT s. If the ball is moving perpendicularly to the field, how much Force is on it?

2) how strong is a field needed to exert a force of $5 \cdot 10^{-11}\text{N}$ on a proton traveling perpendicularly through the field at $2 \cdot 10^8 \text{ m/s}$?

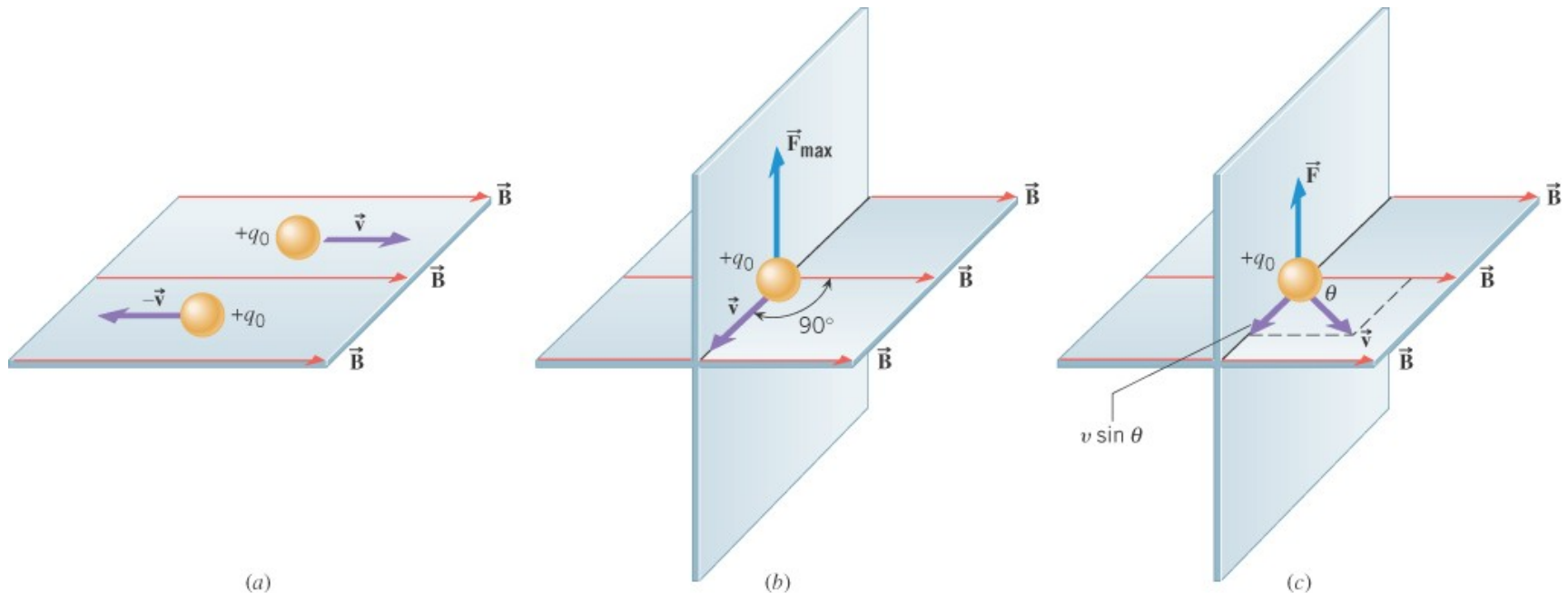
3) What is the direction of the force that a vertical magnetic field, directed upward will exert on the electron traveling eastward in it?

4) what magnetic field would be needed to exert a force of $2 \cdot 10^{-12}\text{N}$ on an electron moving perpendicularly to the field at $2.5 \cdot 10^8\text{m/s}$?
What is its acceleration

21.2 The Force That a Magnetic Field Exerts on a Charge

Example 1 Magnetic Forces on Charged Particles

A proton in a particle accelerator has a speed of 5.0×10^6 m/s. The proton encounters a magnetic field whose magnitude is 0.40 T and whose direction makes an angle of 30.0 degrees with respect to the proton's velocity (see part (c) of the figure). Find (a) the magnitude and direction of the force on the proton and (b) the acceleration of the proton. (c) What would be the force and acceleration of the particle were an electron?



21.2 The Force That a Magnetic Field Exerts on a Charge

$$\begin{aligned} \text{(a)} \quad F &= |q_o|vB \sin \theta = (1.60 \times 10^{-19} \text{ C})(5.0 \times 10^6 \text{ m/s})(0.40 \text{ T}) \sin(30.0^\circ) \\ &= 1.6 \times 10^{-13} \text{ N} \end{aligned}$$

$$\text{(b)} \quad a = \frac{F}{m_p} = \frac{1.6 \times 10^{-13} \text{ N}}{1.67 \times 10^{-27} \text{ kg}} = 9.6 \times 10^{13} \text{ m/s}^2$$

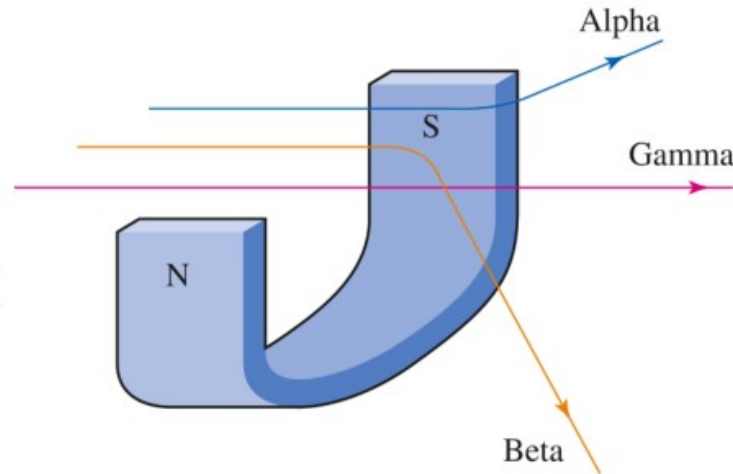
(c) Magnitude is the same, but direction is opposite.

$$a = \frac{F}{m_e} = \frac{1.6 \times 10^{-13} \text{ N}}{9.11 \times 10^{-31} \text{ kg}} = 1.8 \times 10^{17} \text{ m/s}^2$$

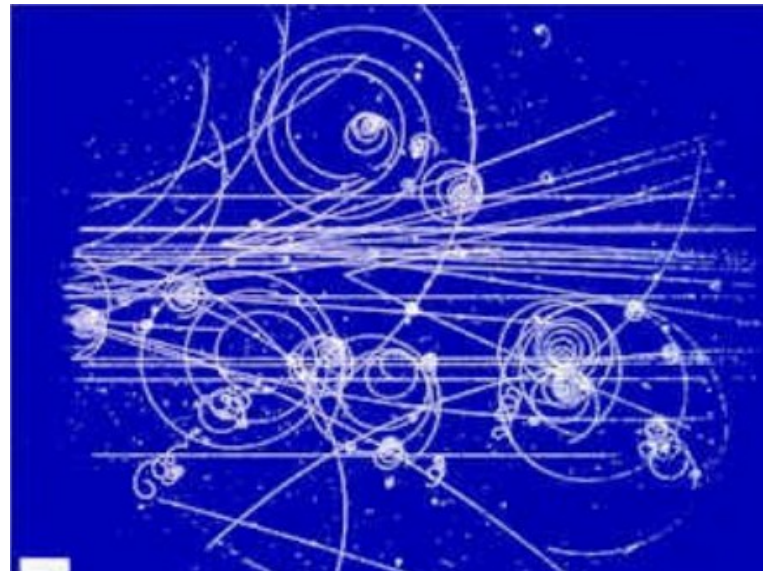
Textbook 27.1

A beam of protons ($q = 1.6 \times 10^{-19} \text{ C}$) moves at $3.0 \times 10^5 \text{ m/s}$ through a uniform 2.0 T magnetic field directed along the positive z – axis (Fig. 27.10□). The velocity of each proton lies in the xz -plane and is directed at 30° to the $+z$ – axis. Find the force on a proton.

- The type of radiation can be determined by passing it through a magnetic field.
- Since each type has a different electric charge, they are deflected differently by the magnet.



**Cloud chamber
radioactivity**

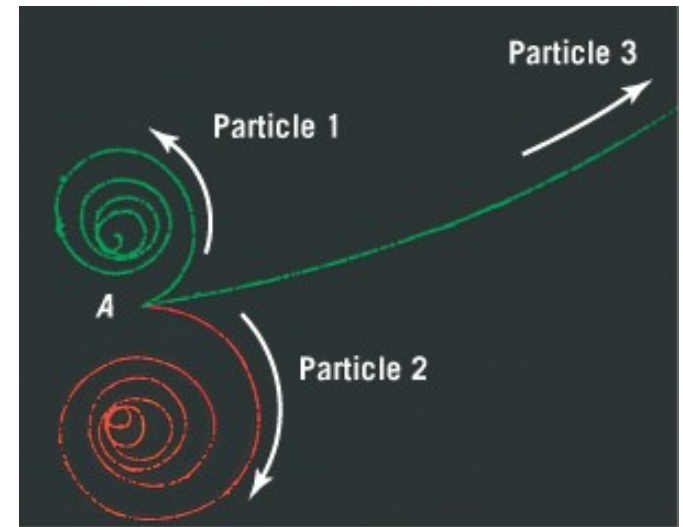


21.3 The Motion of a Charged Particle in a Magnetic Field

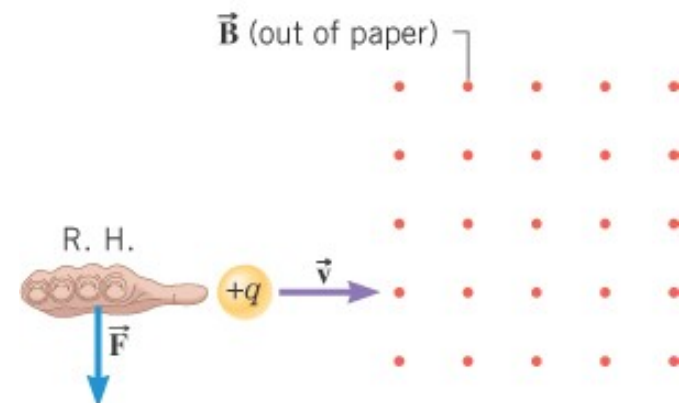
Conceptual Example 4 Particle Tracks in a Bubble Chamber

The figure shows the bubble-chamber tracks from an event that begins at point A. At this point a gamma ray travels in from the left, spontaneously transforms into two charged particles. The particles move away from point A, producing two spiral tracks. A third charged particle is knocked out of a hydrogen atom and moves forward, producing the long track.

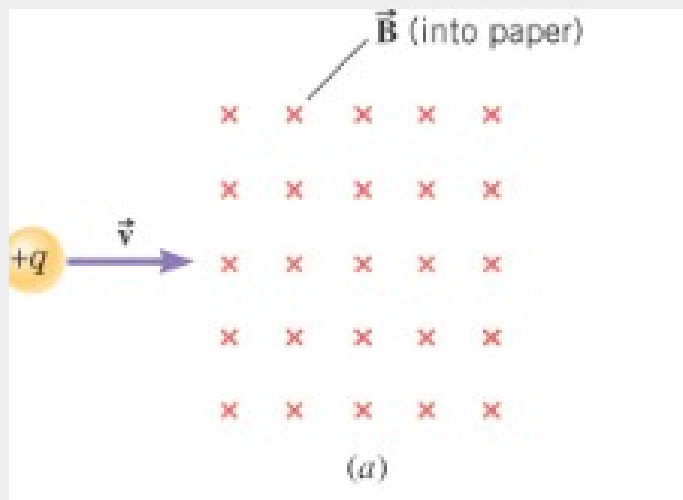
The magnetic field is directed out of the paper. Determine the sign of each particle and which particle is moving most rapidly.



(a)

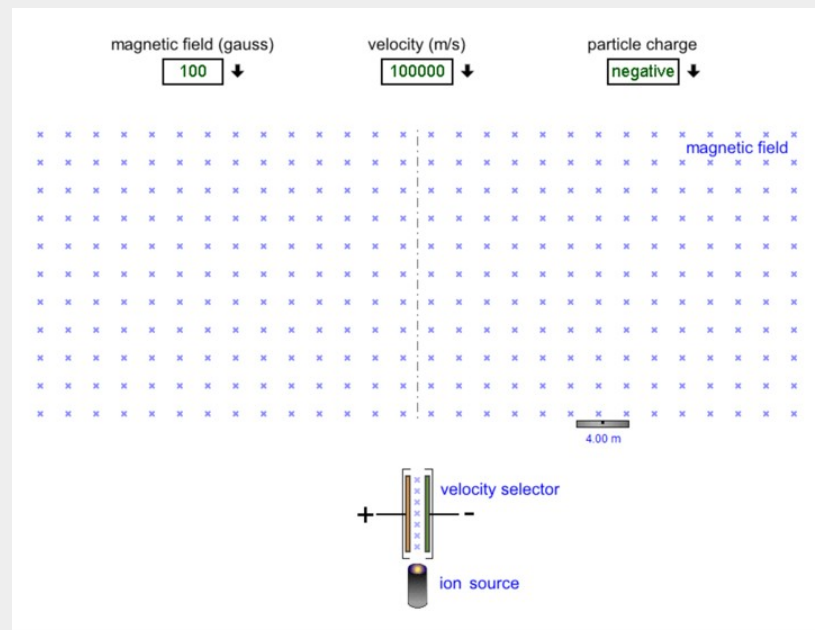


(b)



A moving negative charge placed in an external magnetic field circulates counterclockwise in the plane of the paper. In which direction is the magnetic field pointing?

- A. Into the page
- B. Out of the page
- C. Toward the center of the circle
- D. Tangent to the circle



Application: Cyclotron , mass spectrometer – see next slide

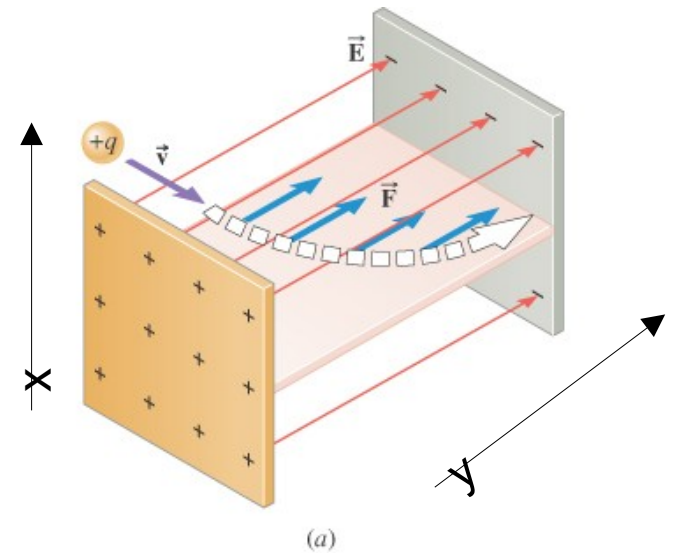
The Motion of a Charged Particle in a Magnetic Field

Charged particle in an electric field
Similar to a projectile motion.

It is moving forward and a constant
Speed $V_x = V_0$

And it is speeding up at constant
Along E ($a = qE/m$)

$V_y = at$.

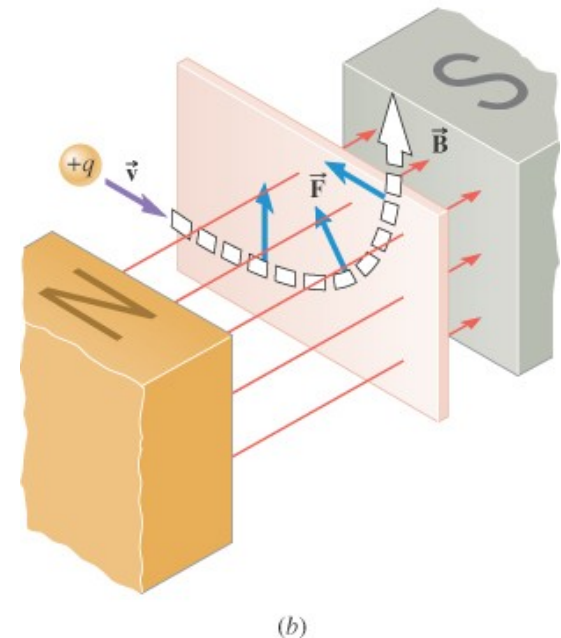


Charged particle in a magnetic field.

The force is perpendicular to velocity and to B

We get a uniform circular motion.

The acceleration is centripetal $= mV^2/R$



Demo exploration physical sciences

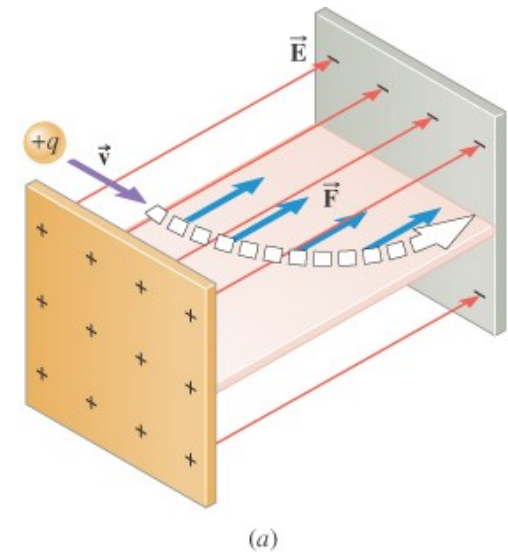
Videos: cathode ray tube and cyclotron

21.3 The Motion of a Charged Particle in a Magnetic Field

The electrical force *can* do work on a charged particle.

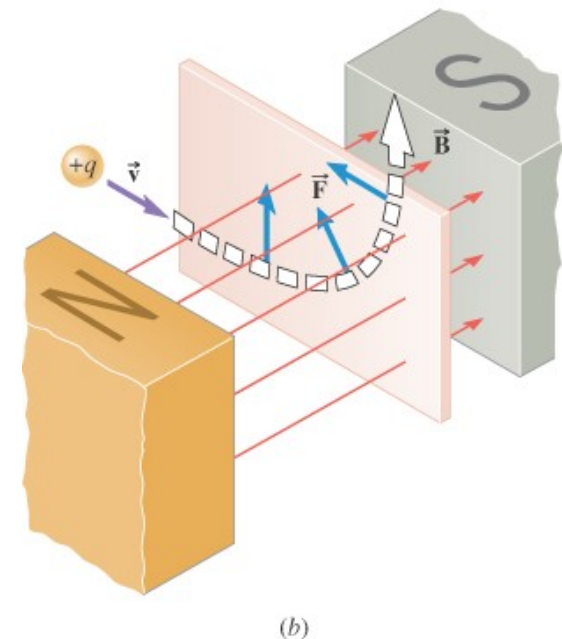
Because the KE increases

Work is done by a force when it
Changes the speed.



The magnetic force *cannot* do work on a charged particle.

The speed is constant.
Force is perpendicular to path



21.3 The Motion of a Charged Particle in a Magnetic Field

The magnetic force always remains perpendicular to the velocity and is directed toward the center of the circular path.

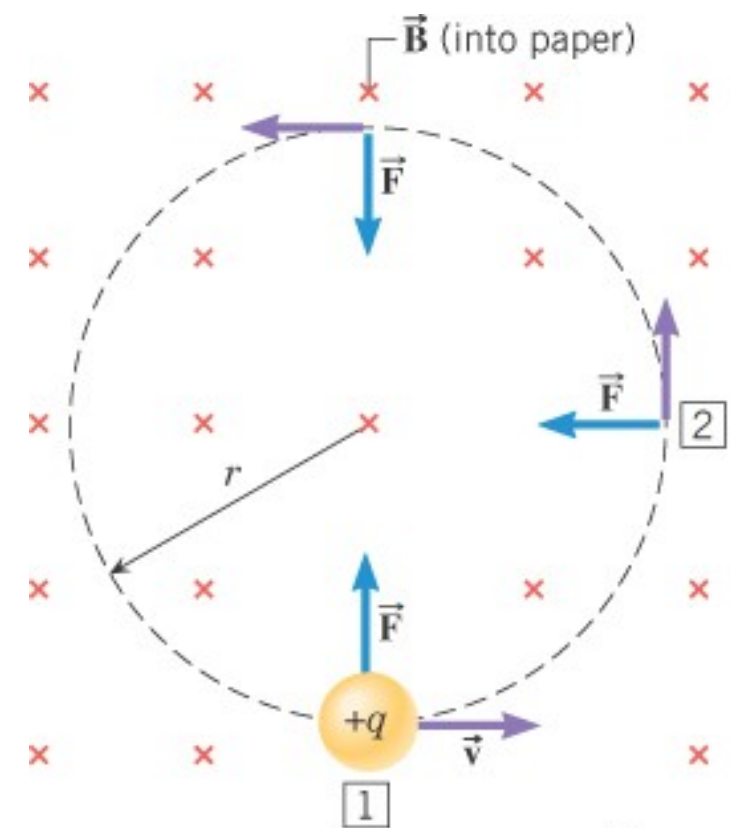
$$F_c = m \frac{v^2}{r}$$

↓

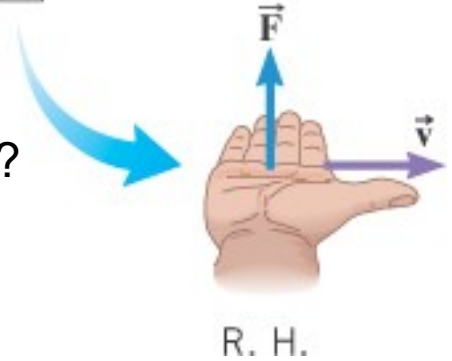
$$qvB = m \frac{v^2}{r}$$

↓

$$r = \frac{mv}{qB}$$



What is the angular frequency?
The frequency?
The period ?



Do it in class. Show slides with cyclotron. Frequency does not depend on radius

Motion of a Charged Particle in a Uniform Magnetic Field

- F_B never has a component parallel to $\mathbf{v}$ and can't change the particle's kinetic energy. The force can change only the direction of $\mathbf{v}$.

- Charged particle moves in a circle in a plane perpendicular to the magnetic field.

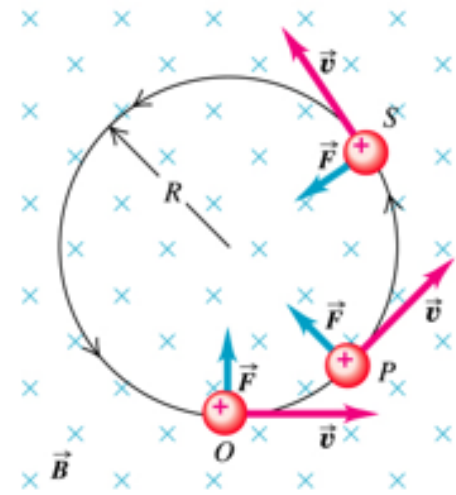
- Start with $\sum F = F_B = ma$
- Then, we have
$$F_B = qvB = \frac{mv^2}{r}$$

- The radius of the circular path: $r = \frac{mv}{qB}$

- The angular speed: $\omega = \frac{v}{r} = \frac{qB}{m}$

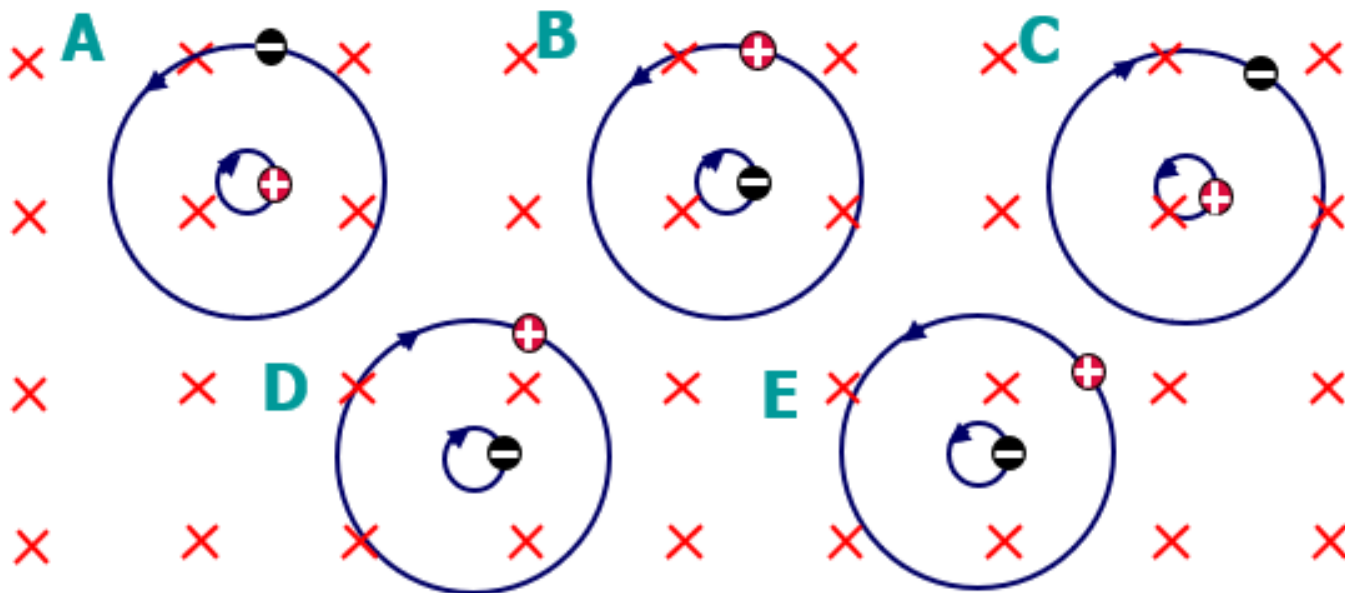
- The period of the motion:

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$



- T and ω do not depend on v of the particle. Fast particles move in large circles and slow ones in small circles, but all particles with the same charge-to-mass ratio take the same time T to complete one round trip.
- The direction of rotation for a positive particle is always counterclockwise, and the direction for a negative particle is always clockwise.

3. The figures shows the circular paths of two particles that travel at the same speed in a uniform magnetic field B , which is directed into the page. One particle is proton; the other is an electron (which is less massive). Which figure is physically reasonable?



$$r = \frac{mv}{qB}$$

First take a positive charge. Is the motion CCW or CW
What about a negative charge?

Example. Find the radius of curvature for a fast electron with speed $v = 2 \times 10^7$ m/s in a) the Earth's magnetic field at the surface, and b) in a 10 T magnetic field.



Example. Find the radius of curvature for a fast electron with speed $v = 2 \times 10^7$ m/s in a) the Earth's magnetic field at the surface, and b) in a 10 T magnetic field.

a) $B_{\text{Earth}} \sim 0.5 \text{ gauss} = 0.5 \times 10^{-4} \text{ T}$ Or 0.5 Gauss

$$r = mv/(qB) = (9.11 \times 10^{-31})(2 \times 10^7)/[(1.6 \times 10^{-19})(0.5 \times 10^{-4})]$$
$$= \mathbf{2.3 \text{ m}}$$

b) $B = 10 \text{ T}$

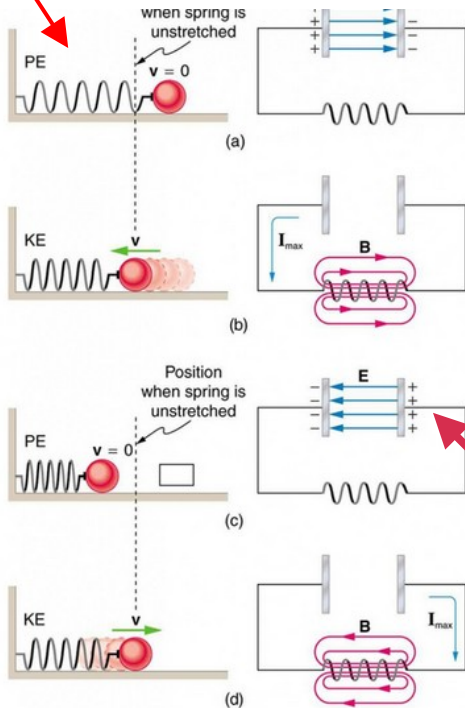
$$r = mv/(qB) = (9.11 \times 10^{-31})(2 \times 10^7)/[(1.6 \times 10^{-19})(10)]$$
$$= \mathbf{1.1 \times 10^{-5} \text{ m} = 11 \text{ } \mu\text{m}}$$

Applications: Cyclotron Motion

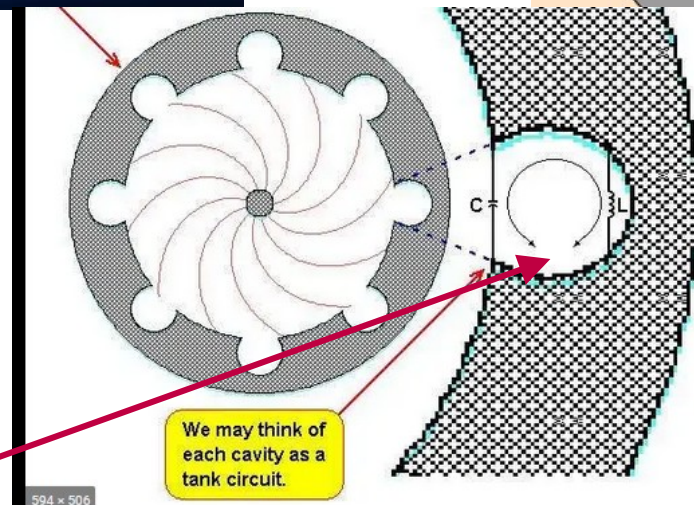
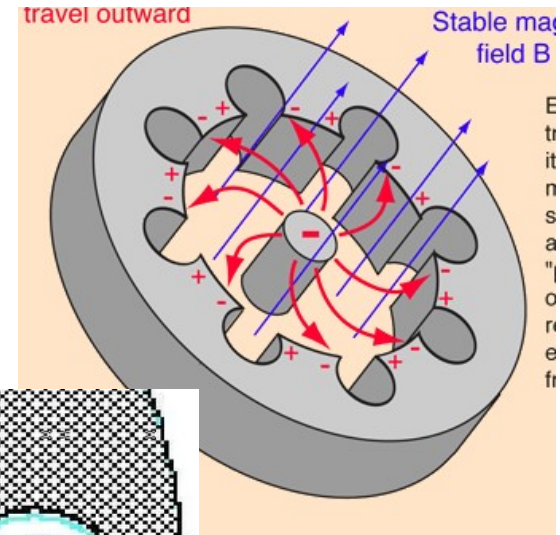
Microwave ovens



simple harmonic oscillator system



Textbook 27.3. Magnetron

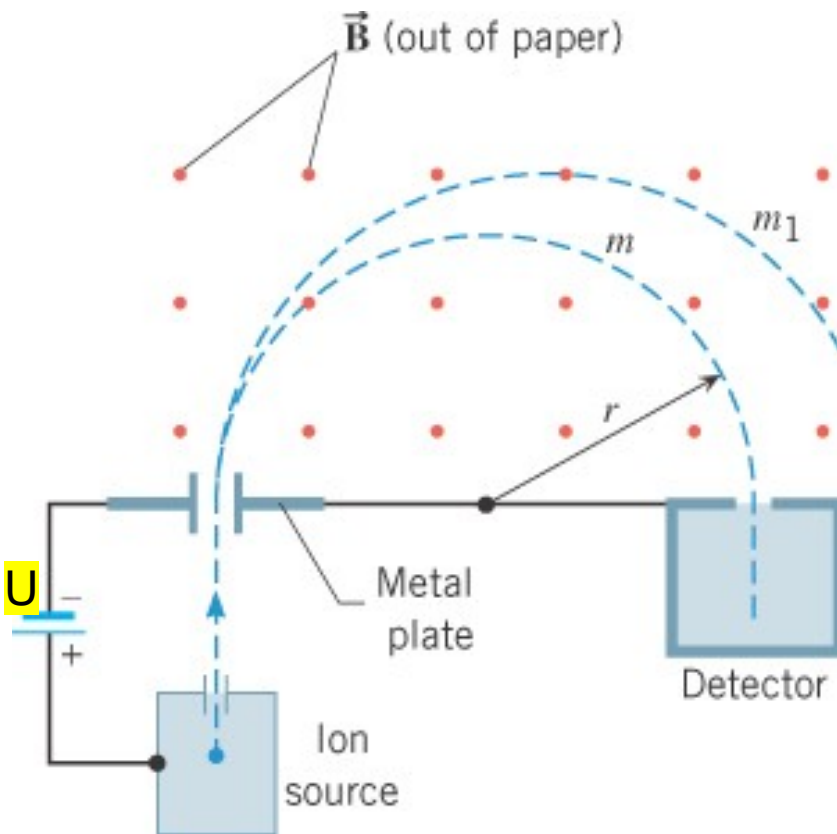


An LC circuit, also called a resonant circuit, tank circuit, or tuned circuit, is an electric circuit consisting of an inductor, represented by the letter L , and a capacitor, represented by the letter C , connected together.

Simple **Mass Spectrometer**.

Technique similar to Thomson experiment

Discovery of isotopes for Neon.



$$\frac{1}{2}mv^2 = qU$$

EQ1

First the charged particle is Accelerated in a voltage U
PE = KE. $v = \sqrt{2qU/m}$

$$r = \frac{mv}{qB}$$

EQ2

The particle is then deflected by the Magnetic field.

$$m = \left(\frac{qr^2}{2U} \right) B^2$$

EQ1 and EQ2

$$r = \left(\frac{1}{B} \right) \sqrt{\frac{2mU}{q}}$$

For a given B: Larger mass $\rightarrow$ larger r

We can find the mass of an isotope.
Change B until you hear a "beep"
In your detector

The Mass Spectrometer

We can count the isotopes.

Change B until you hear a “beep”

Then count the number of beeps

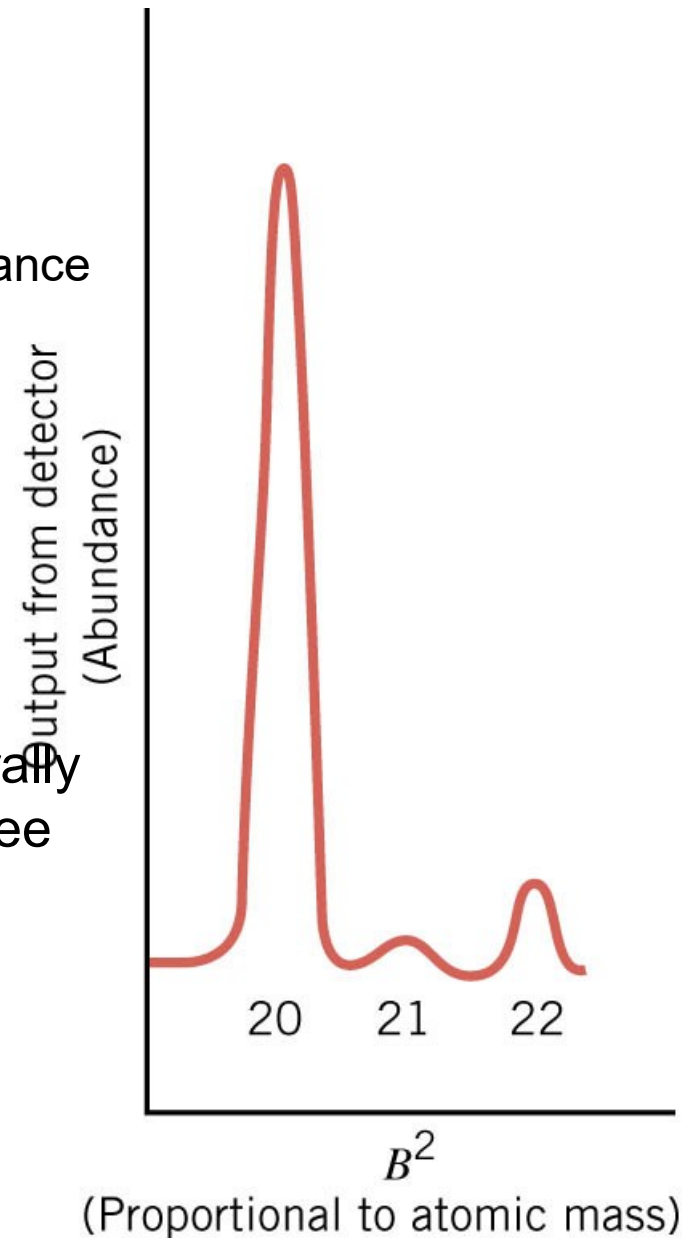
We can use this device to find the relative abundance of isotopes. For a given sensor (r) we can vary B and count the isotopes,

$$m = \left(\frac{qr^2}{2V} \right) B^2$$

The mass spectrum of naturally occurring neon, showing three isotopes.

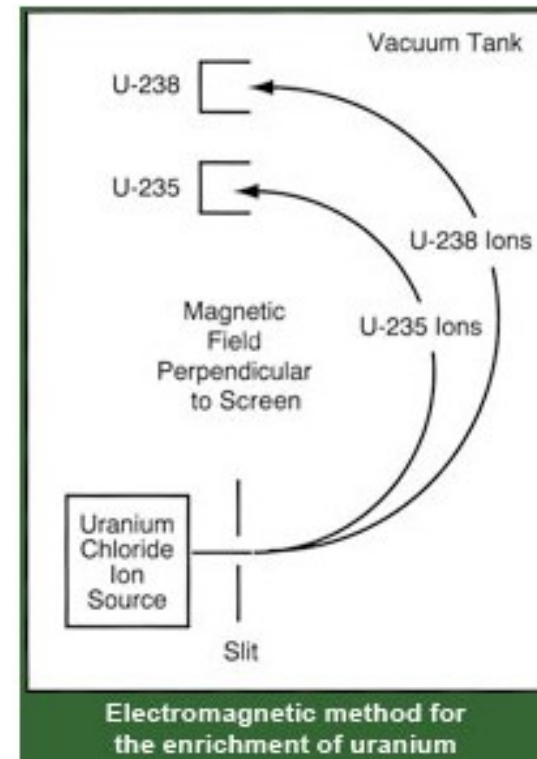
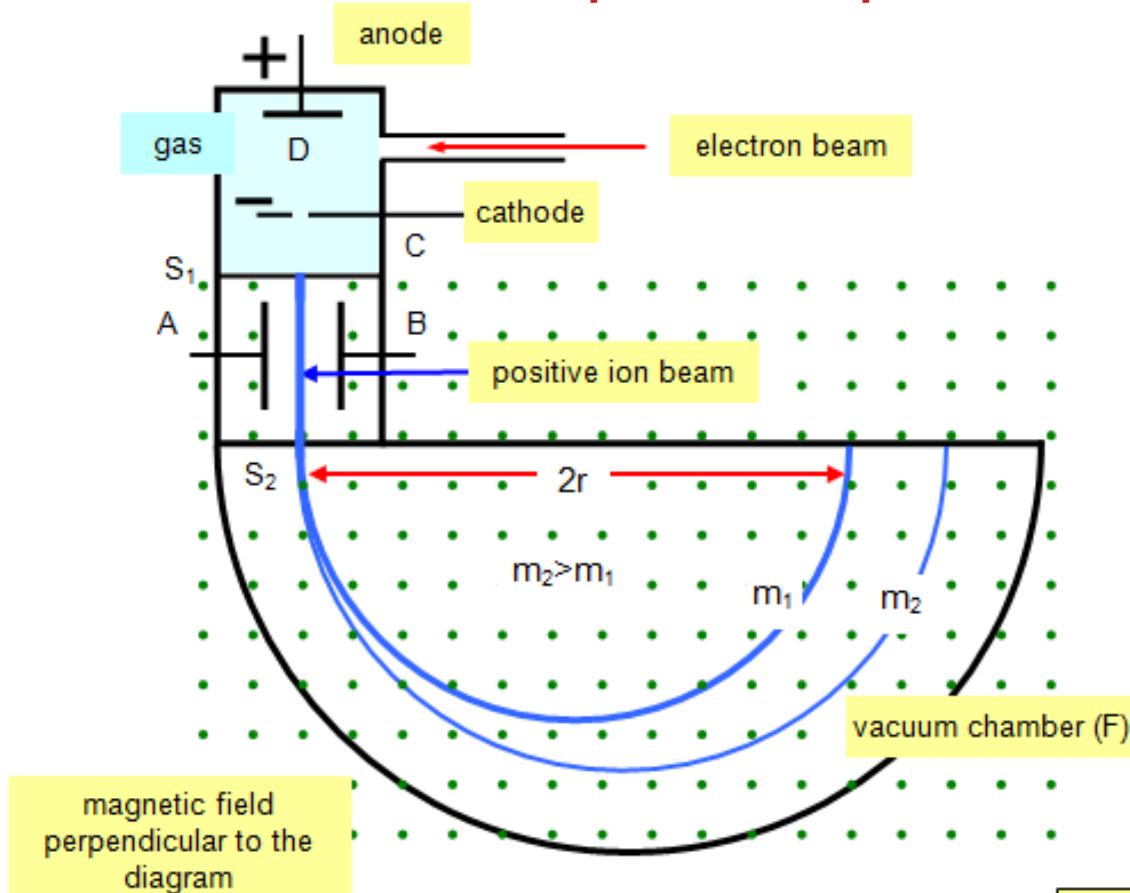
Large mass → needs large B
To get in the detector.

Larger mass → more inertia
→ larger R.



We can use the technique to separate isotopes.

$$m = \left(\frac{qr^2}{2U} \right) B^2$$



Application: Calutron used during WWII
To enrich uranium

1) What is the mass of particle bearing a single elementary charge if it travels in a circle whose radius of 35cm when it is going at 0.90 times the speed of light in a magnetic field of 8.0 Ts ? (the speed of light is $3 \times 10^8 \text{ m/s}$ or 186,000 miles/s)
Use equation slide 61

2) below. The particles are positive? Negative? Neutral

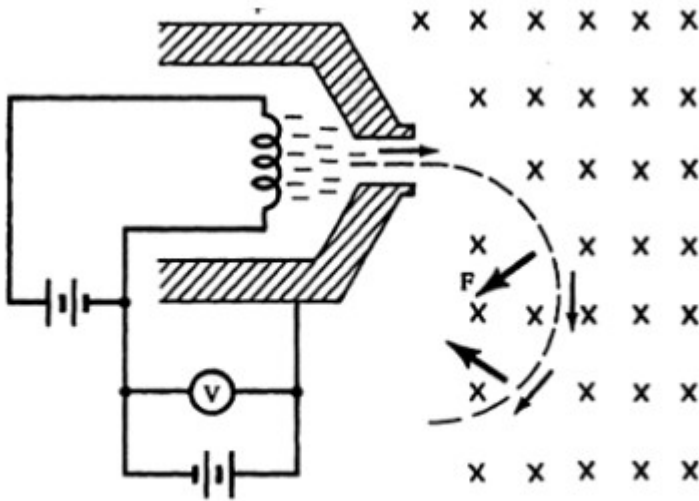


FIGURE 11-23

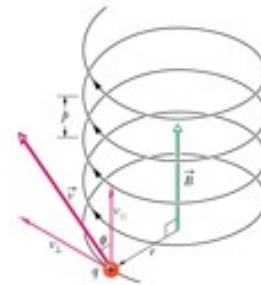
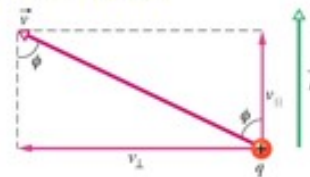
3) a positive charge (charge is e) enters a magnetic field region $B = 25 \text{ mTs}$
It was accelerated before through a potential $U = 150 \text{ V}$
The particle describes a circle of radius 7.1cm/ What is its mass?
Use equation slide 68

Motion of a Charged Particle in Magnetic Field

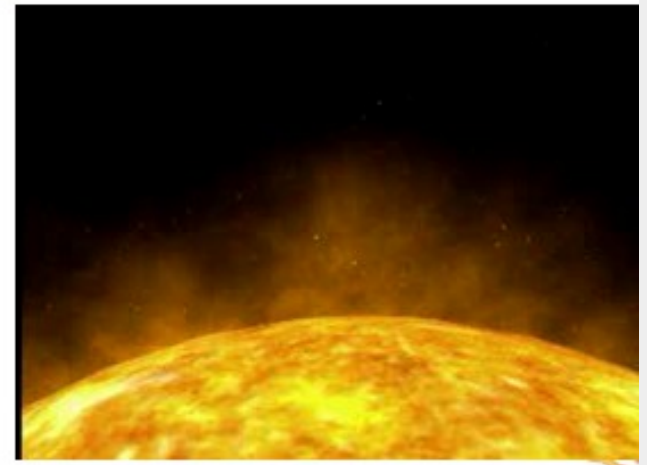
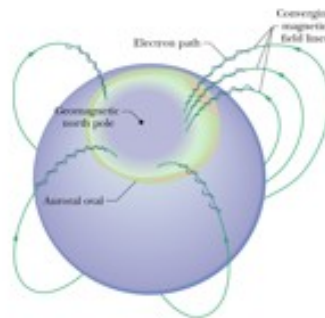
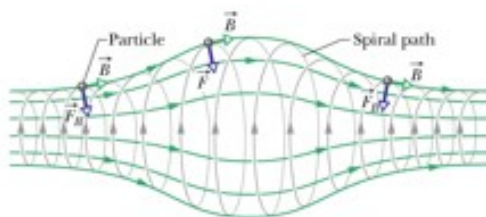
- Circle Paths: $\mathbf{v}$ is perpendicular to $\mathbf{B}$ (uniform);
- Helical Paths: $\mathbf{v}$ has a component parallel to $\mathbf{B}$.

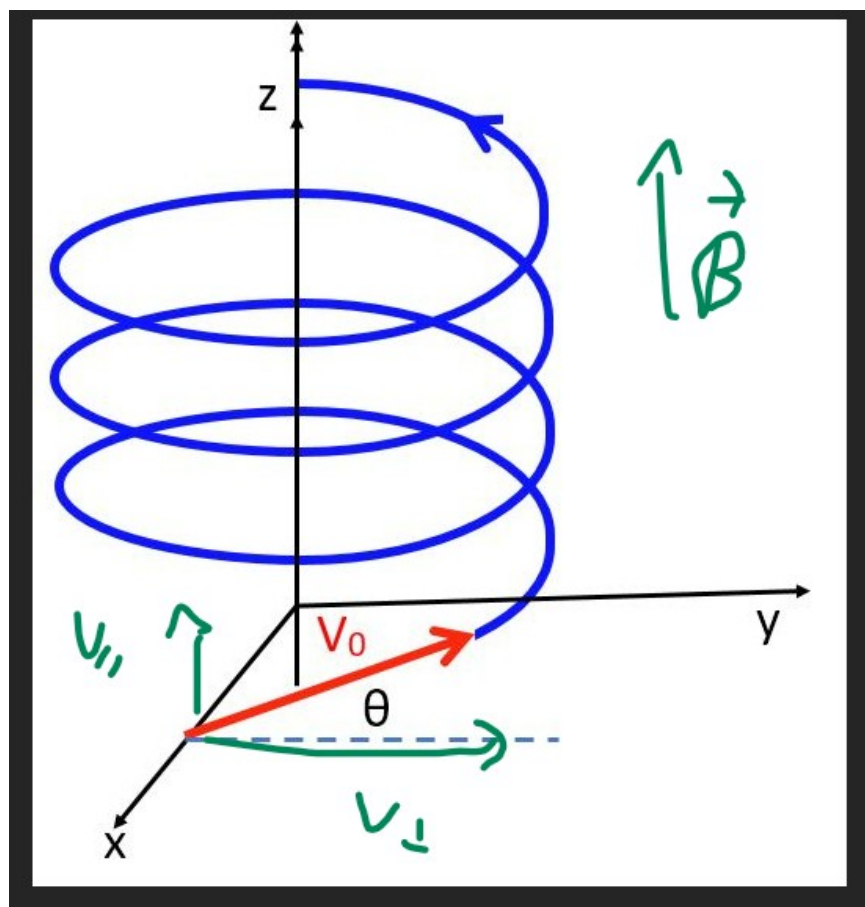
$$v_{\parallel} = v \cos \phi$$

$$v_{\perp} = v \sin \phi$$



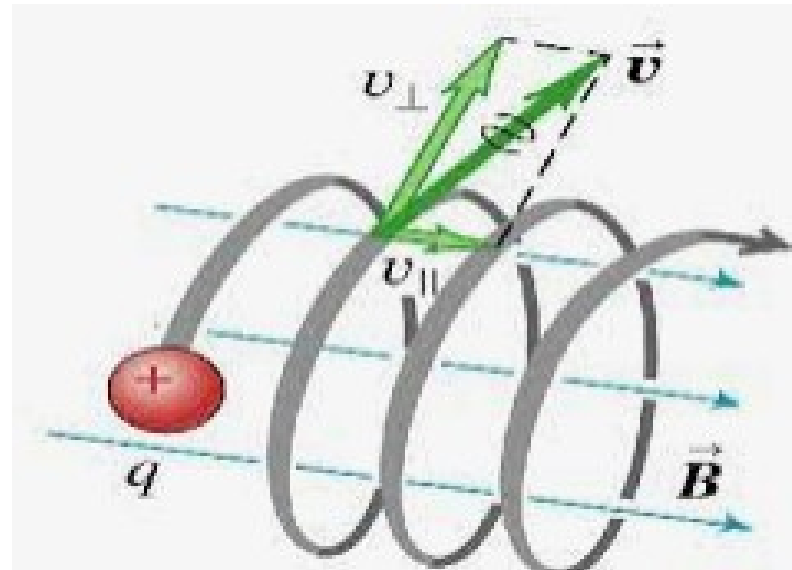
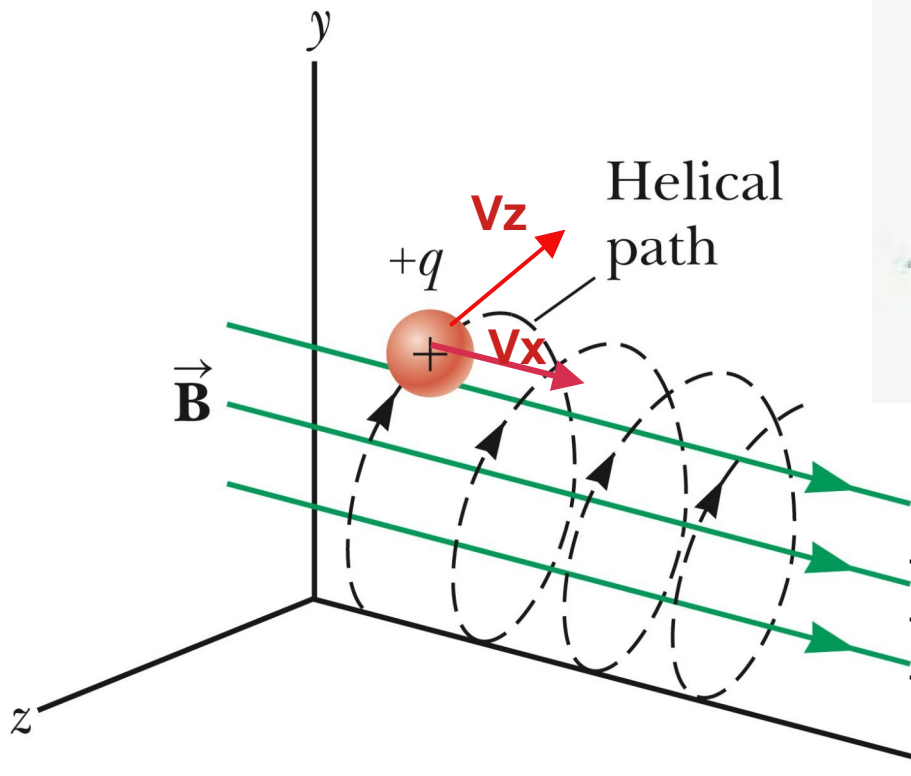
- Motion in a nonuniform magnetic field: strong at the ends and weak in the middle;
 - Magnetic bottle
 - Aurora





If there is a component of the velocity along $\vec{B}$ ($v_{\parallel}$) and 1 component Perpendicular, 2 motions are superimposed :Forward ($v_{\parallel}$ stays the same) And circular. The path is helical.

Simulation particles trapped In Earth field.

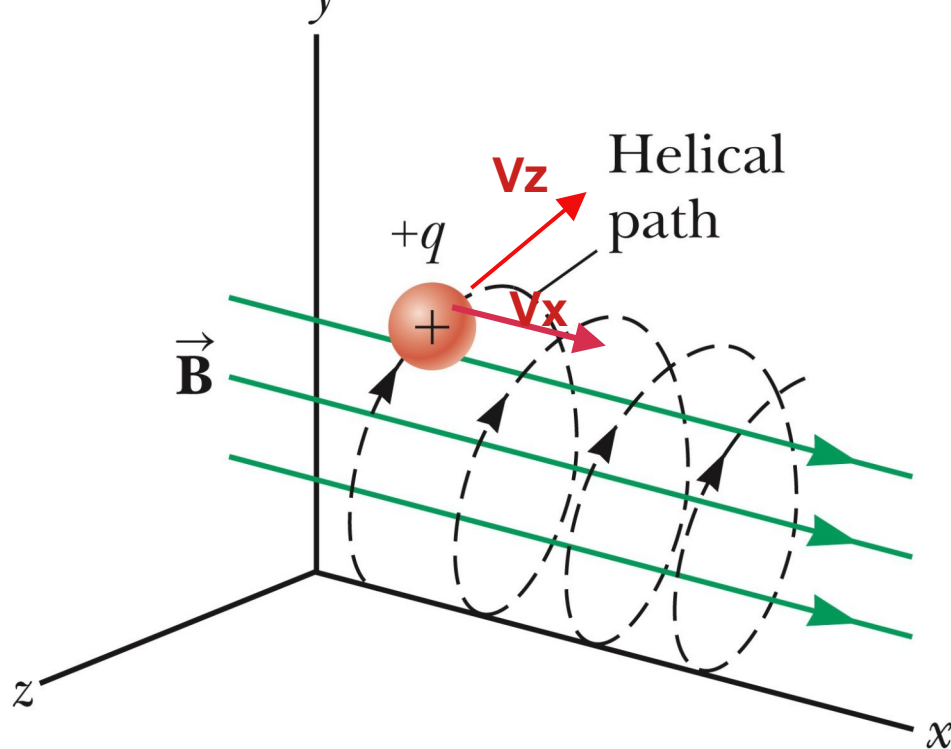


Textbook 27.4

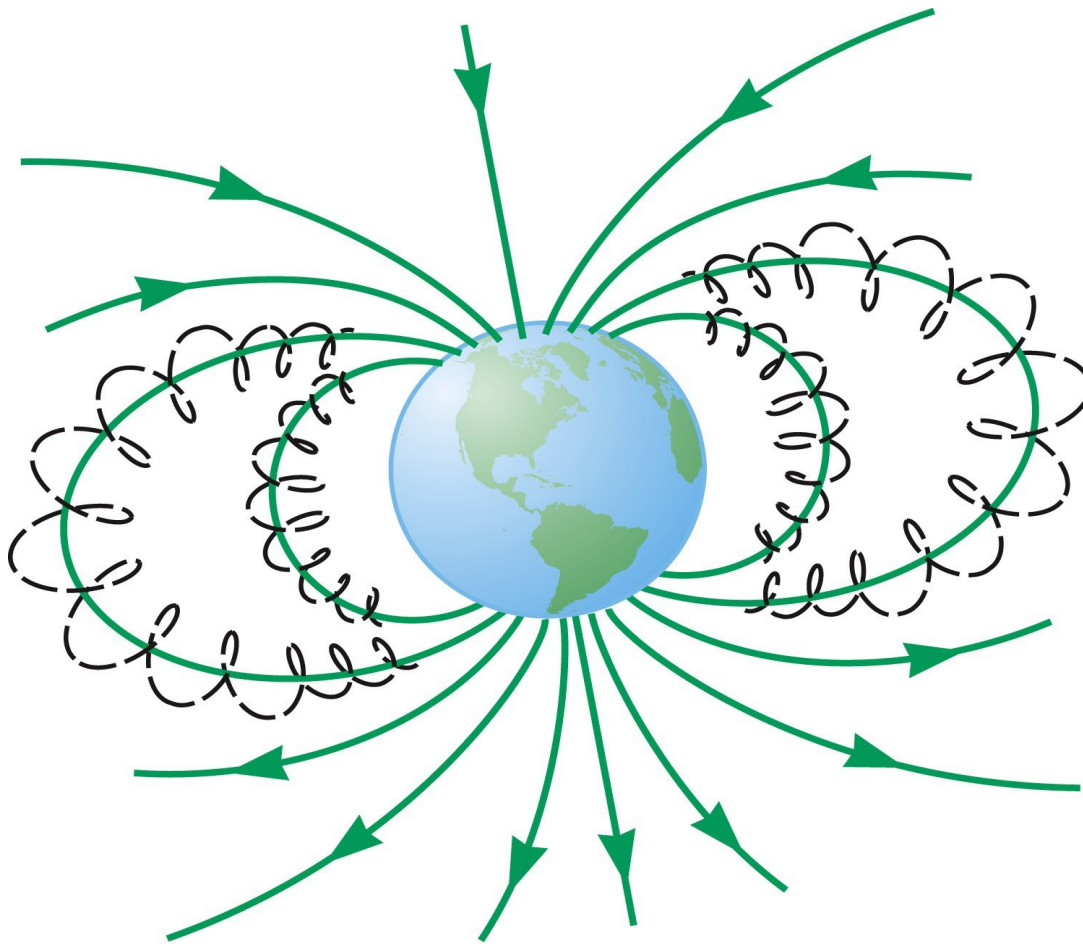
The magnetic field is along the x -axis

The velocity of the proton has 2 components

v_x (along $\vec{B}$) and v_z perpendicular to $\vec{B}$.



In a situation like that shown in Fig. 27.18, the charged particle is a proton ($q = 1.60 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) and the uniform, 0.500 T magnetic field is directed along the x -axis. At $t=0$ the proton has velocity components $v_x = 1.50 \times 10^5 \text{ m/s}$, $v_y = 0$, and $v_z = 2.00 \times 10^5 \text{ m/s}$. Only the magnetic force acts on the proton. (a) At $t=0$, find the force on the proton and its acceleration. (b) Find the radius of the resulting helical path, the angular speed of the proton, and the pitch of the helix (the distance traveled along the helix axis per revolution).



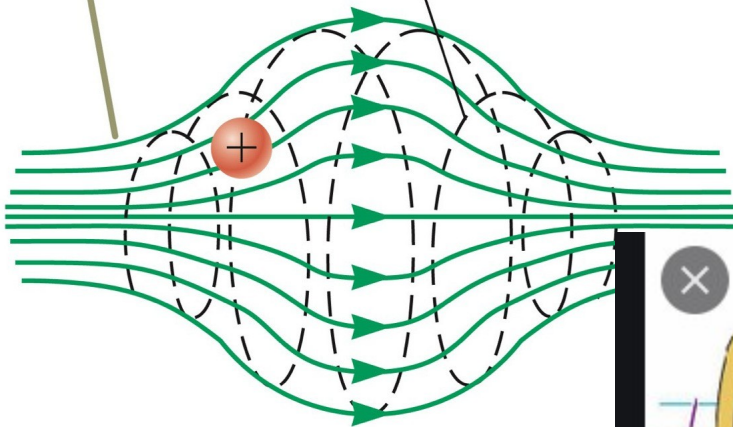
Van Allen radiation belts.
Discovered in 1958



See exploration of physical sciences
Particle in a magnetic field and in the magnetic field of the Earth

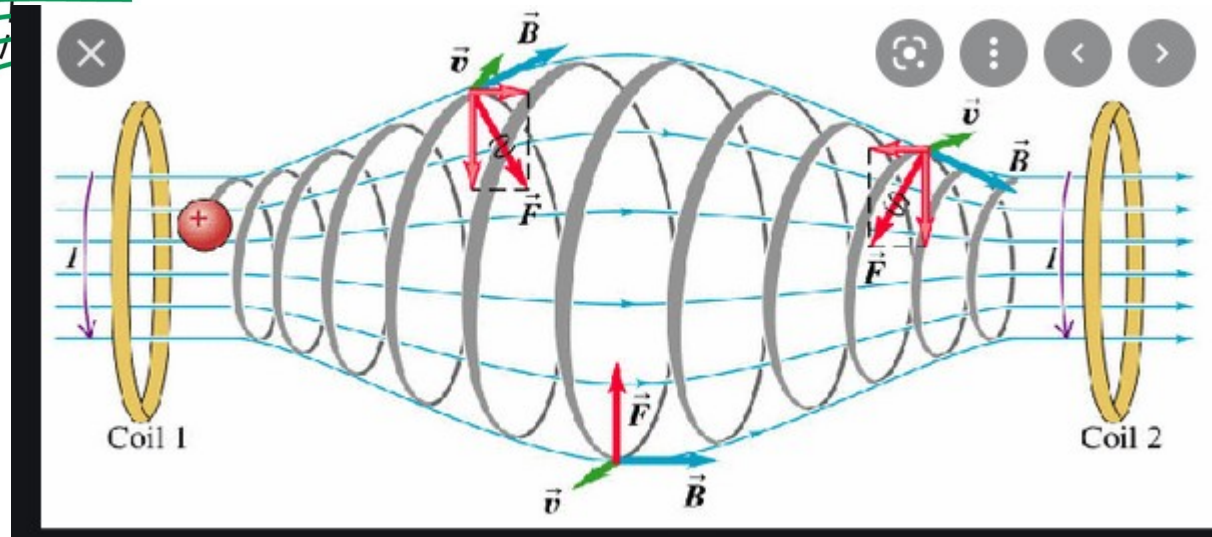
The magnetic force exerted on the particle near either end of the bottle has a component that causes the particle to spiral back toward the center.

Path of particle



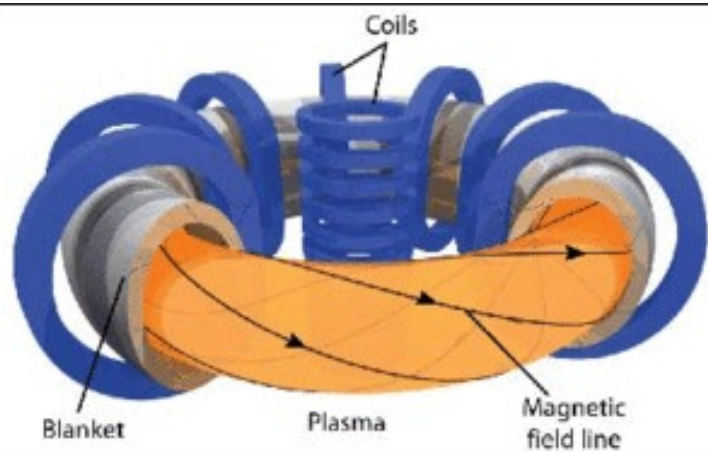
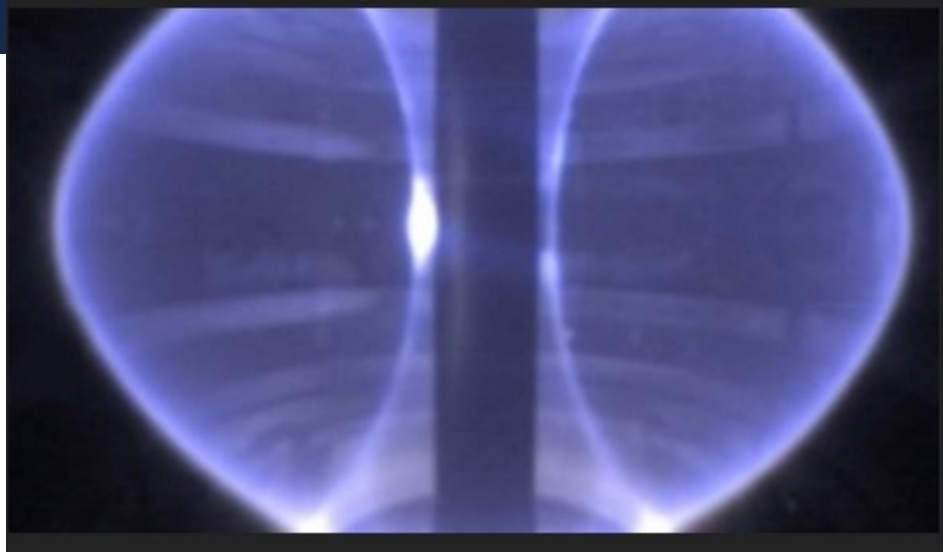
If the magnetic field is not Uniform, the radius change.
 $R = (V_m)/(qB)$
 If B increases $\rightarrow R$ decreases.

This is a magnetic bottle.
 Used to contain hot plasma
 That can not be contained
 In physical container.

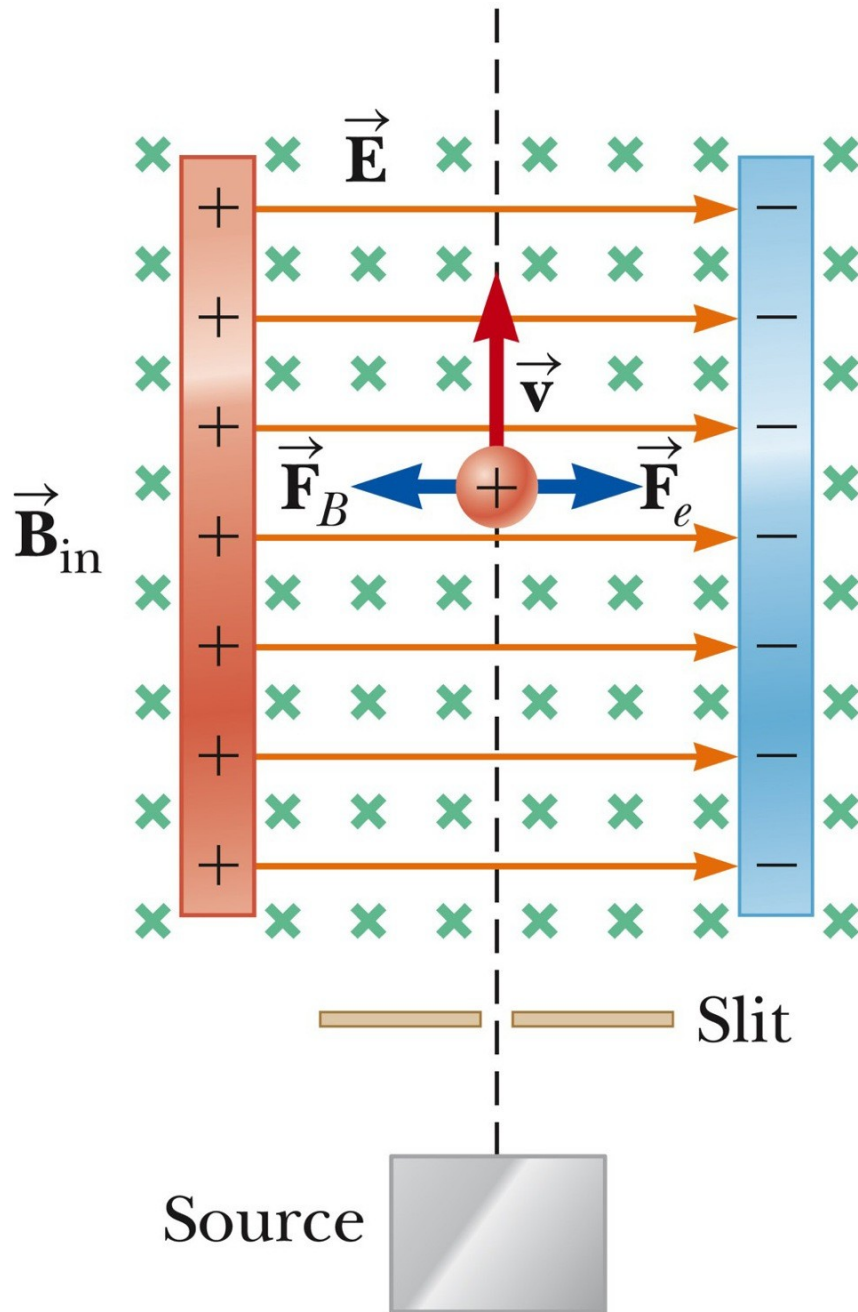


Applications: Cyclotron Motion

Plasma heating



Tokamak



Velocity selector

$$V = E/B$$

Simulation Physics

$$E = 5 \text{ N/C}$$

$$B = 1 \text{ T}$$

Mass spectrometer (Chem) to separate isotopes

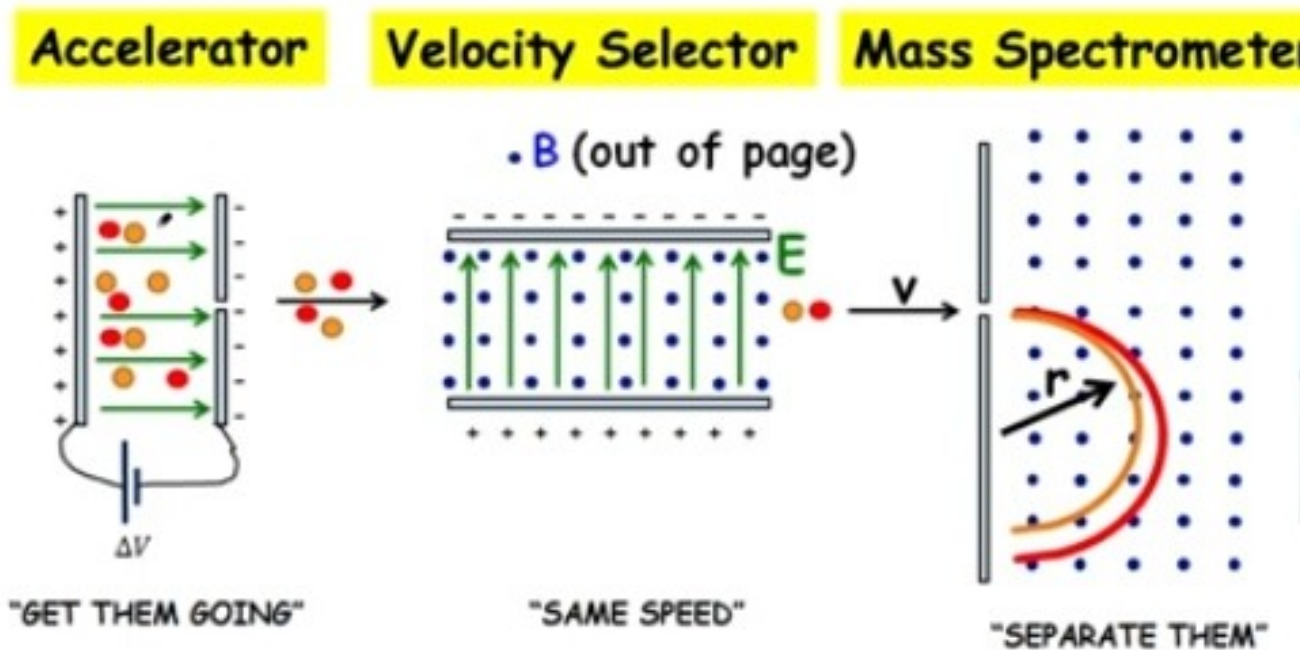
1) speed up isotopes in E

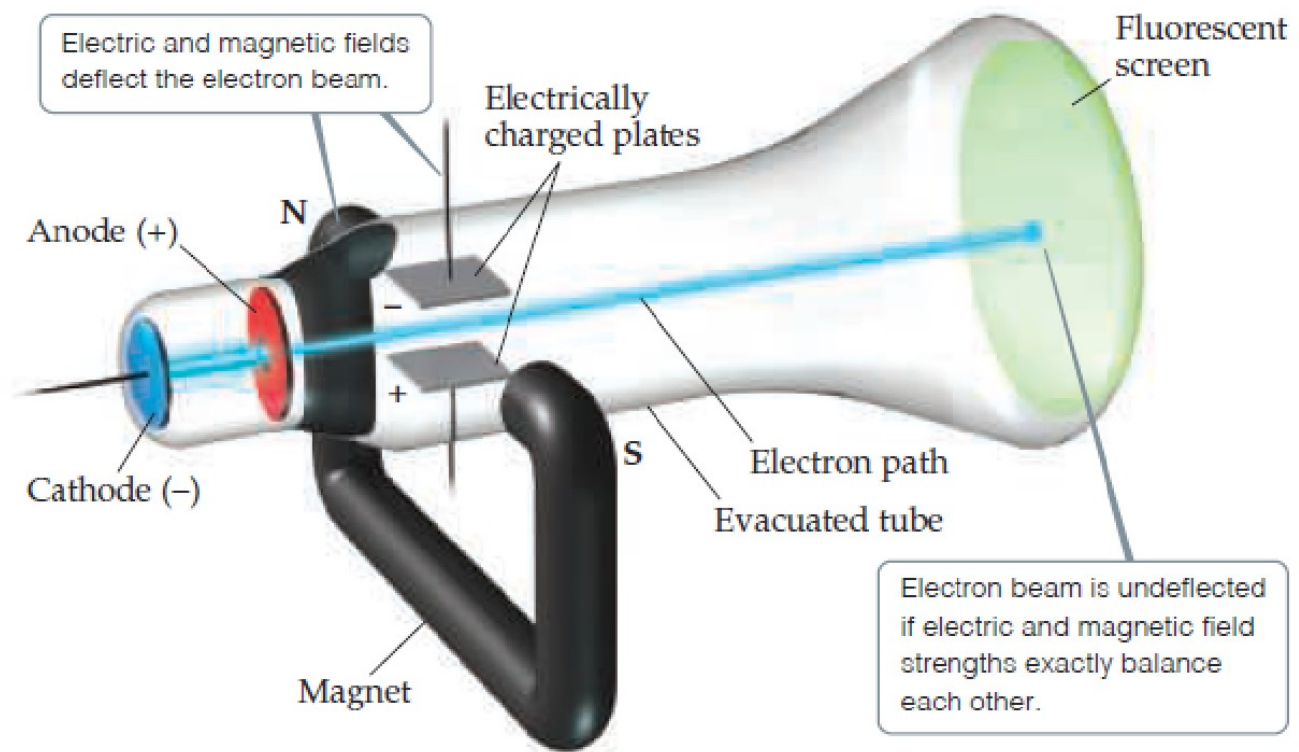
(speed given by about $qU = 0.5mv^2$)

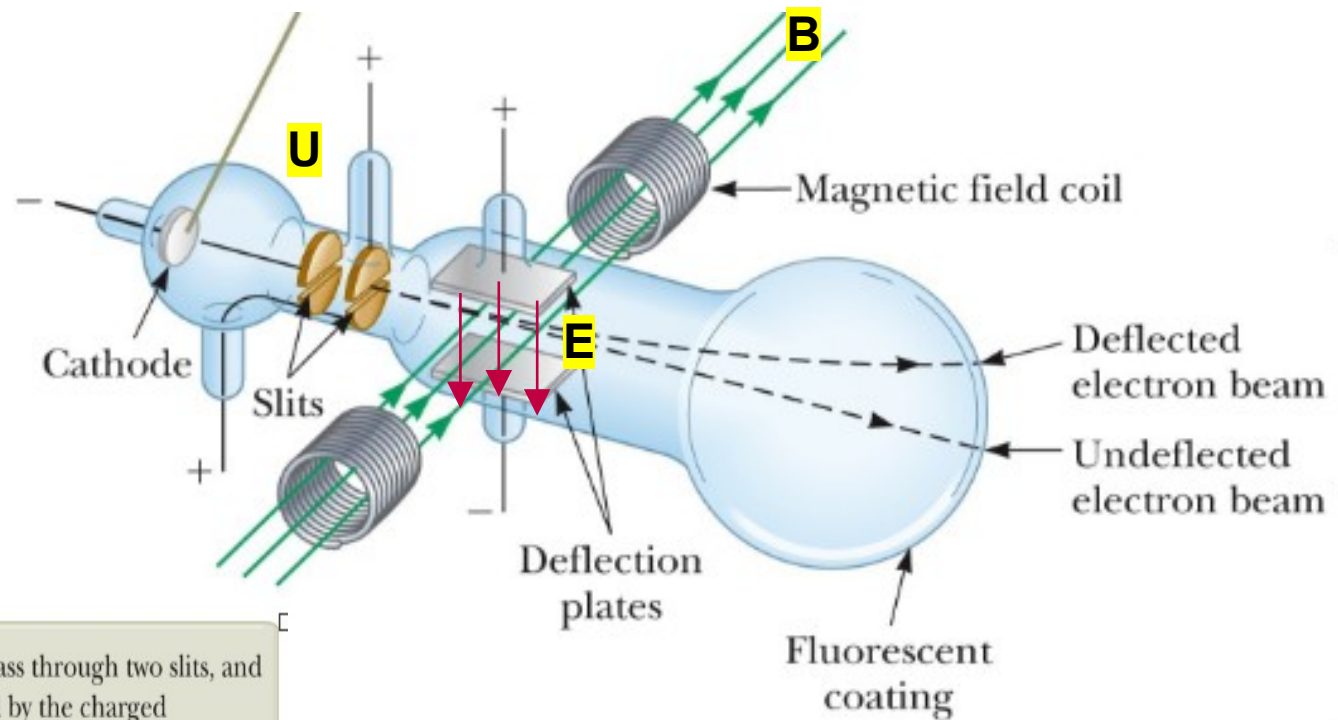
2) Speed selector. Use E and B

($qE = qvB$ or $E = vB$ or $v = E/B$)

3) curve the isotopes with B. $\longrightarrow r = \frac{mv}{qB}$







Electrons are accelerated from the cathode, pass through two slits, and are deflected by both an electric field (formed by the charged deflection plates) and a magnetic field (directed perpendicular to the electric field). The beam of electrons then strikes a fluorescent screen.

Speed $v = E/B$ $e/m = E^2/(2UB^2)$

Explain how it is done v is the speed of the electrons.

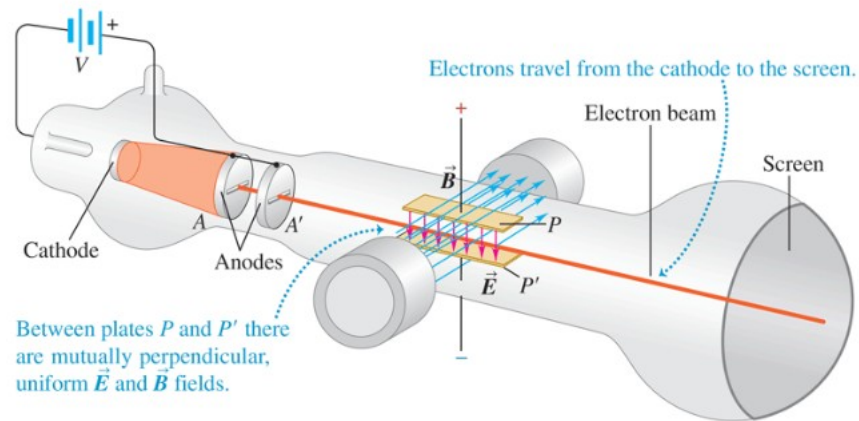
U is the voltage used to speed them up $qU = KE$ and $qVB = qE$

JJ Thomson experiment 1897 (1906 Nobel Prize) →

finding the ratio m/q for an electron

Then the charge was determined by Robert Millikan and Harvey Fletcher in 1923 to find the charge of an electron

Textbook 27.5 do it easy



Thomson's apparatus for measuring the ratio e/m for the electron.

Example 27.5 An e/m demonstration experiment

WITH VARIATION PROBLEMS

You set out to reproduce Thomson's e/m experiment with an accelerating potential of 150 V and a deflecting electric field of magnitude 6.0×10^6 N/C. (a) How fast do the electrons move? (b) What magnetic-field magnitude will yield zero beam deflection? (c) With this magnetic field, how will the electron beam behave if you increase the accelerating potential above 150 V?

$$eU = \frac{1}{2}mv^2$$

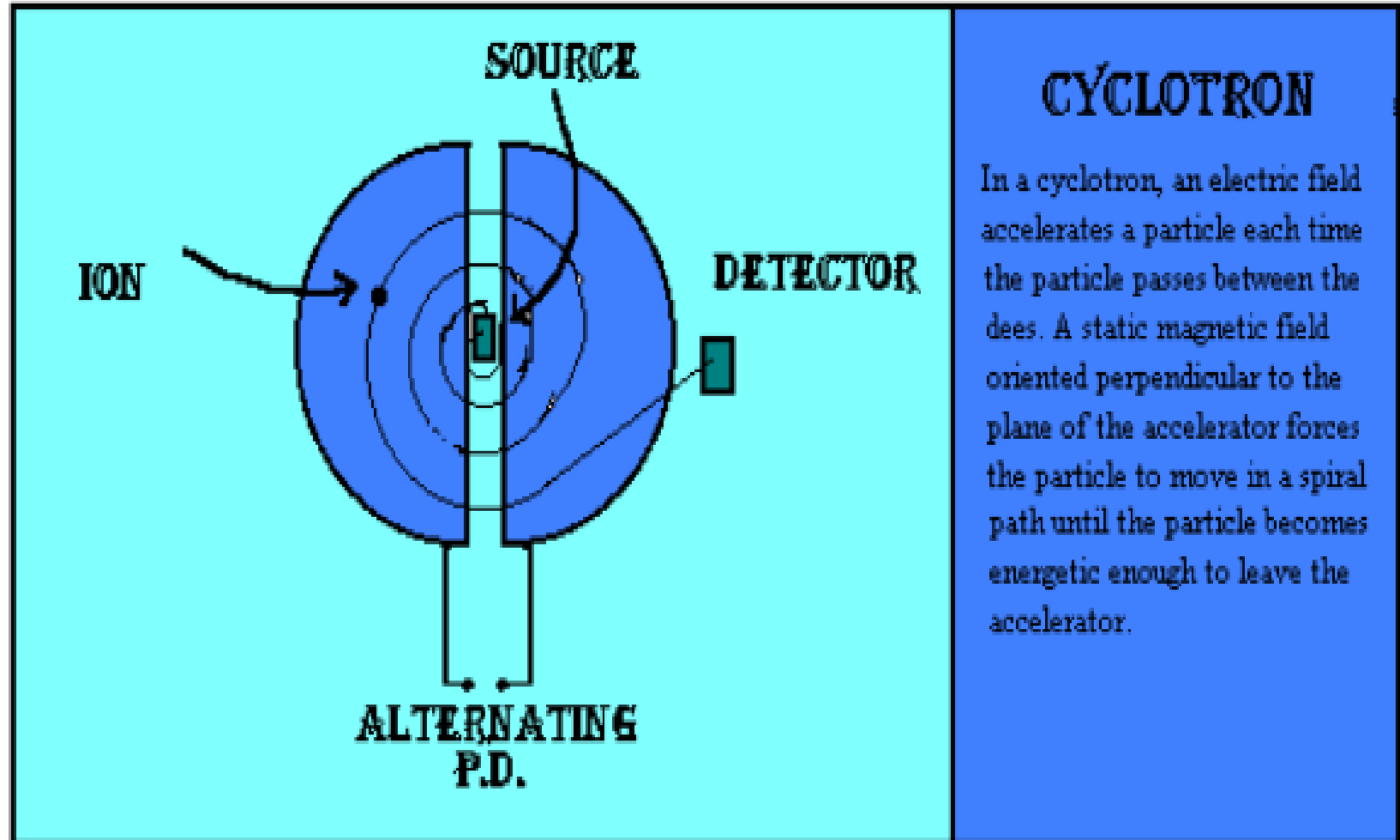


$$B = \frac{E}{v} \quad 0.825 \text{ T}$$

$$v = \sqrt{\frac{2eU}{m}}$$

$$F_{\text{magnetic}} = ev_{\text{new}}B > F_{\text{electric}}$$

- **Result:** The magnetic force overpowers the electric force, bending the beam **toward the magnetic force direction**. In Thomson's setup, if the electric field pushes electrons up, the magnetic field pushes down (or vice versa). With a stronger magnetic force, the beam curves **down** (assuming standard orientation—electric field up, magnetic field into the page, electrons moving right).

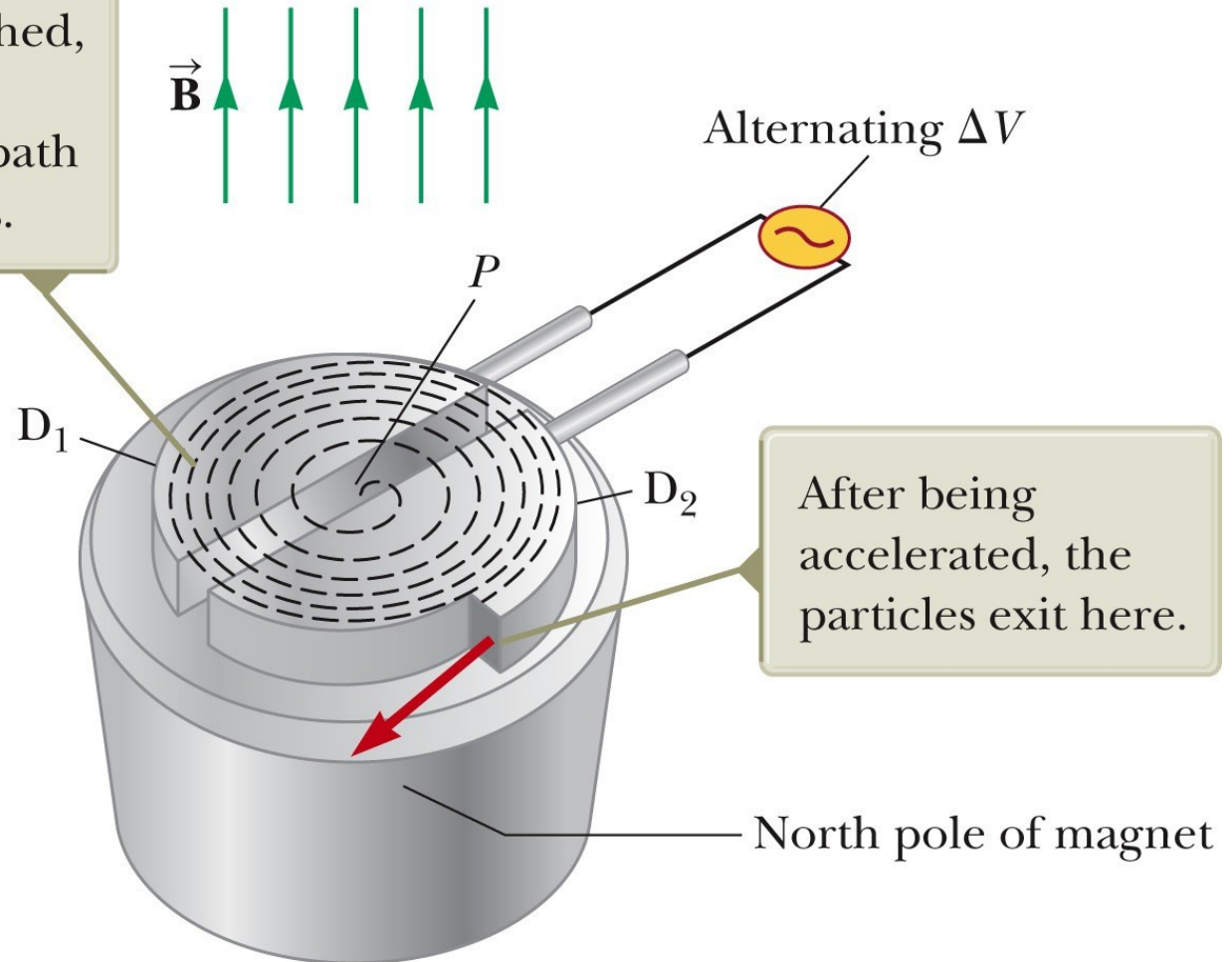


CYCLOTRON

In a cyclotron, an electric field accelerates a particle each time the particle passes between the dees. A static magnetic field oriented perpendicular to the plane of the accelerator forces the particle to move in a spiral path until the particle becomes energetic enough to leave the accelerator.

See PET scan . Because the period does not depend on the speed
Its easy to synchronize a pulsating E. See video LHC .

The black, dashed, curved lines represent the path of the particles.

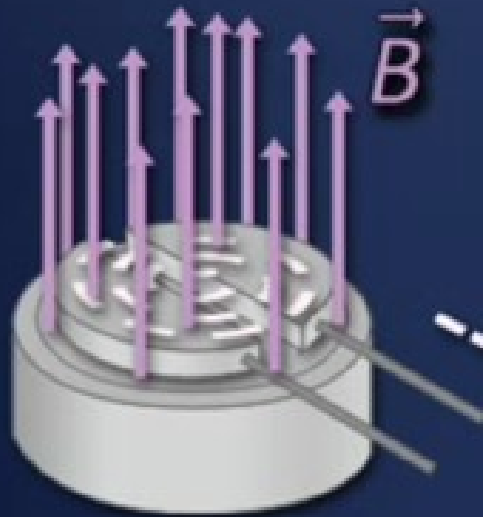


Example 5

a

© Cengage Learning. All Rights Reserved.

Applications: Cyclotron Motion



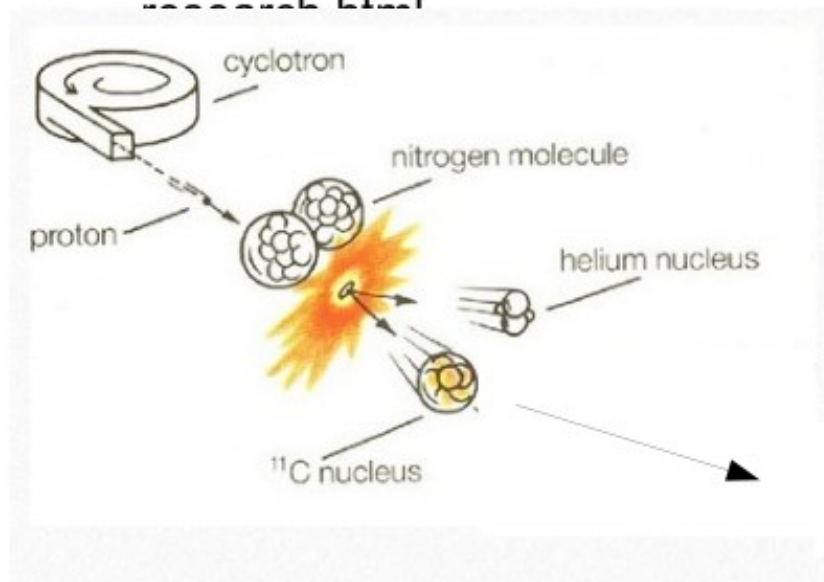
Cyclotron particle accelerator



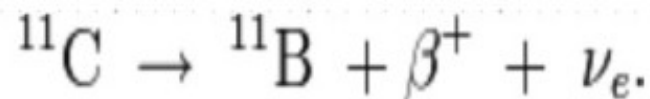
Used for PET scan

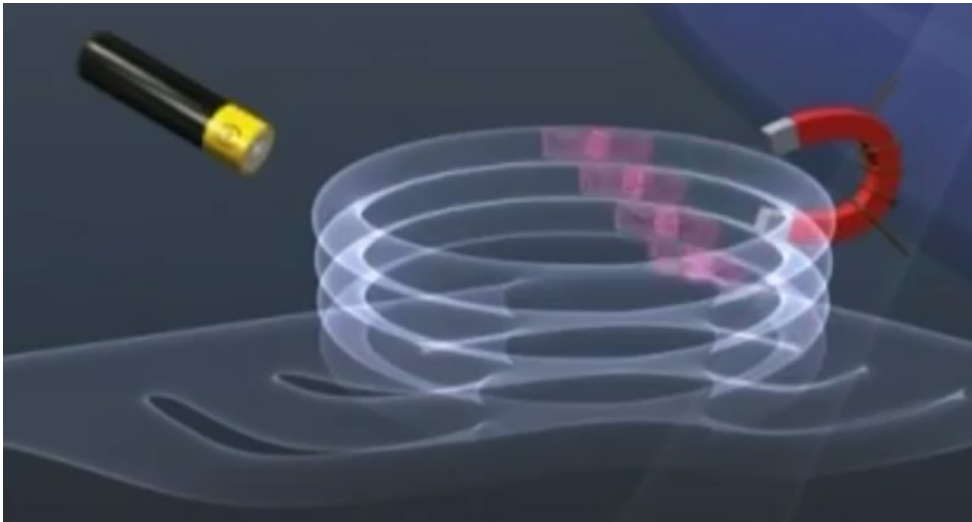
Radionuclide	Half-life	Chemical Form	Nuclear Reaction(s)
^{11}C	20 min	$^{11}\text{CO}_2$ / $^{11}\text{CH}_4$	$^{14}\text{N}(\text{p},\alpha)^{11}\text{C}$
^{13}N	10 min	$^{13}\text{NH}_4^+$ / $^{13}\text{NO}_x$	$^{16}\text{O}(\text{p},\alpha)^{13}\text{N}$
^{15}O	2 min	$^{15}\text{O}_2$	$^{15}\text{N}(\text{p},\text{n})^{15}\text{O}$ $^{14}\text{N}(\text{d},\text{n})^{15}\text{O}$
^{18}F	110 min	$^{18}\text{F}^-$ or $^{18}\text{F}_2$	$^{18}\text{O}(\text{p},\text{n})^{18}\text{F}$ $^{20}\text{Ne}(\text{d},\alpha)^{18}\text{F}$

<http://www.imperial.ac.uk/people/a.gee/>
research.html

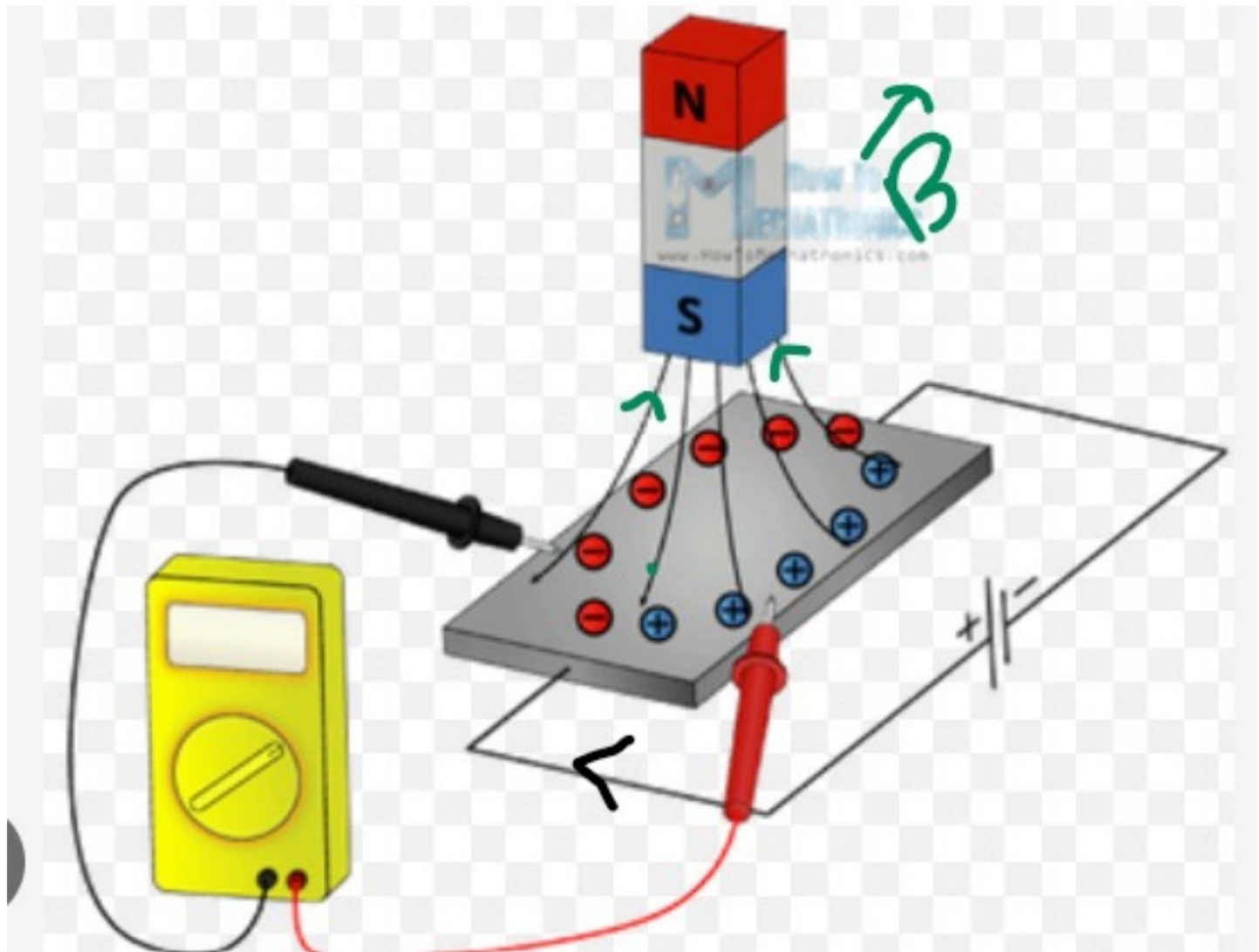


The radioisotopes used for PET scan have to be made in the hospitals using a cyclotron. (remember a magnetic field is used to trap protons around a circular path. An electric field is used to speed them up.)

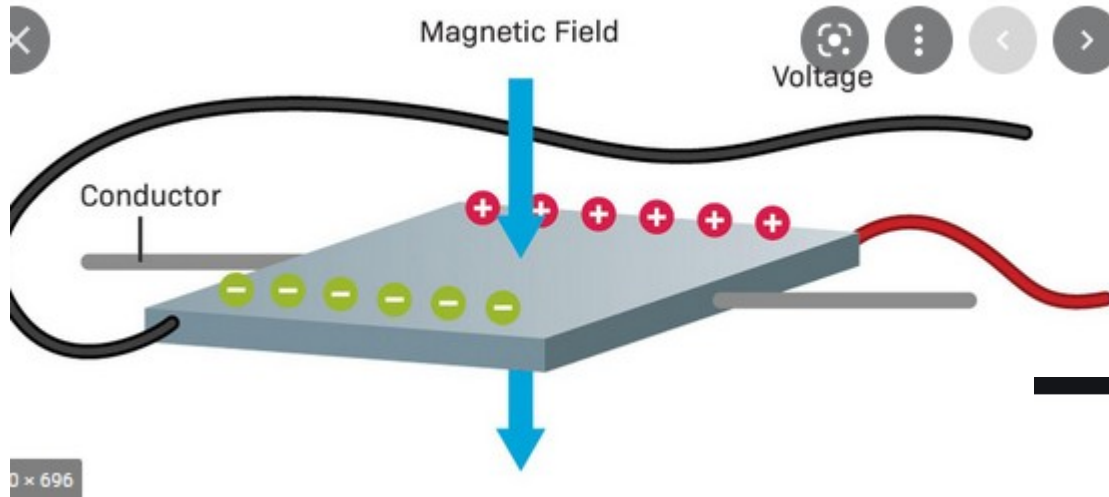




Video . LHC



Show animation



Application: Hall effect.

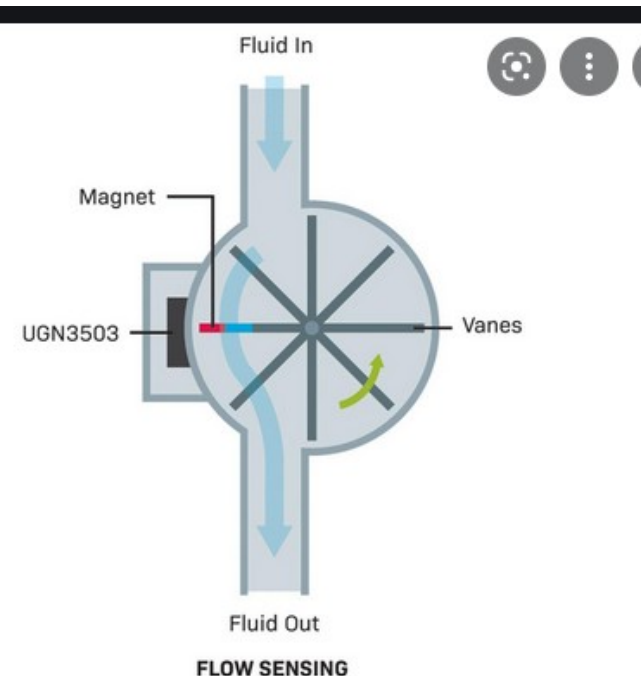
Used in sensors to:

- measure magnetic fields
- rotational speed
- (How fast you row in a rowing machine)
- proximity sensor
- current sensor
- leak sensor

See the link below. Really cool with arduino

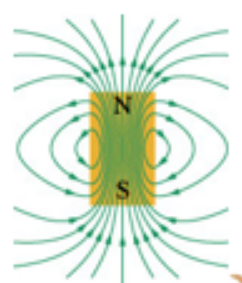
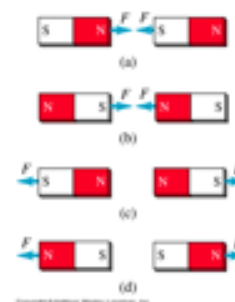
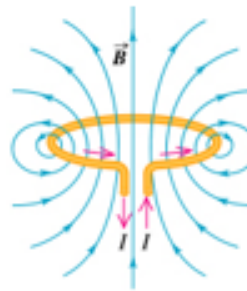
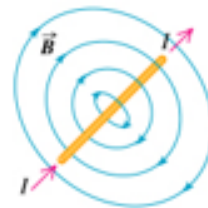
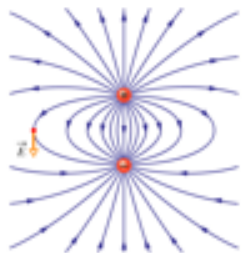
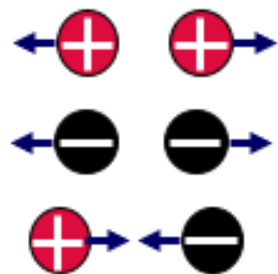
https://diyodemag.com/education/the_classroom_the_ugn3503_hall_effect_sensor

animation



Electric Field & Magnetic Field

- Electric forces acting at a distance through electric field.
 - Vector field, $\mathbf{E}$.
 - Source: electric charge.
 - Positive charge (+) and negative charge (-).
 - Opposite charges attract, like charges repel.
 - Electric field lines visualizing the direction and magnitude of $\mathbf{E}$.
- Magnetic forces acting at a distance through Magnetic field.
 - Vector field, $\mathbf{B}$
 - Source: **moving** electric charge (current or magnetic substance, such as permanent magnet).
 - North pole (N) and south pole (S)
 - Opposite poles attract, like poles repel.
 - Magnetic field lines visualizing the direction and magnitude of $\mathbf{B}$.



Copyright © 2008 Wiley-Interscience, Inc.

Motion of a Charged Particle in a Uniform Electric Field and Magnetic Field

- Charged particle in both electric field and magnetic field

$$\vec{F} = q \vec{E} + q \vec{v} \times \vec{B}$$

- Velocity Selector:

$$qE = qvB$$

$$v = \frac{E}{B}$$

- The Mass Spectrometer:

$$r = \frac{mv}{qB} \quad \frac{m}{q} = \frac{rB_0}{v} \quad \frac{m}{q} = \frac{rB_0 B}{E}$$

<http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/maspec.html#c3>

- The Cyclotron:

$$T = \frac{2\pi m}{|q|B} \quad f = f_{osc} = \frac{1}{T} \quad |q|B = 2\pi m f_{osc}$$

For PET scan
 Magnetron (microwave oven)
 Plasma container (fusion)
 LHC (particle accelerator)
 Astrophysics $f \rightarrow B$

Summary

- Magnetism involves **moving electric charge**

The magnetic force on a moving charge depends on the charge q , the magnetic field $\vec{B}$, and the charge's velocity: $\vec{v}$

$$\vec{F}_B = q\vec{v} \times \vec{B}$$



- Charges undergo circular or spiral motion in uniform magnetic fields
- Forces on electric currents result in torques that tend to align current loops with the magnetic field

The source of magnetism is moving charges

Electricity → creates a magnetic field that circulates.

What Magnetism is Really About

Moving Electric Charge

- Moving charge responds to magnetic fields
- Moving charge causes magnetic fields

21.2 The Force That a Magnetic Field Exerts on a Moving Charge The direction of the magnetic force acting on a charge moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$ is perpendicular to both $\vec{v}$ and $\vec{B}$. For a positive charge the direction can be determined with the aid of Right-Hand Rule No. 1 (see below). The magnetic force on a moving negative charge is opposite to the force on a moving positive charge.

Right-Hand Rule No. 1: Extend the right hand so the fingers point along the direction of the magnetic field $\vec{B}$ and the thumb points along the velocity $\vec{v}$ of the charge. The palm of the hand then faces in the direction of the magnetic force $\vec{F}$ that acts on a positive charge.

$$B = \frac{F}{|q_0|v \sin \theta} \quad (21.1)$$

The magnitude B of the magnetic field at any point in space is defined according to Equation 21.1, where F is the magnitude of the magnetic force on a test charge, $|q_0|$ is the magnitude of the test charge, and v is the magnitude of the charge's velocity, which makes an angle θ with the direction of the magnetic field. The SI unit for the magnetic field is the tesla (T). Another, smaller unit for the magnetic field is the gauss; $1 \text{ gauss} = 10^{-4} \text{ tesla}$. The gauss is not an SI unit.

21.3 The Motion of a Charged Particle in a Magnetic Field When a charged particle moves in a region that contains both magnetic and electric fields, the net force on the particle is the vector sum of the magnetic and electric forces.

A magnetic force does no work on a charged particle moving as in Figure 21.9b, because the direction of the force is always perpendicular to the motion of the particle. Being unable to do work, the magnetic force cannot change the kinetic energy, and hence the speed, of the particle; however, the magnetic force does change the direction in which the particle moves.

When a particle of charge q (magnitude $= |q|$) and mass m moves with speed v perpendicular to a uniform magnetic field of magnitude B , the magnetic force causes the particle to move on a circular path that has a radius given by Equation 21.2.

$$r = \frac{mv}{|q|B} \quad (21.2)$$

21.4 The Mass Spectrometer The mass spectrometer is an instrument for measuring the abundance of ionized atoms or molecules that have different masses. The atoms or molecules are ionized (charge $= +e$), accelerated to a speed v by a potential difference V , and sent into a uniform magnetic field of magnitude B . The magnetic field causes the particles (each with a mass m) to move on a circular path of radius r . The relation between m and B is given by Equation 1.

$$m = \left(\frac{er^2}{2V} \right) B^2 \quad (1)$$

21.5 The Force on a Current in a Magnetic Field An electric current, being composed of moving charges, can experience a magnetic force when placed in a magnetic field of magnitude B . For a straight wire that has a length L and carries a current I , the magnetic force has a magnitude that is given by Equation 21.3, where θ is the angle between the directions of the current and the magnetic field. The direction of the force is perpendicular to both the current and the magnetic field and is given by Right-Hand Rule No. 1.

$$F = ILB \sin \theta \quad (21.3)$$

21.6 The Torque on a Current-Carrying Coil Magnetic forces can exert a torque on a current-carrying loop of wire and thus cause the loop to rotate. When a current I exists in a coil of wire with N turns, each of area A , in the presence of a magnetic field of magnitude B , the coil experiences a net torque that has a magnitude given by Equation 21.4, where ϕ is the angle between the direction of the magnetic field and the normal to the plane of the coil. The quantity NIA is known as the magnetic moment of the coil.

$$\tau = NIAB \sin \phi \quad (21.4)$$